



On bootstrap tokens, secure tokens, and volume ownership



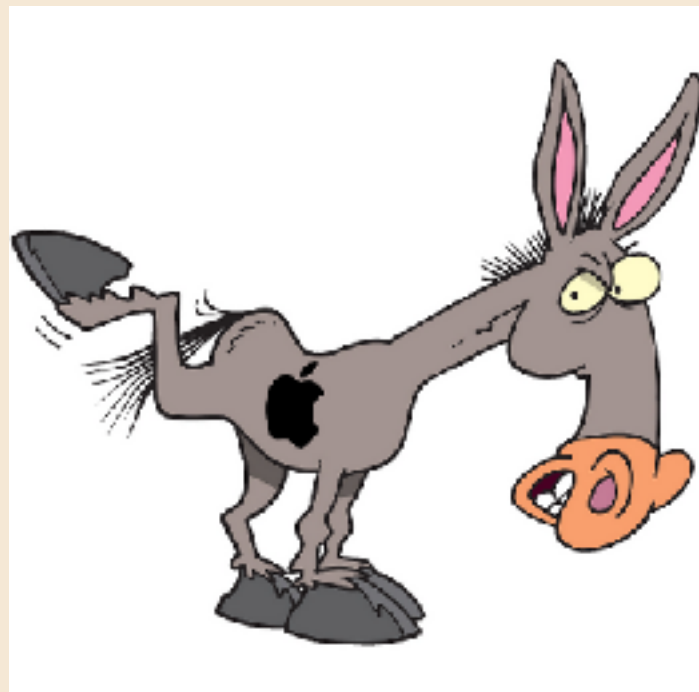
dataJAR
Beyond Device Management



Ben Toms

dataJAR | Head of Innovation and Platform | ben@datajar.co.uk





macmule

macmule.com |  macmule |  @macmuleblog



Secure Tokens



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Secure Tokens

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Bootstrap Tokens



Secure Tokens

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Bootstrap Tokens

Volume ownership

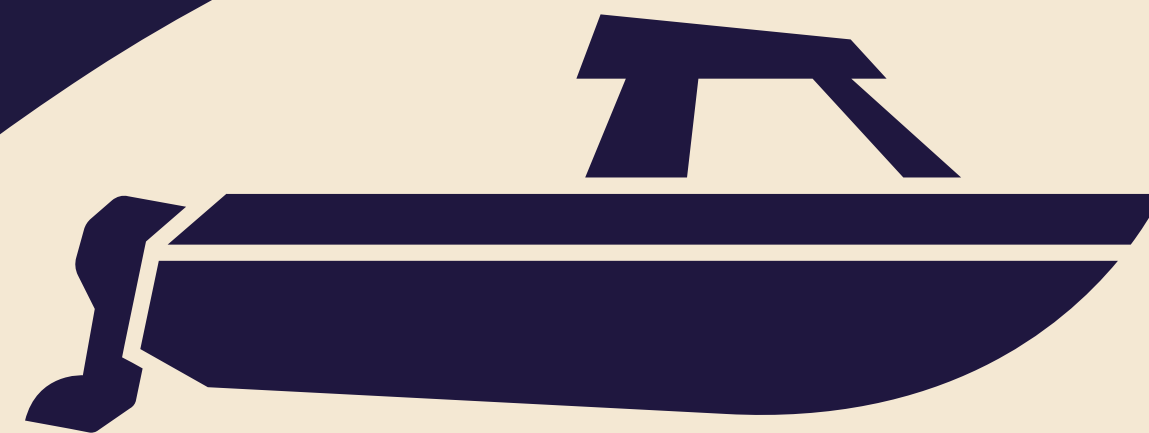


Secure Tokens

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Bootstrap Tokens

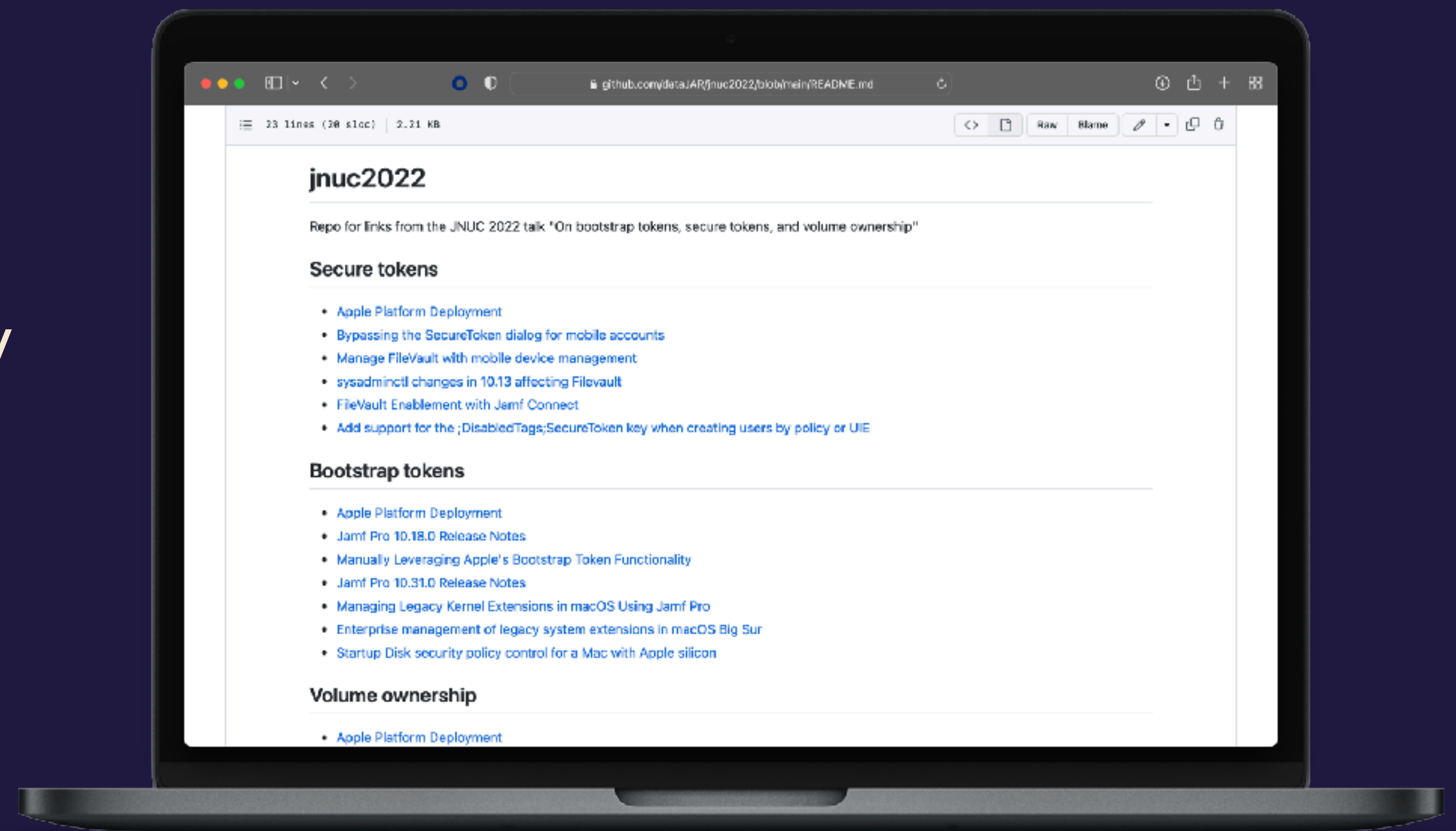
Volume ownership



Recap



[https://github.com/dataJAR/
jnuc2022](https://github.com/dataJAR/jnuc2022)

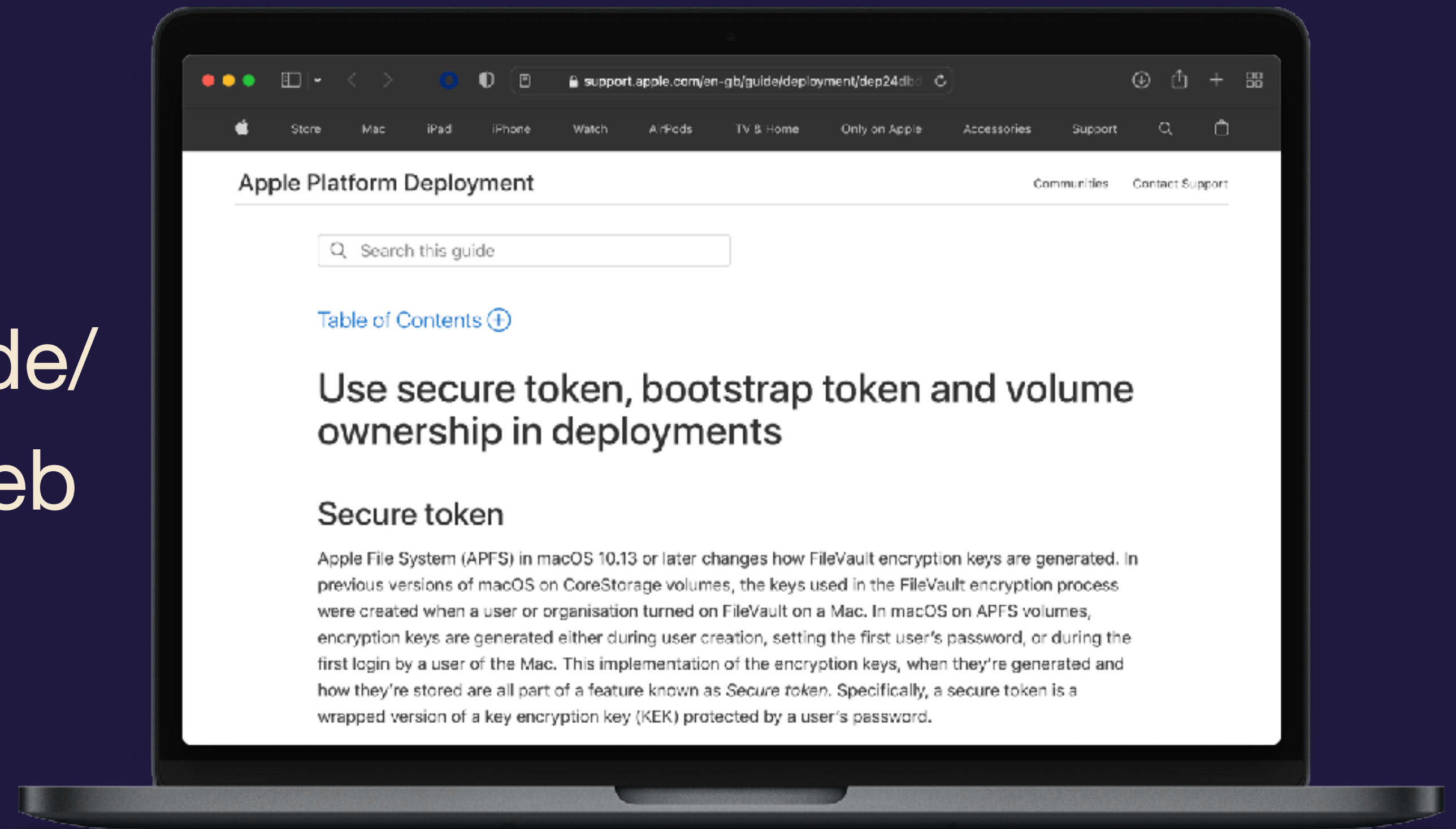


Secure tokens



Secure Tokens

<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>



Secure Tokens



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Secure Tokens

“Apple File System (APFS) in macOS 10.13 or later changes how FileVault encryption keys are generated.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Secure Tokens

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“In macOS on APFS volumes, encryption keys are generated either during user creation, setting the first user’s password, or during the first login by a user of the Mac.”



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Secure Tokens

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Secure Tokens

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Secure Tokens

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Secure Tokens



...I did my own research



Secure Tokens

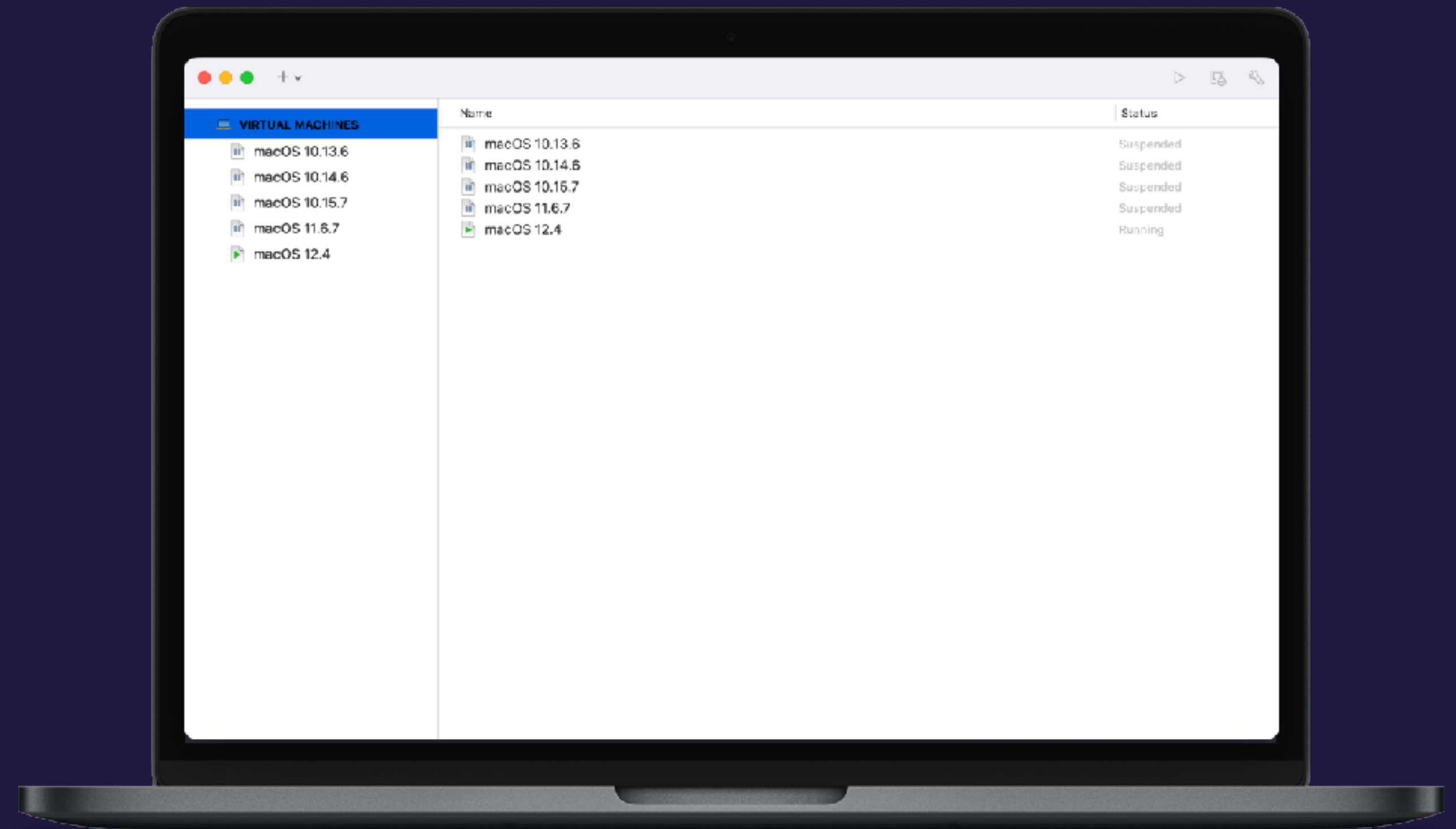
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Secure Tokens

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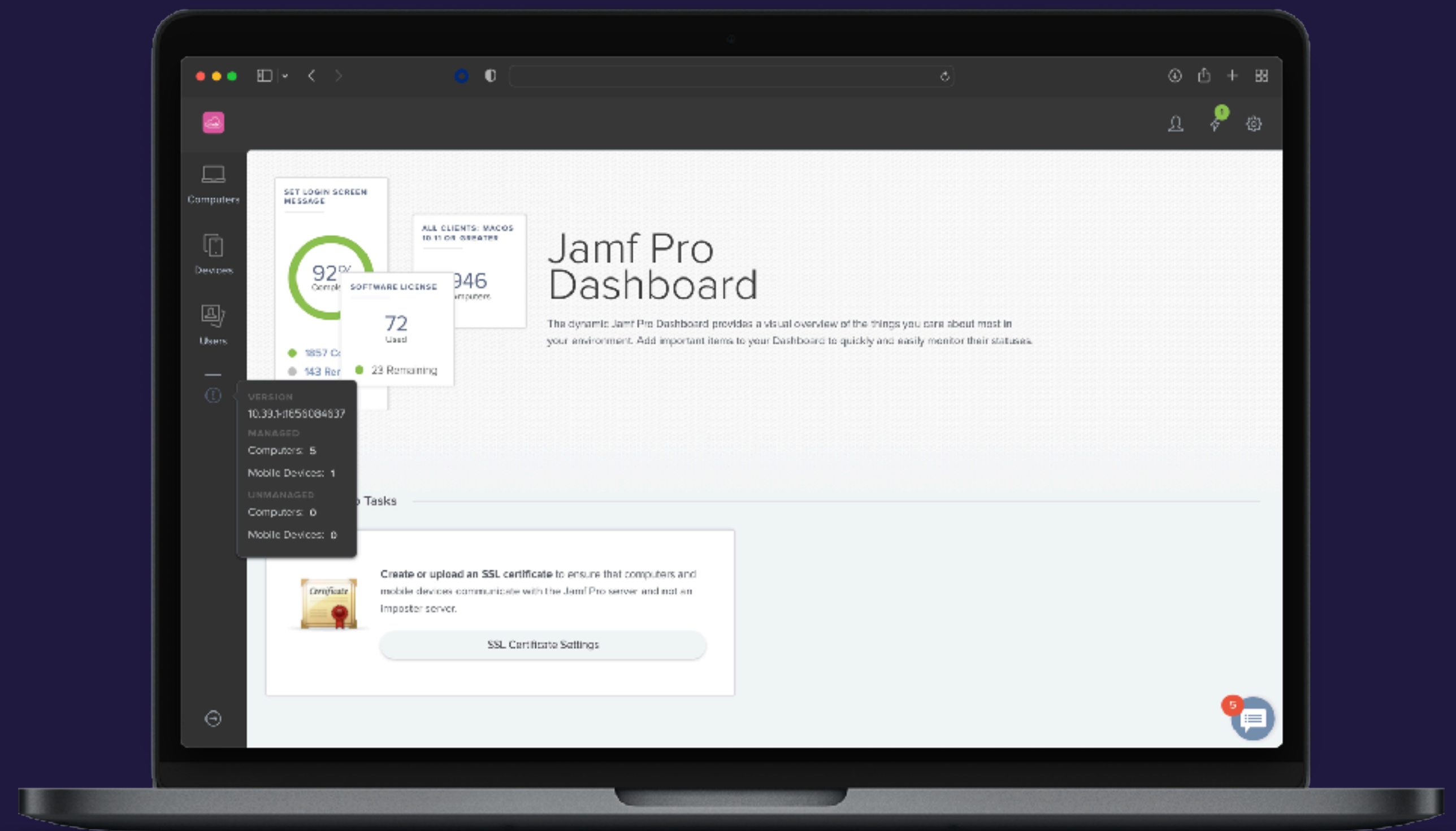
- Intel Virtual Machines



Secure Tokens

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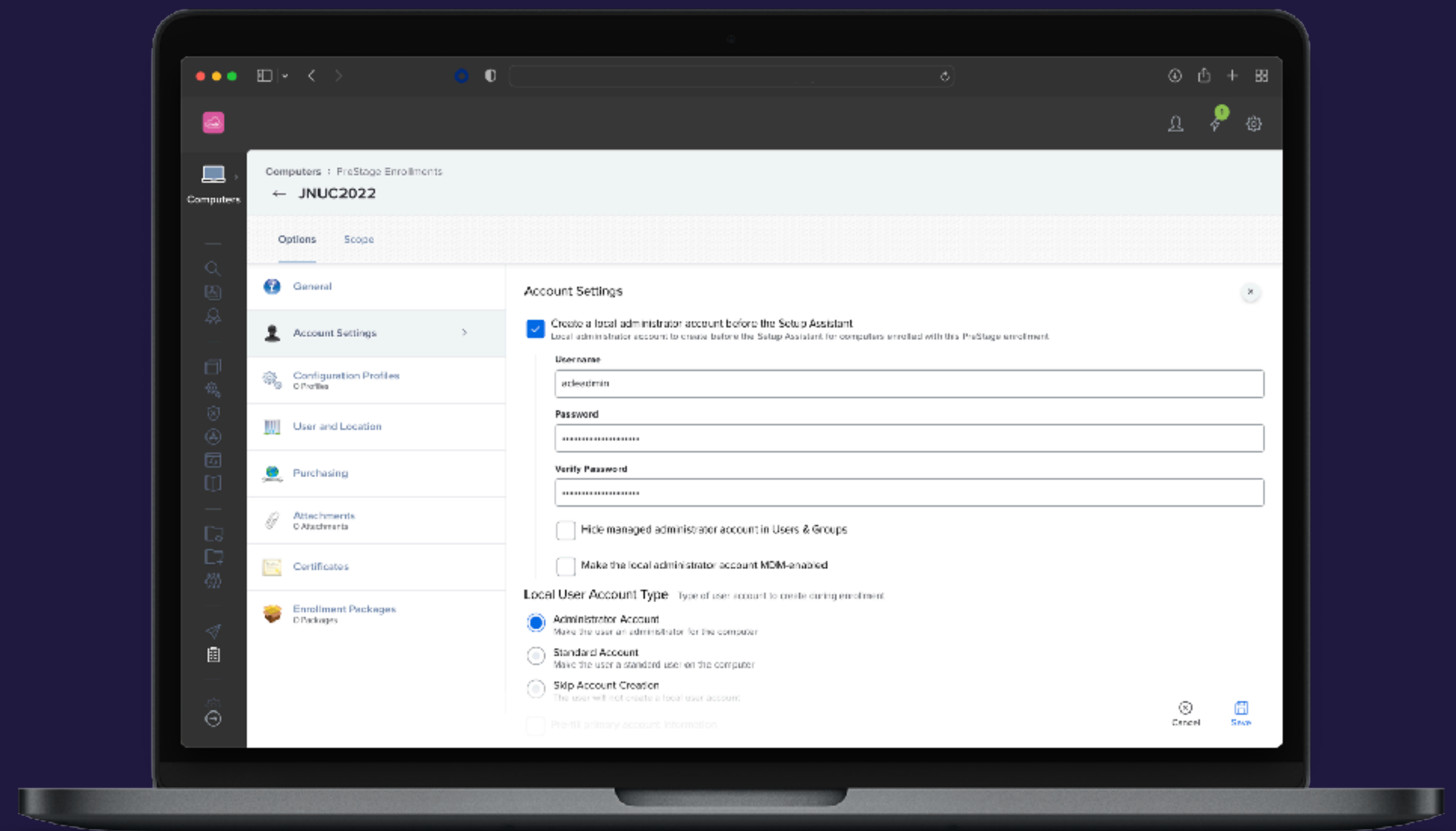
- Intel Virtual Machines
- Jamf Pro 10.39.1



Secure Tokens

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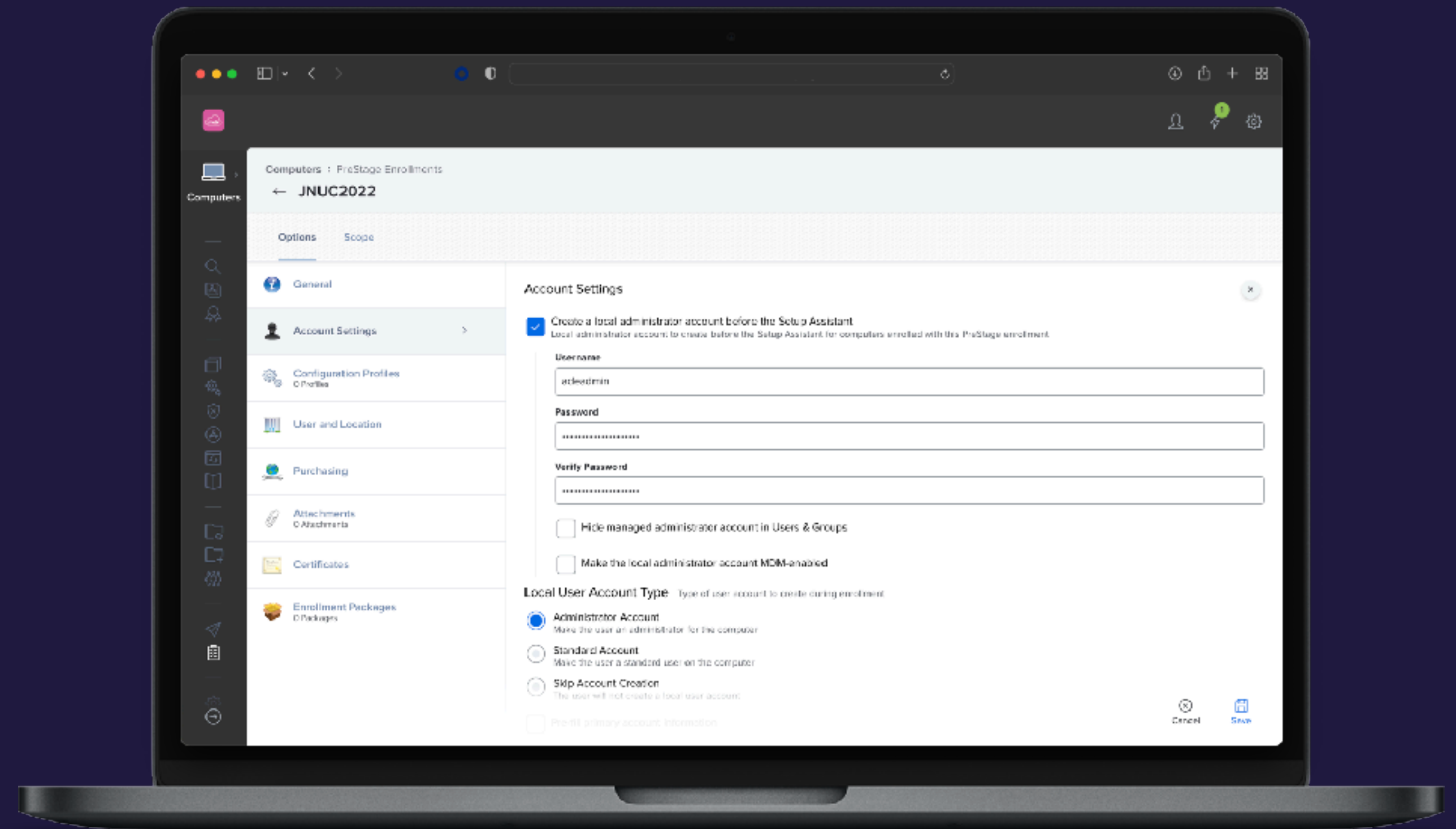
- Intel Virtual Machines
- Jamf Pro 10.39.1
- Enrolled into Jamf Pro via ADE



Secure Tokens

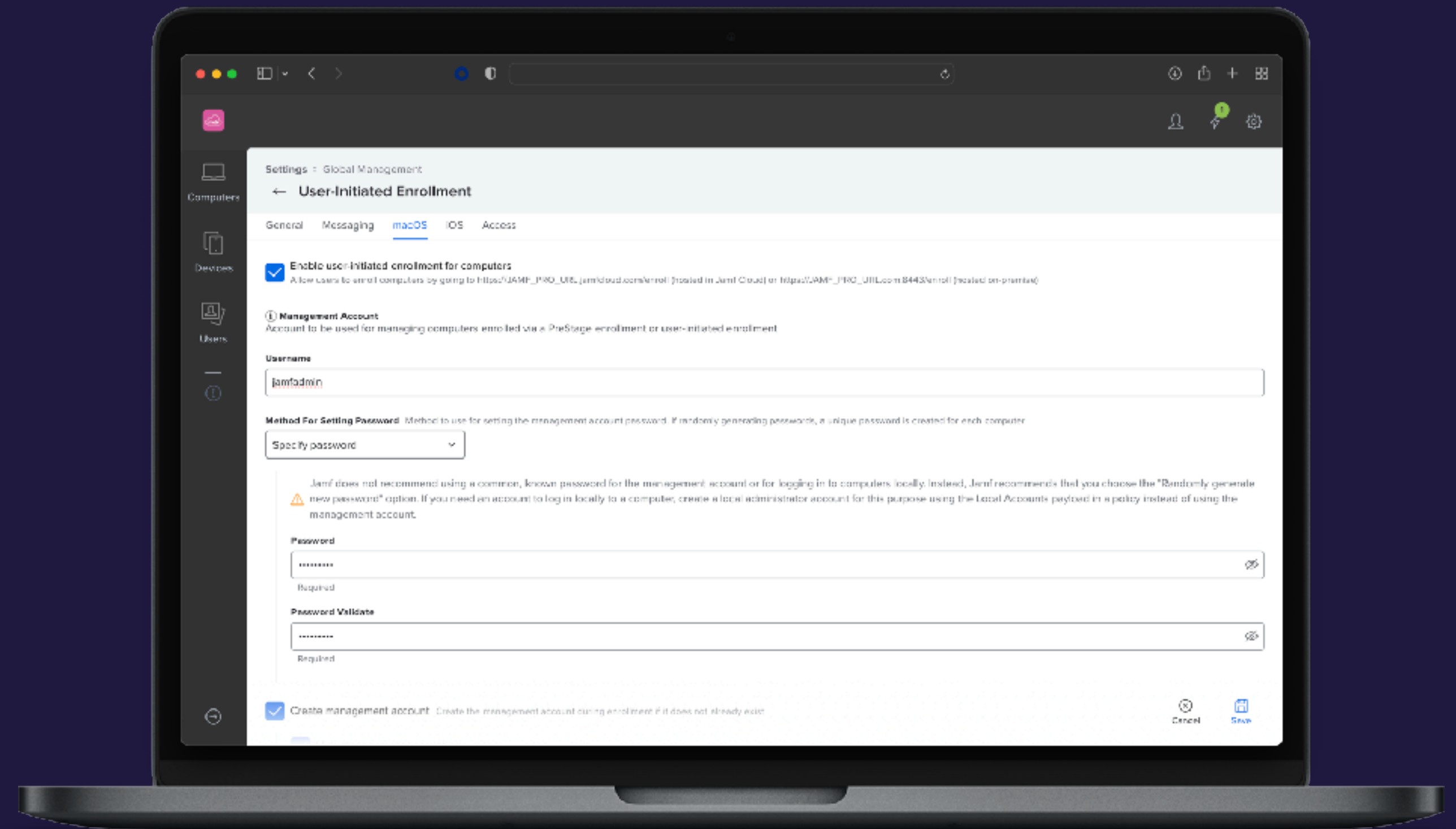
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- Intel Virtual Machines
- Jamf Pro 10.39.1
- Enrolled into Jamf Pro via ADE
- adeadmin created via prestage



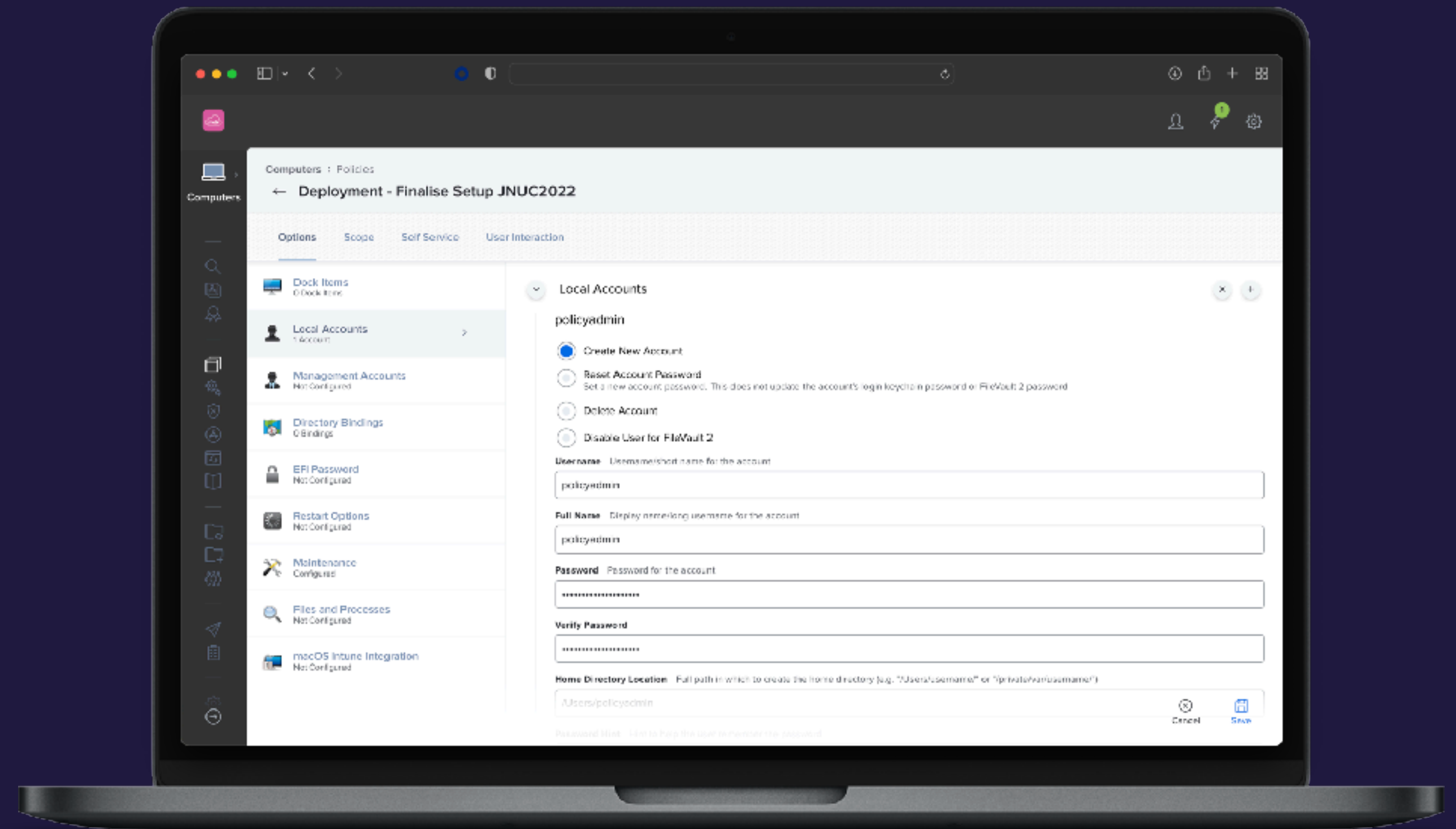
Secure Tokens

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Secure Tokens

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- Enrolled into Jamf Pro via ADE
- adeadmin created via prestage
- jamfadmin created via UIE
- policyadmin created via a policy on the enrolment complete trigger



Secure Tokens

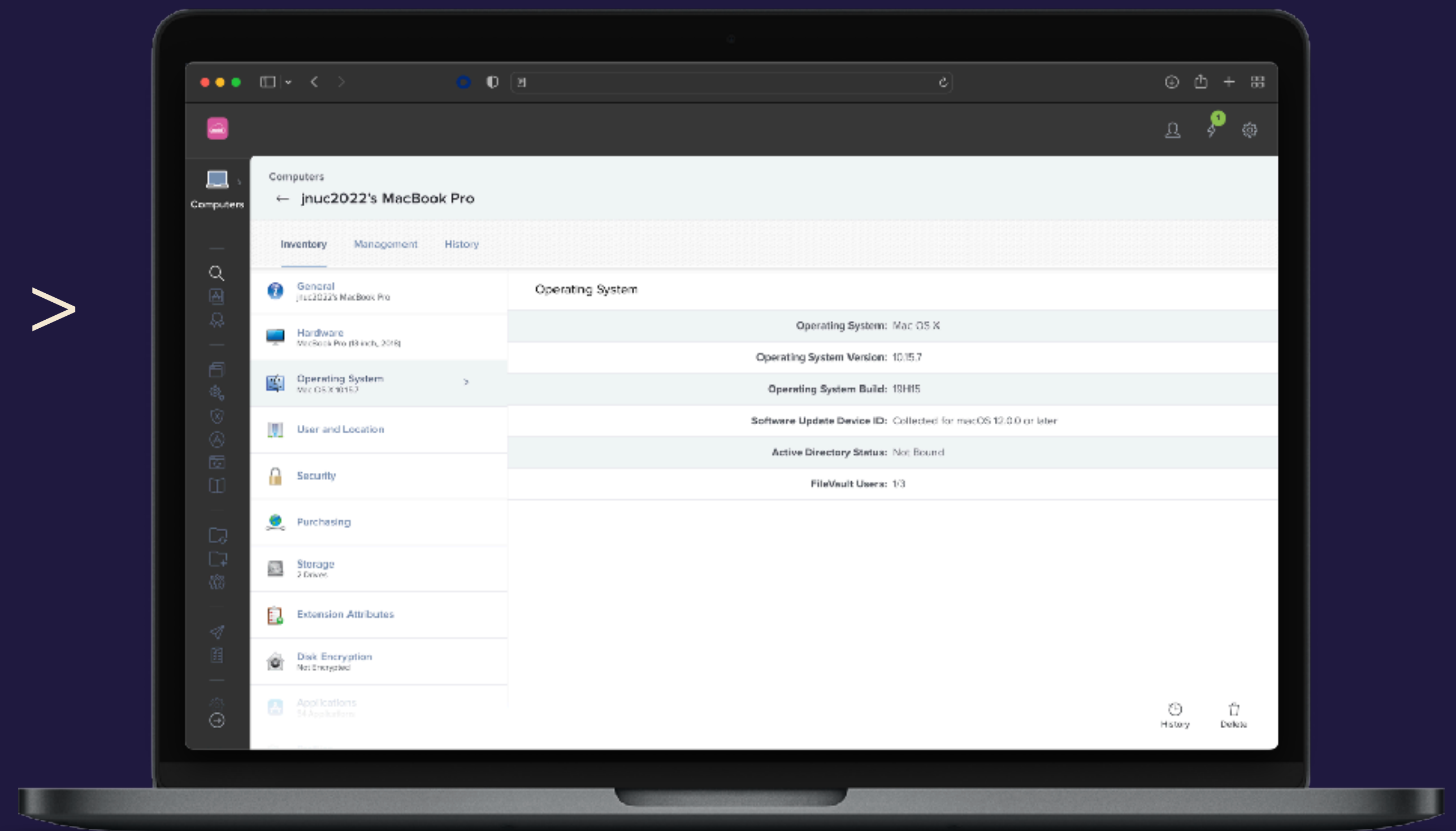
- Intel Virtual Machines
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- adeadmin created via prestage
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- policyadmin created via a policy on the enrolment complete trigger
- jnuc2022 user created at setup assistant



Secure Tokens

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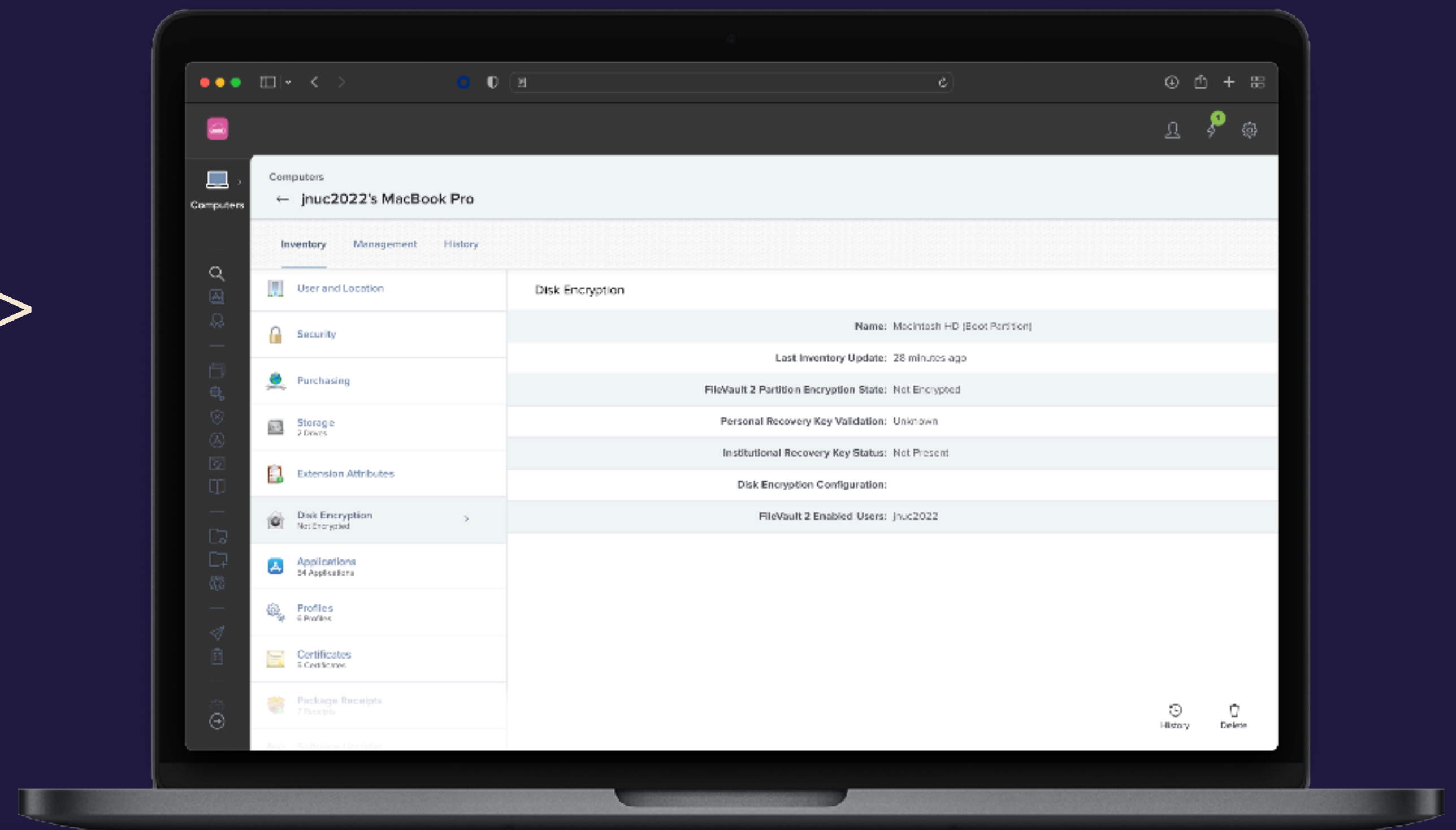
Inventory > Operating System >
FileVault Users



Secure Tokens

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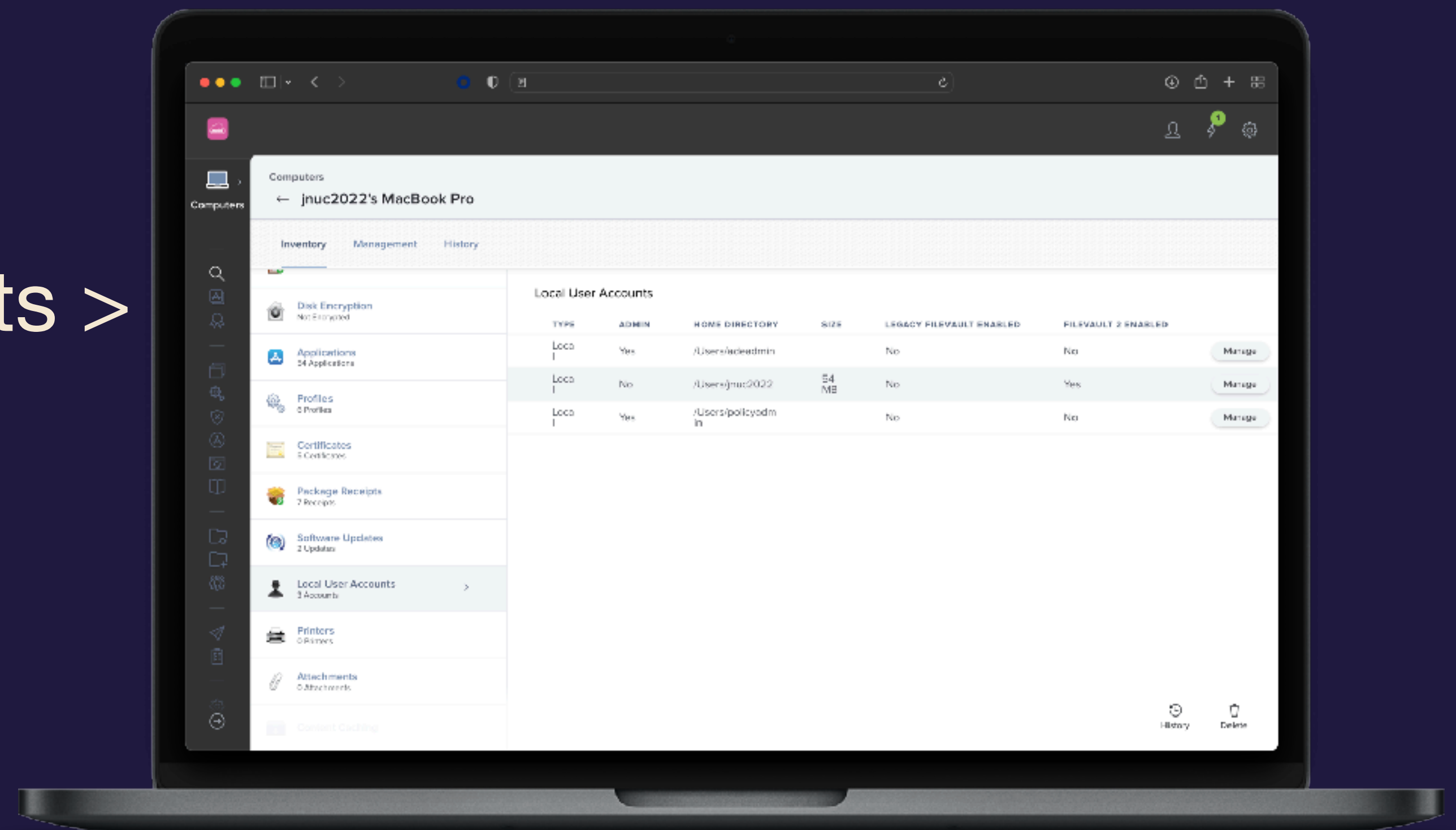
Inventory > Disk Encryption >
FileVault 2 Enabled Users



Secure Tokens

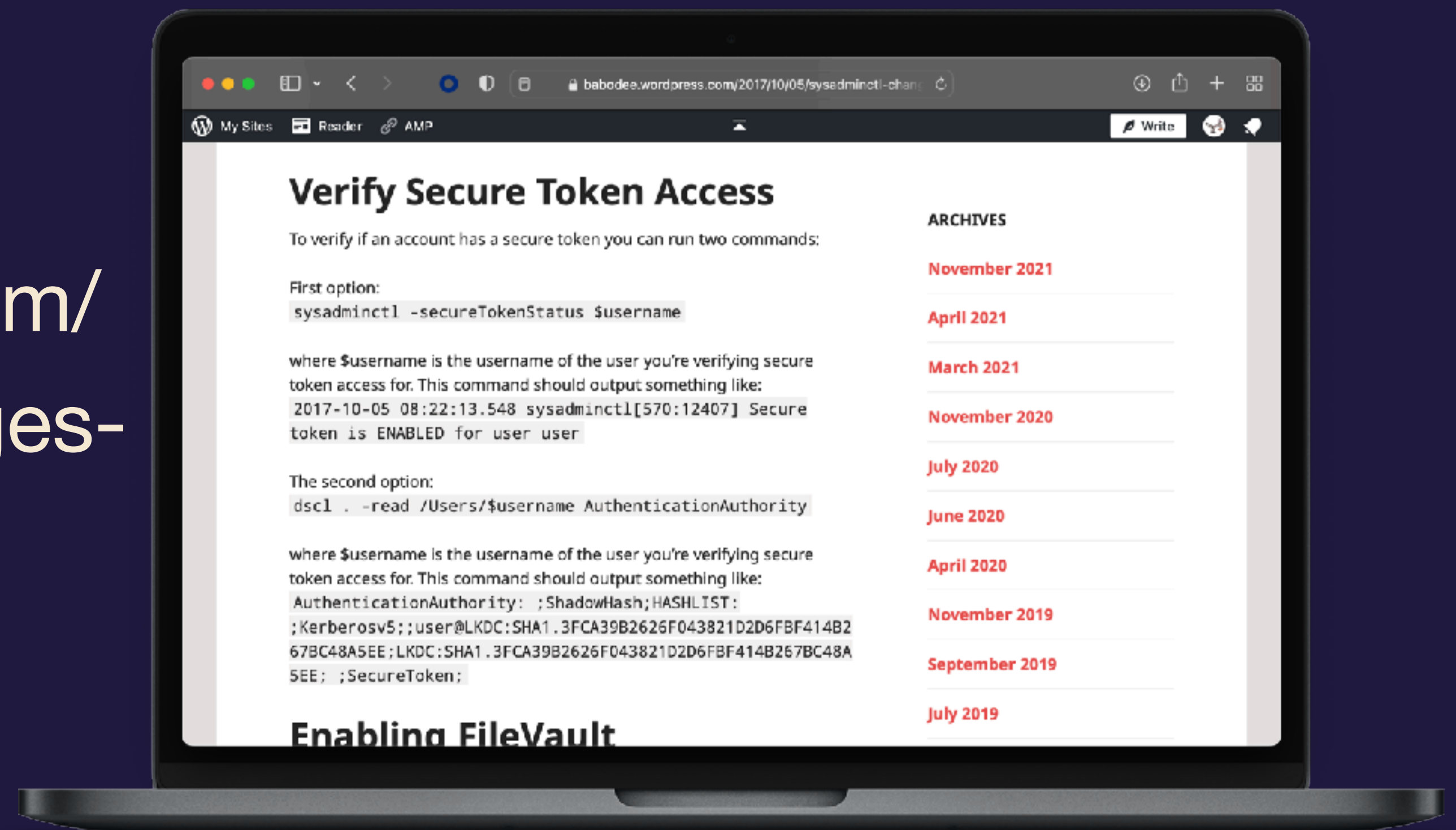
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Inventory > Local User Accounts >
FileVault 2 Enabled



Secure Tokens

<https://babodee.wordpress.com/2017/10/05/sysadminctl-changes-in-10-13/>



Secure Tokens

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High Sierra - Mojave
No secure tokens



Secure Tokens

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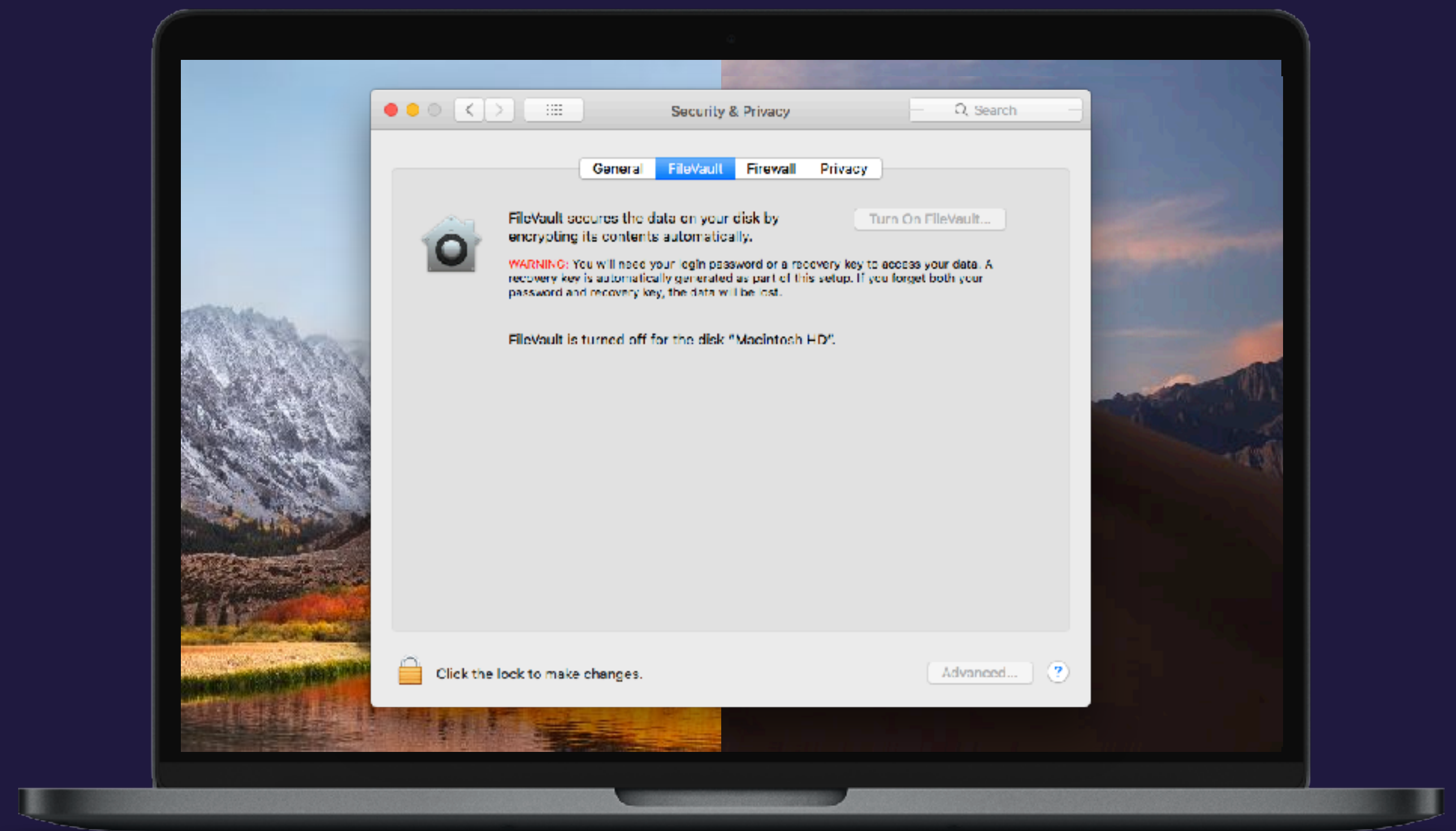
High Sierra - Mojave
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Secure Tokens

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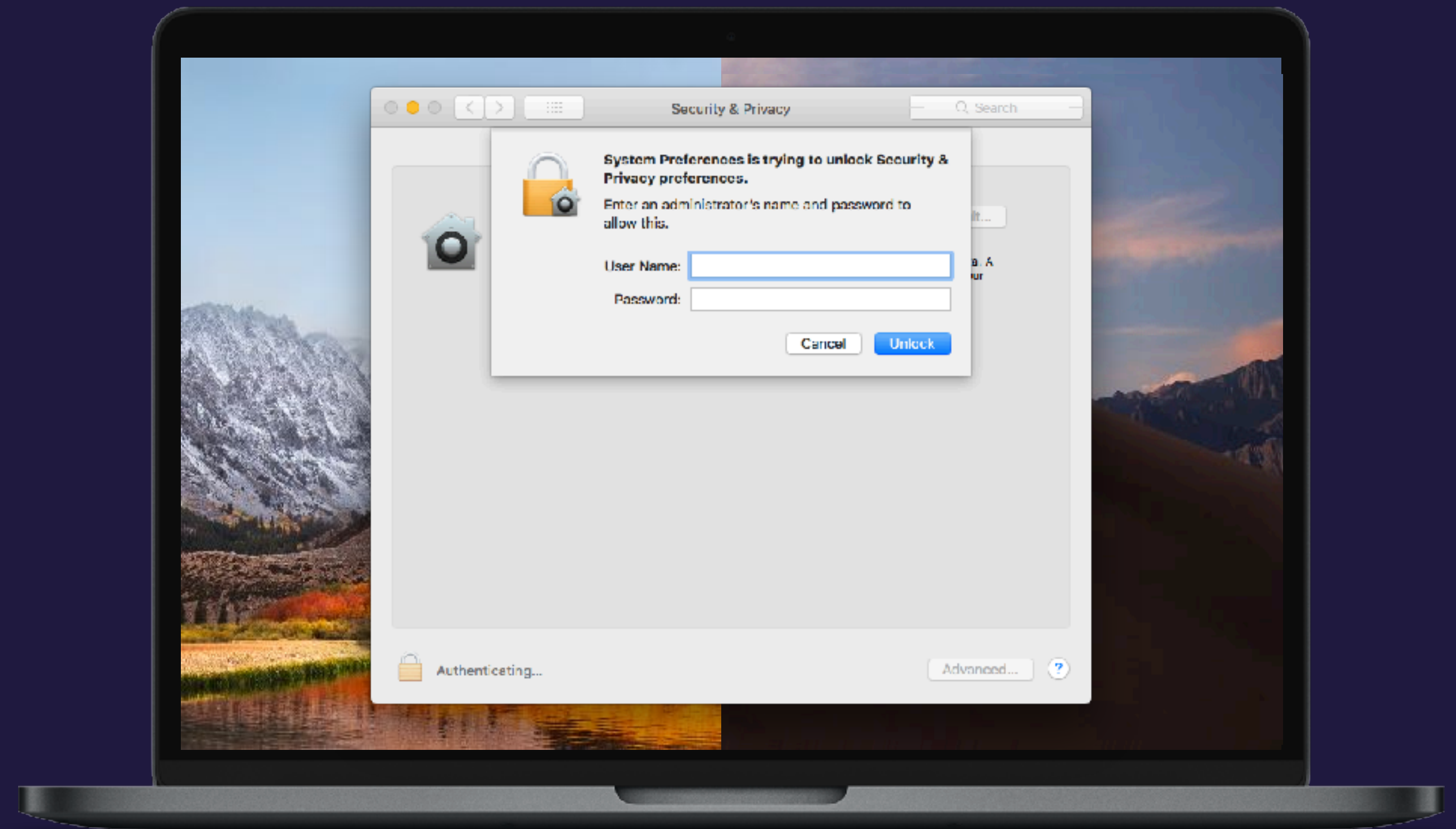
High Sierra - Mojave
Standard User



Secure Tokens

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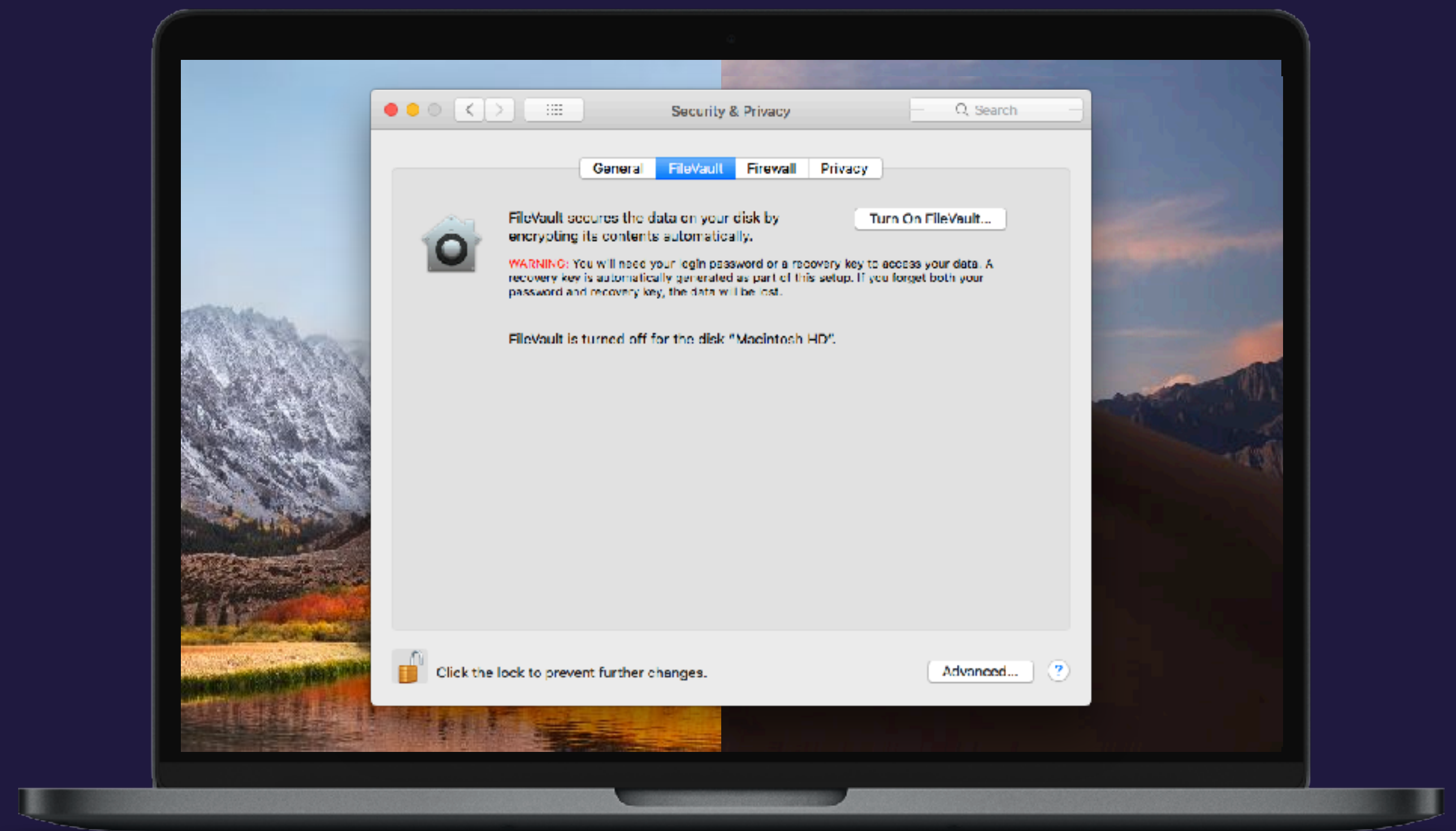
High Sierra - Mojave
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Secure Tokens

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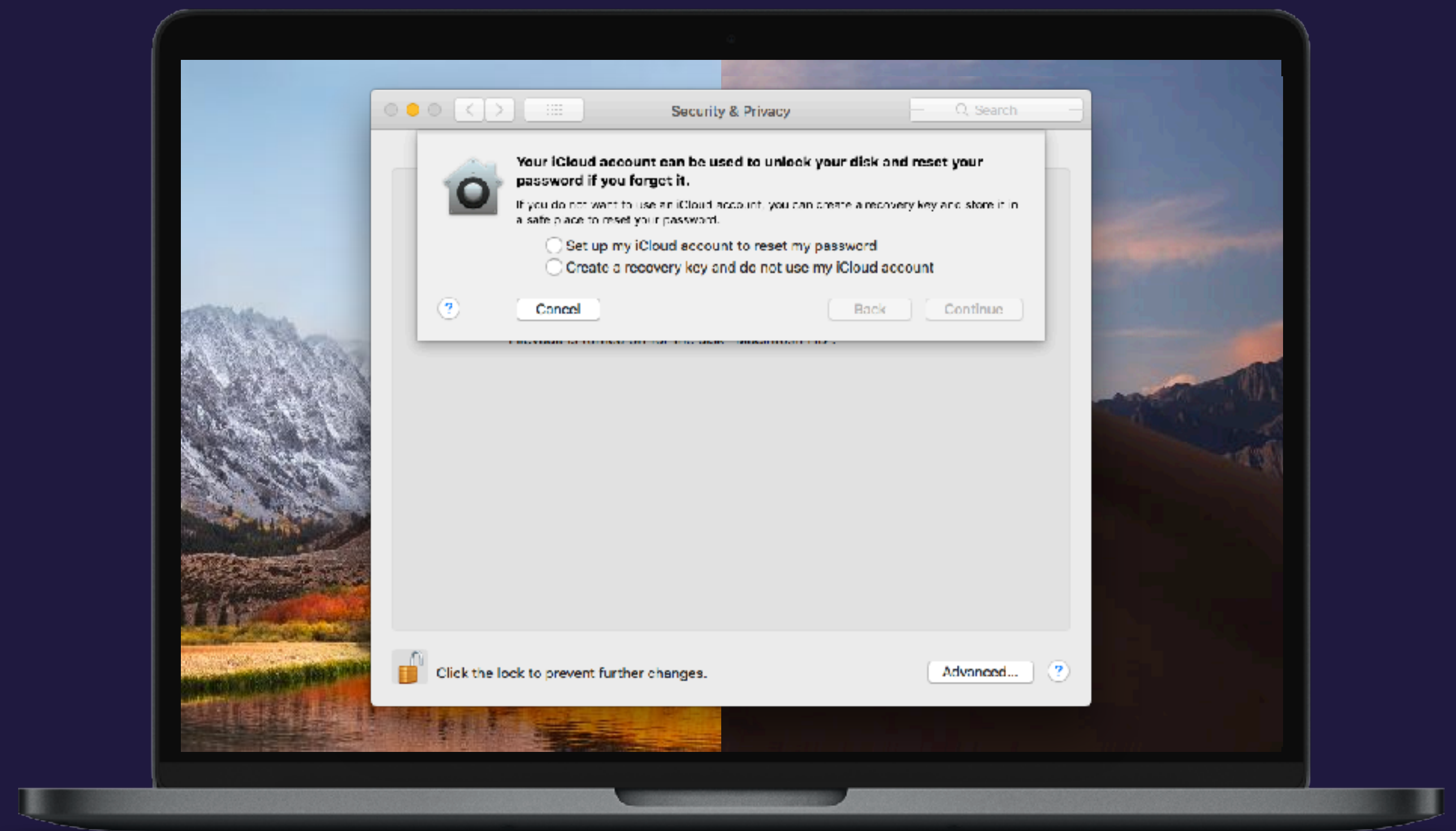
High Sierra - Mojave
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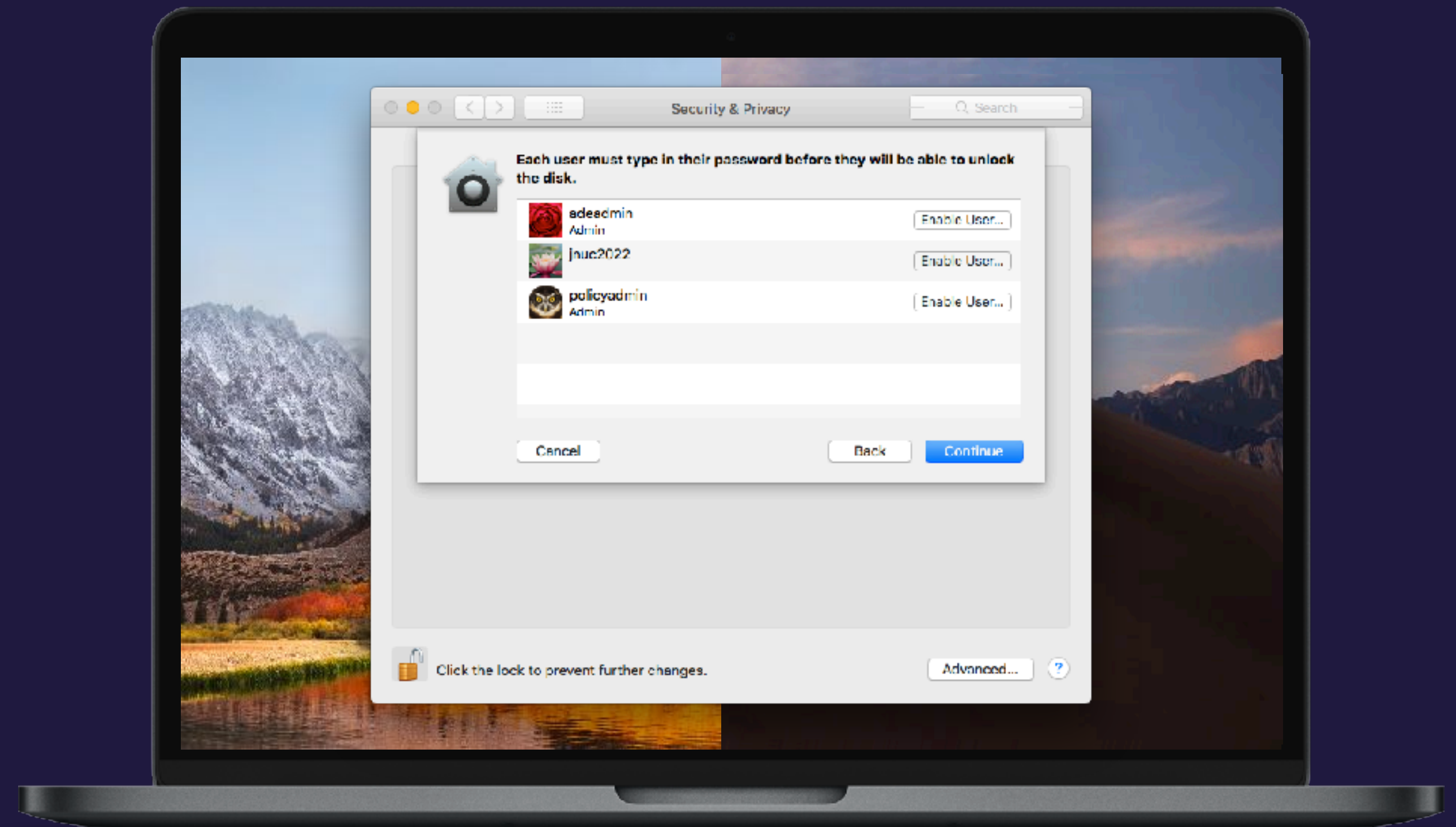
High Sierra - Mojave
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Secure Tokens

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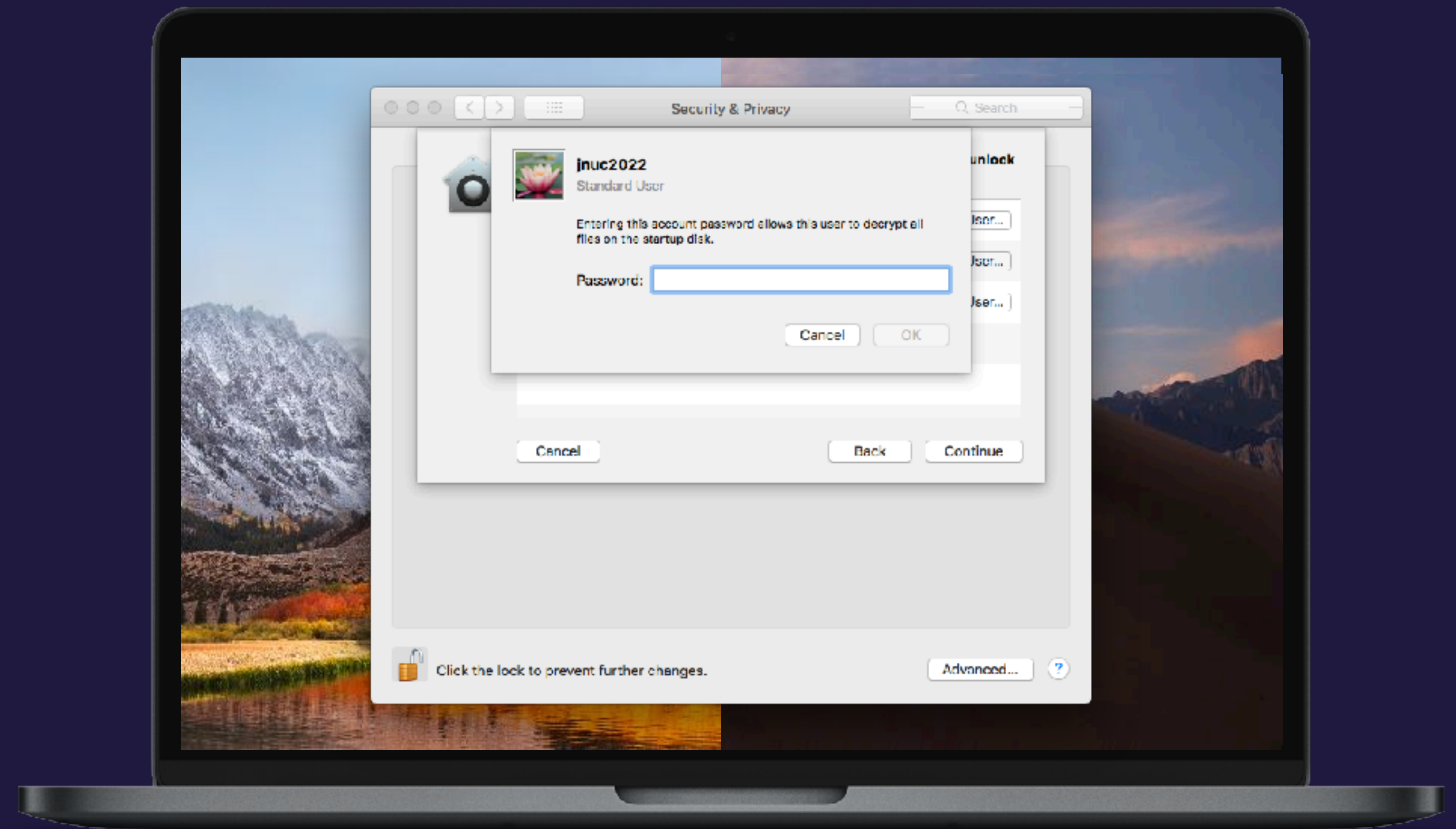
High Sierra - Mojave
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Secure Tokens

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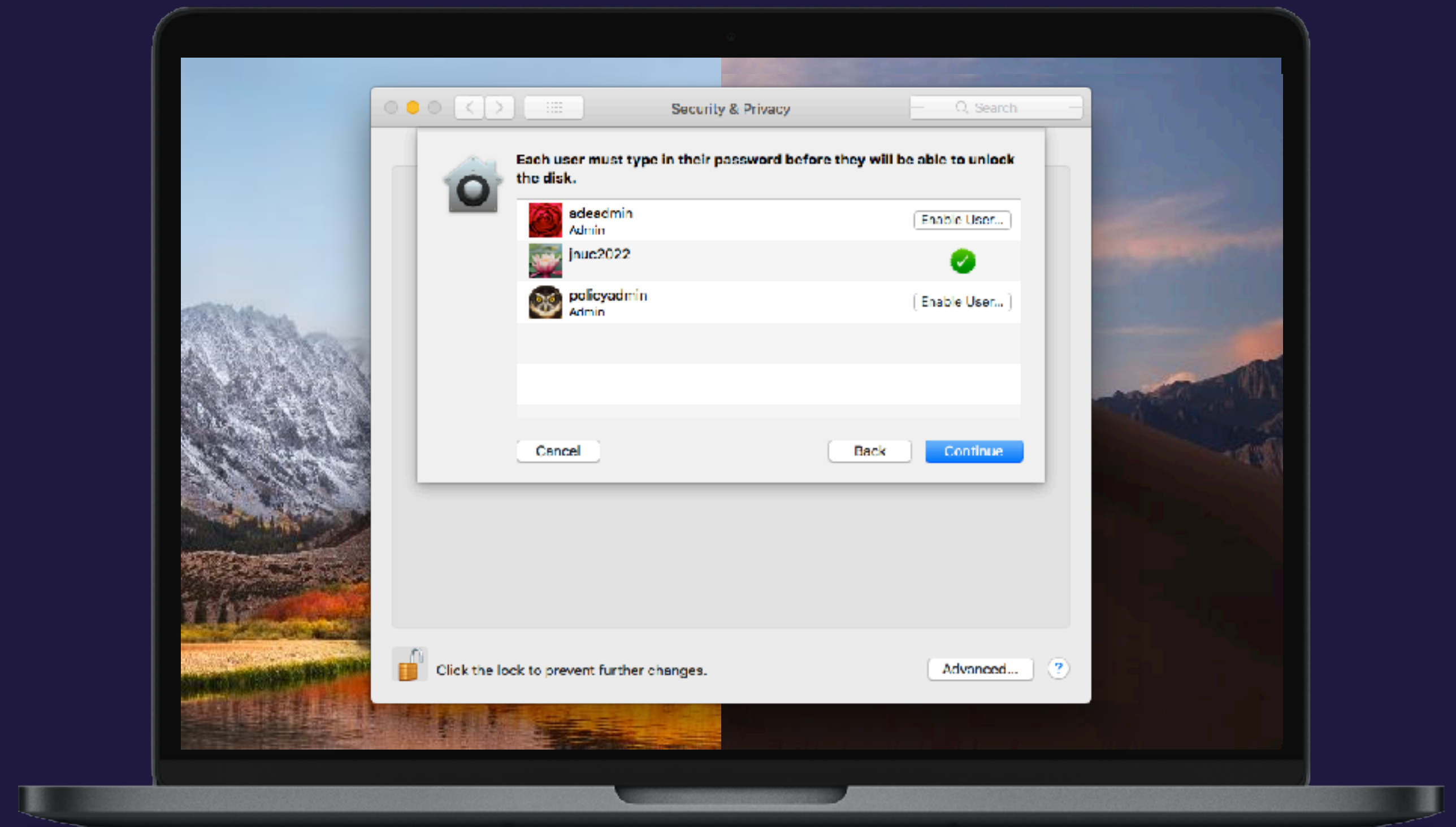
High Sierra - Mojave
Standard User



Secure Tokens

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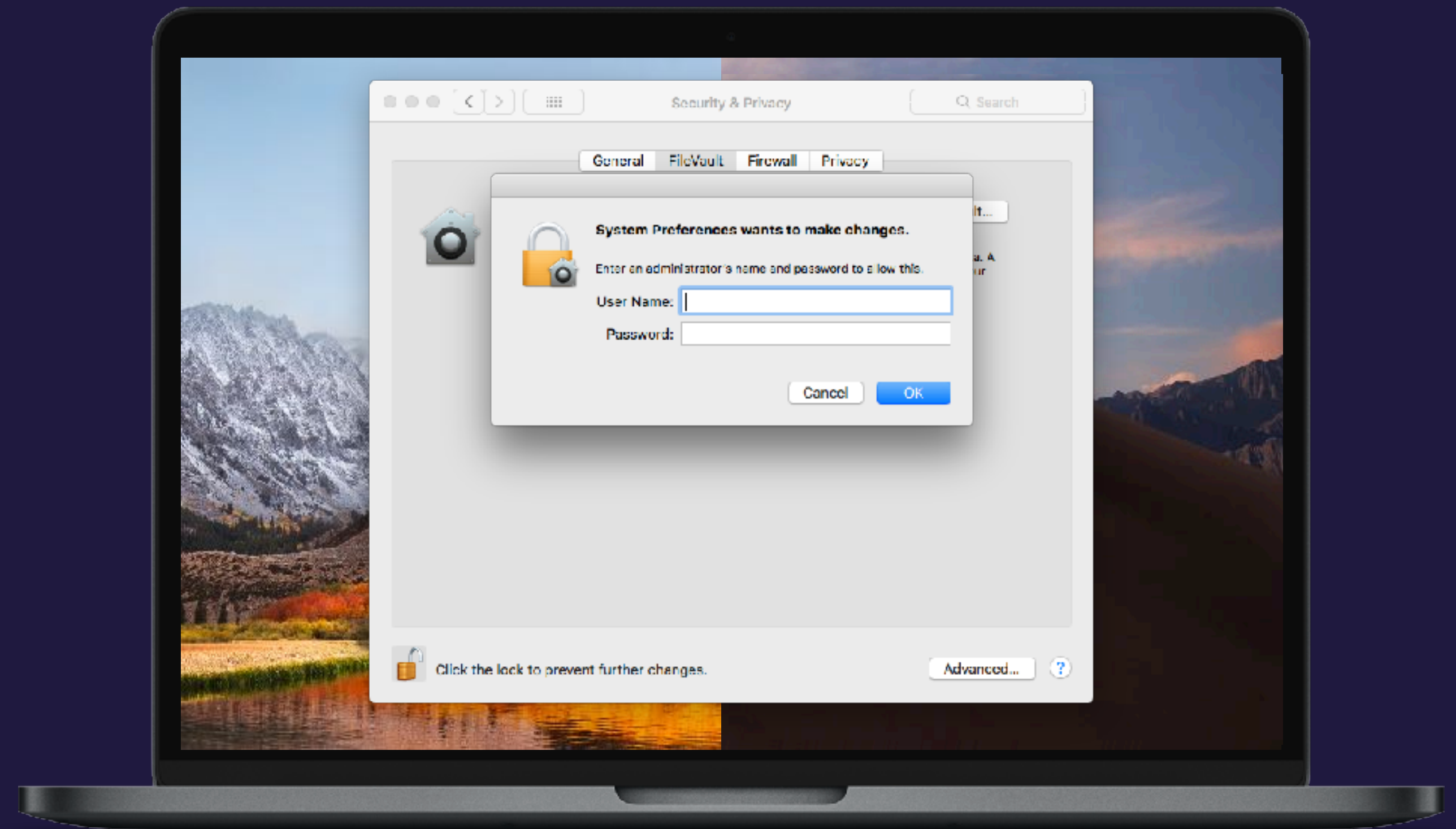
High Sierra - Mojave
Standard User



Secure Tokens

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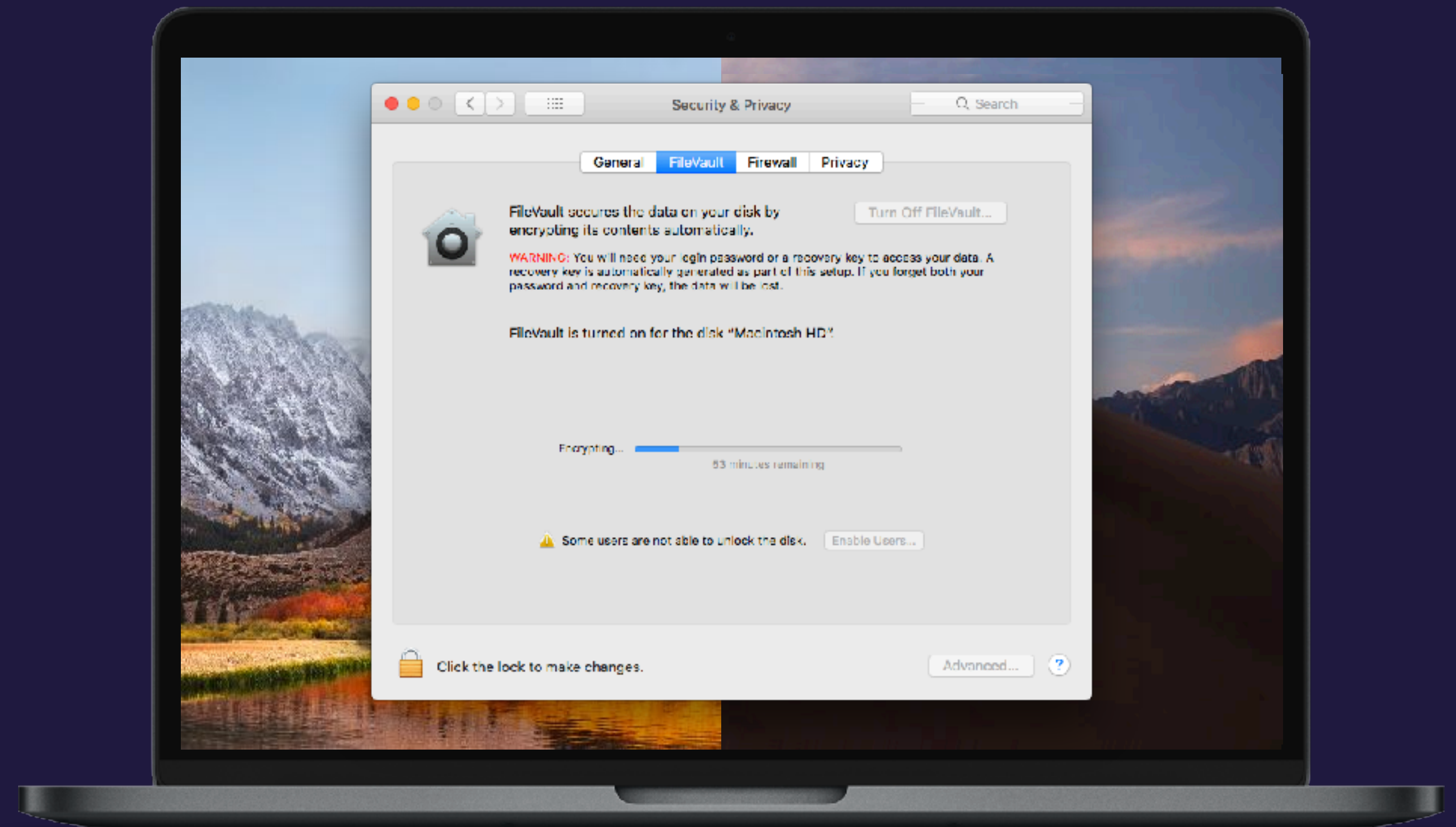
High Sierra - Mojave
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Secure Tokens

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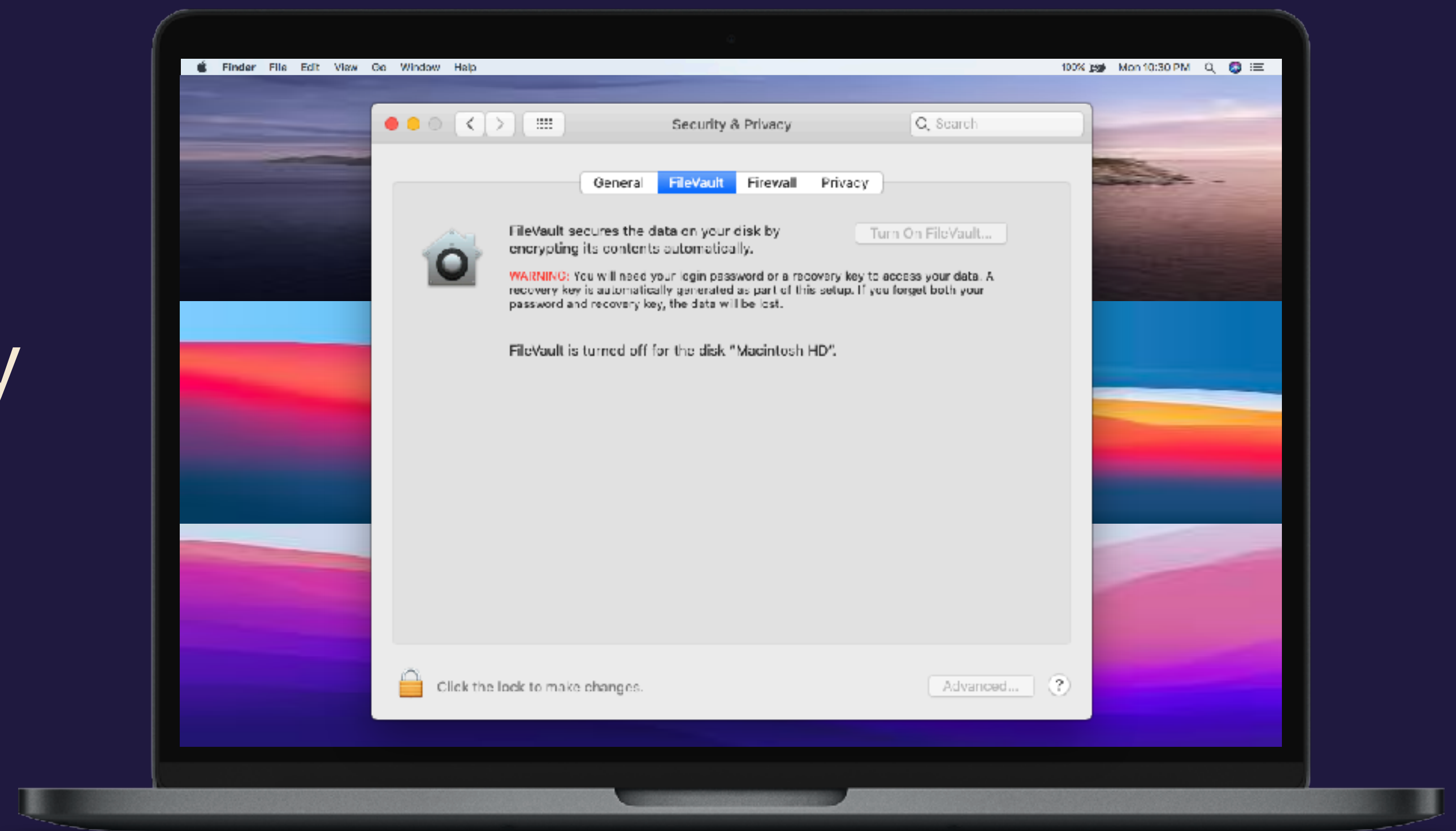
High Sierra - Mojave Standard User



Secure Tokens

dataJAR
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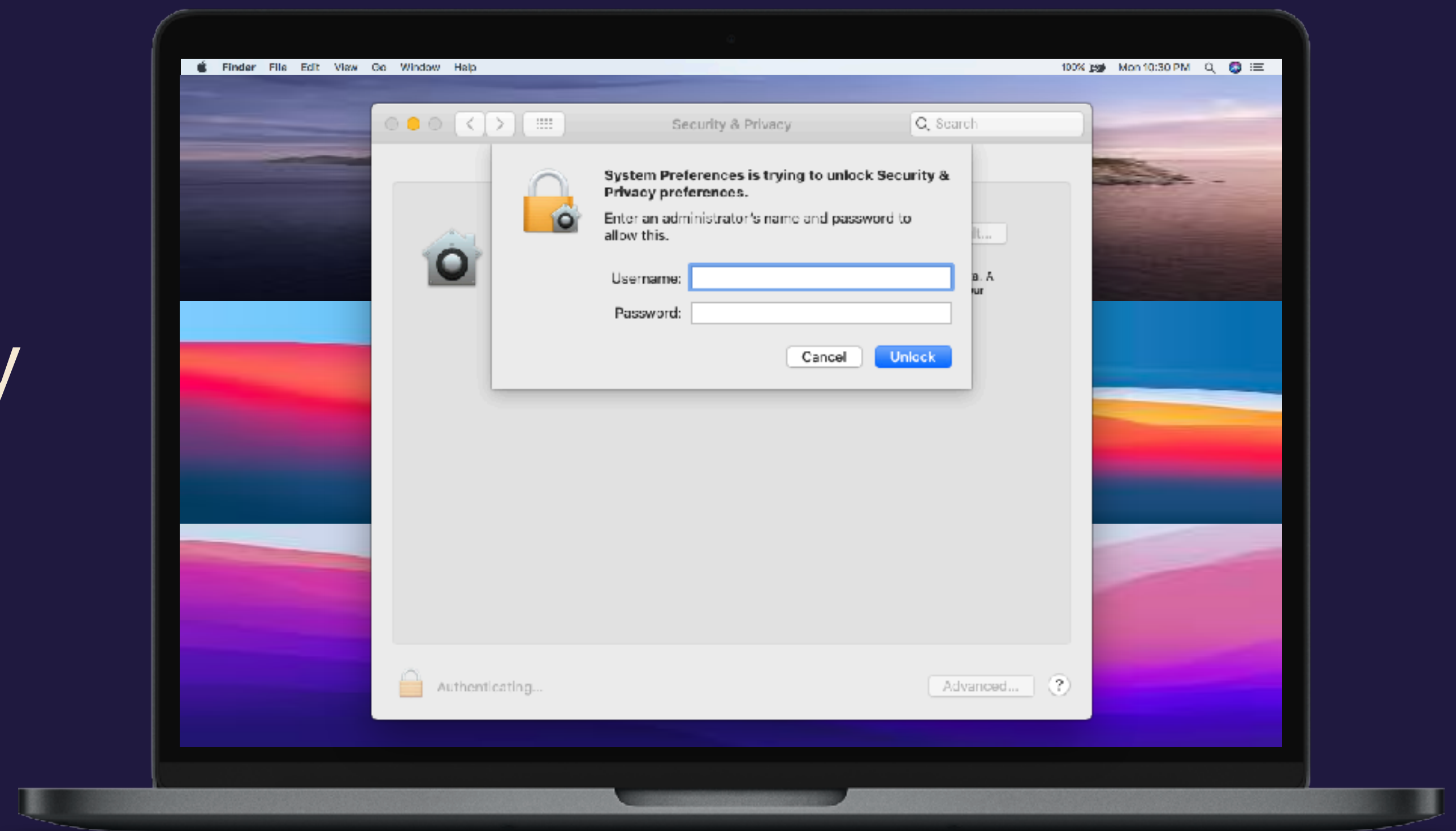
Catalina - Big Sur - Monterey
Standard User



Secure Tokens

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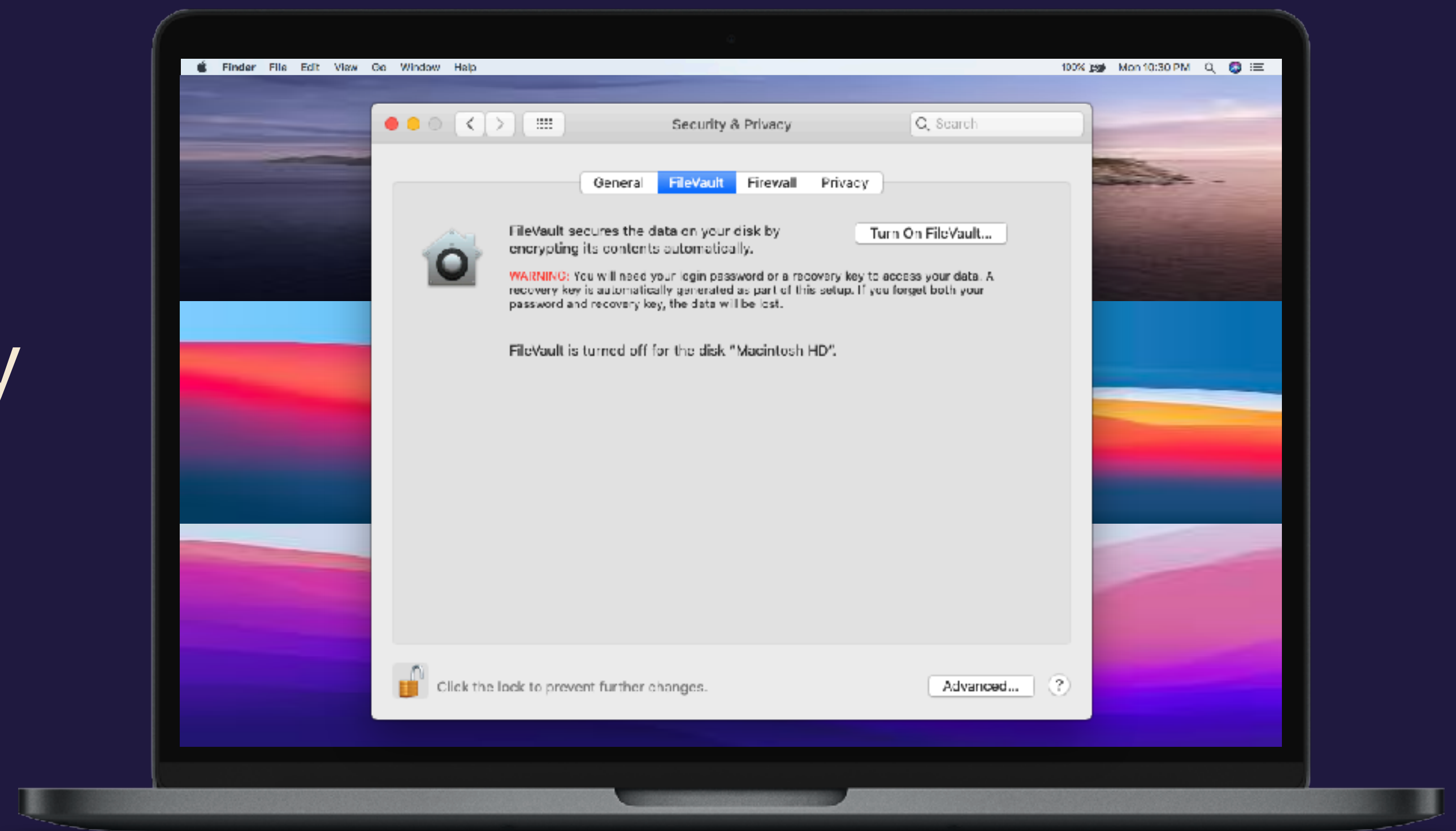
Catalina - Big Sur - Monterey
Standard User



Secure Tokens

dataJAR
Beyond Device Management

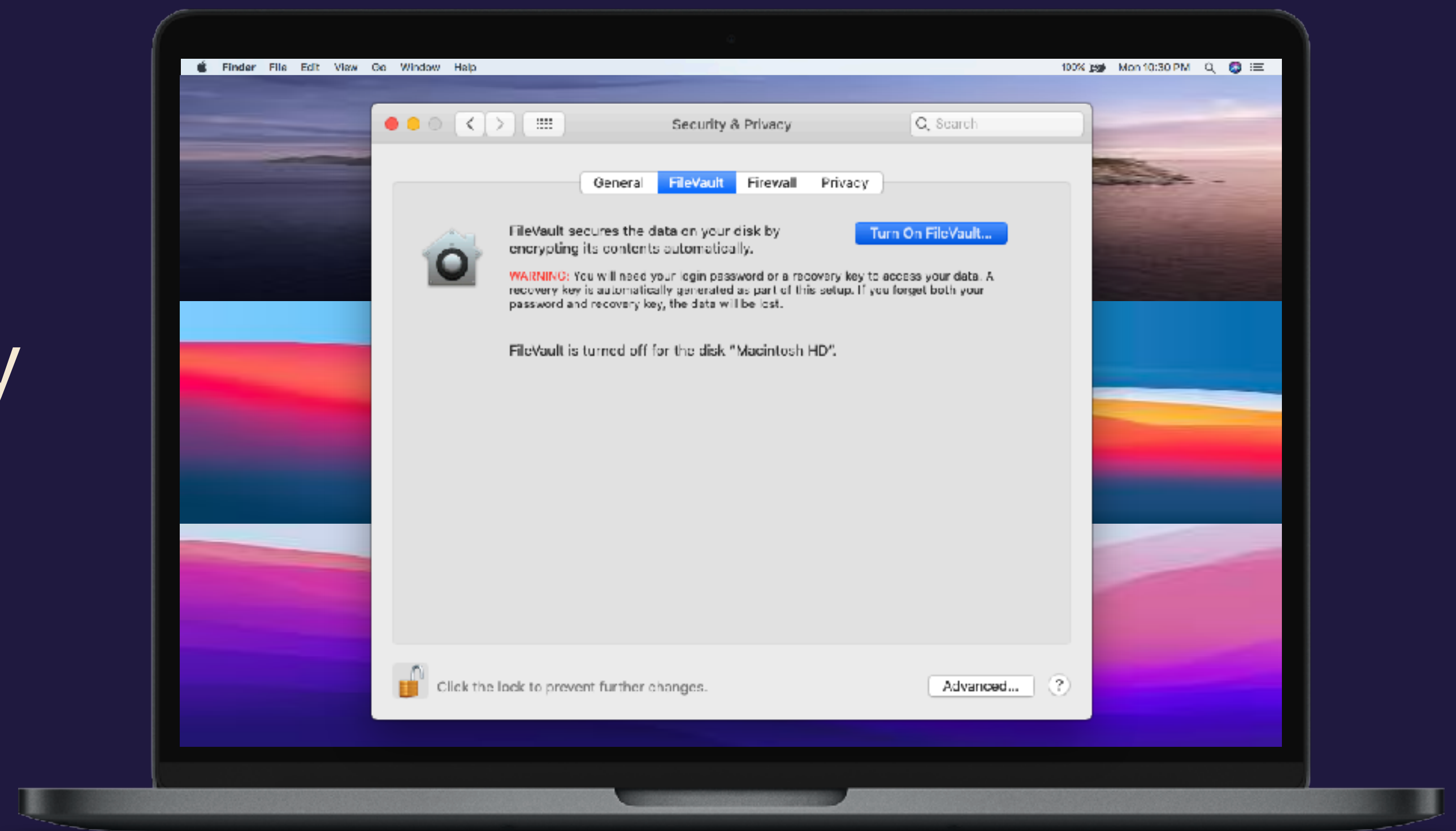
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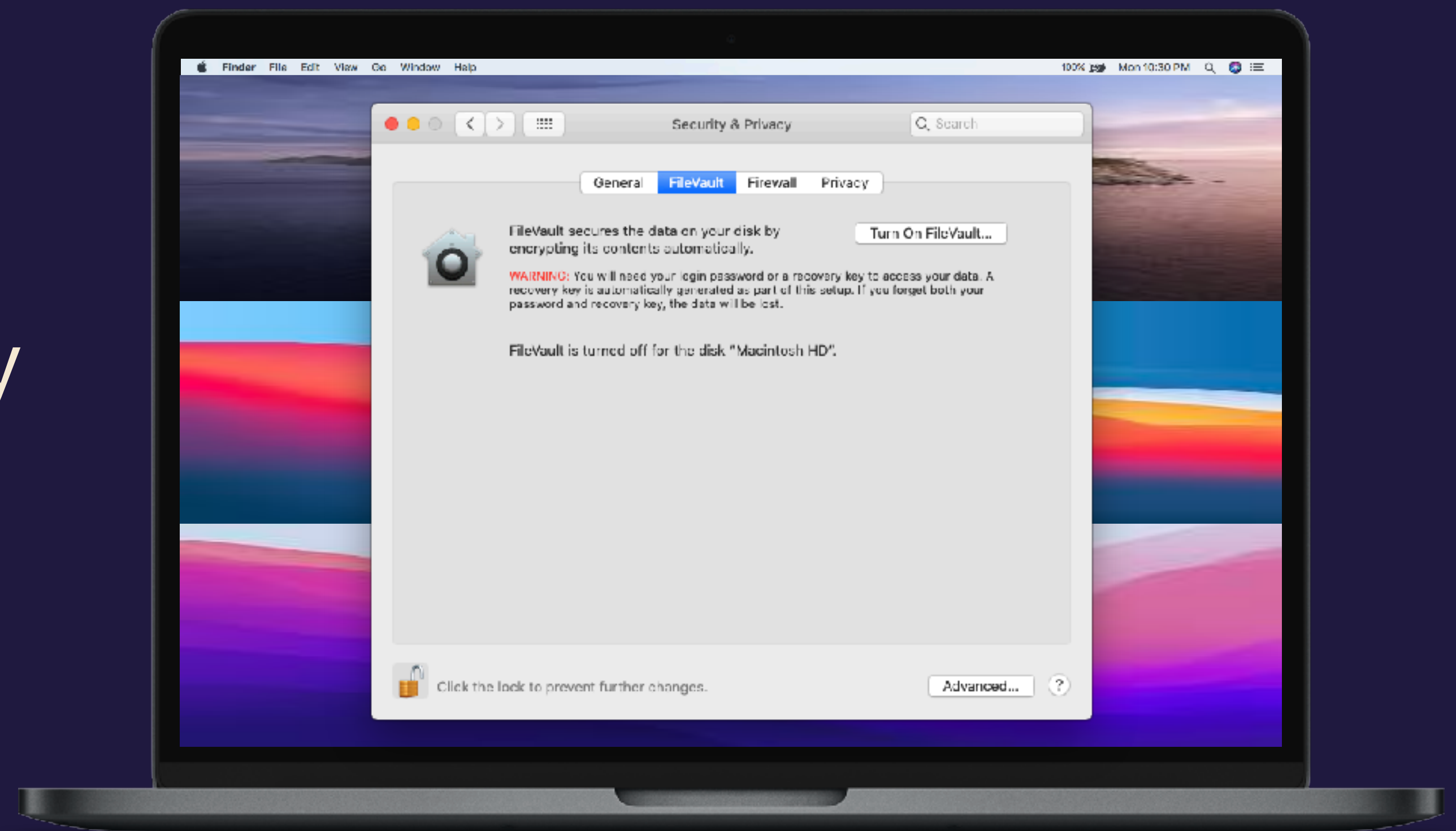
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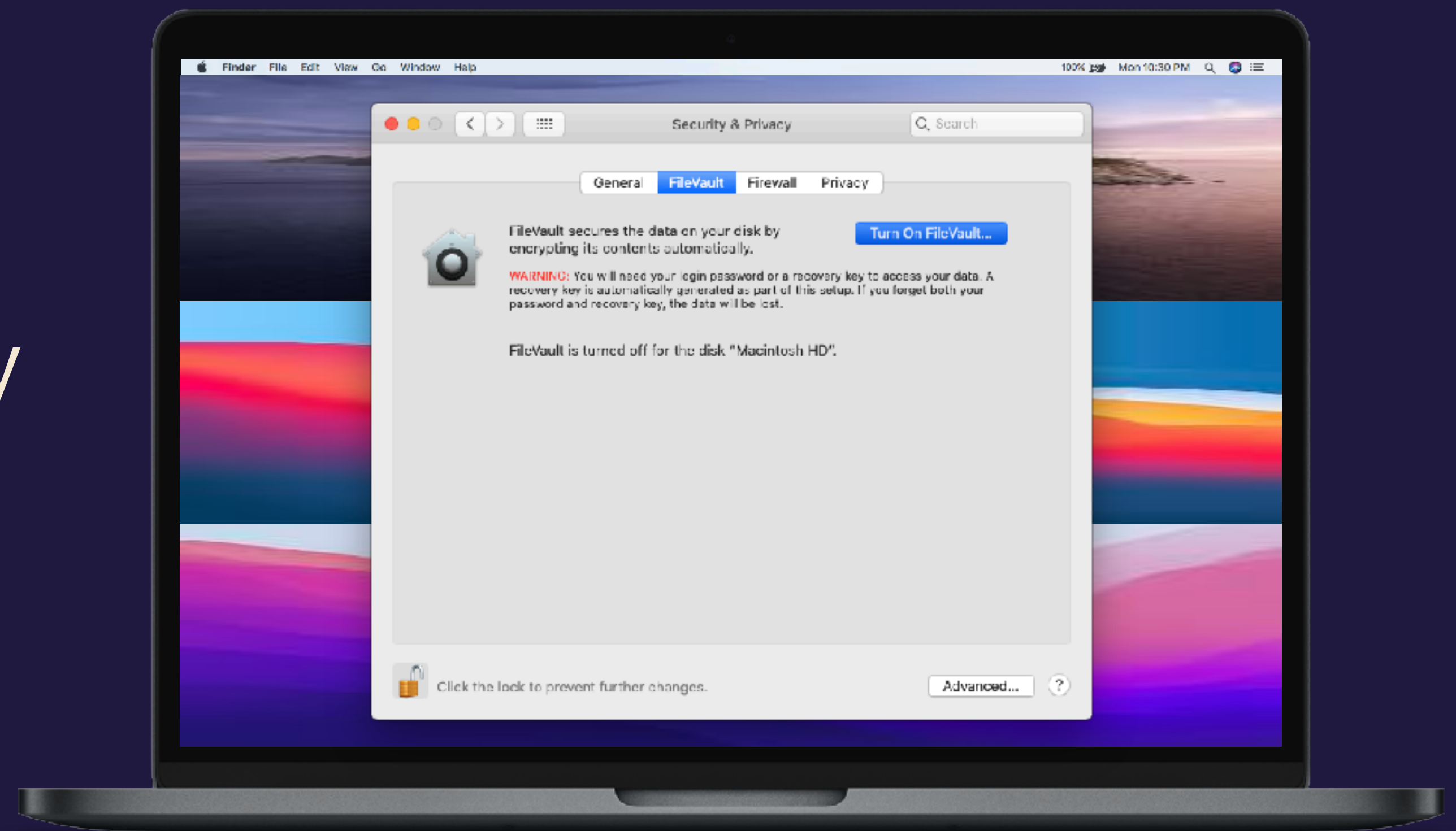
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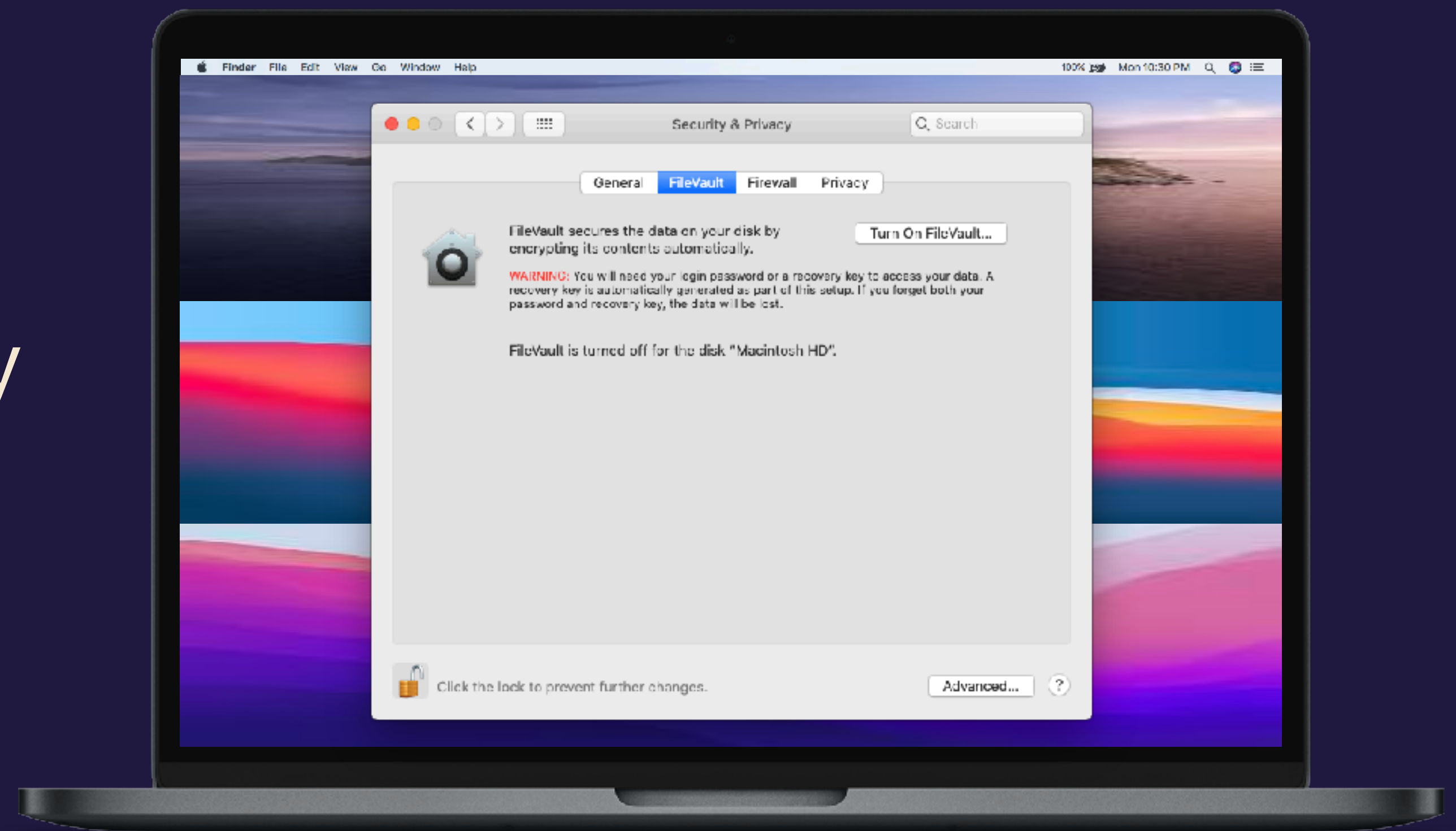
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Secure Tokens

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Secure Tokens

“In macOS 11 or later, setting the initial password for the very first user on the Mac results in that user being granted a secure token. In some workflows, that may not be the desired behaviour, as previously, granting the first secure token would have required the user account to log in.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Secure Tokens

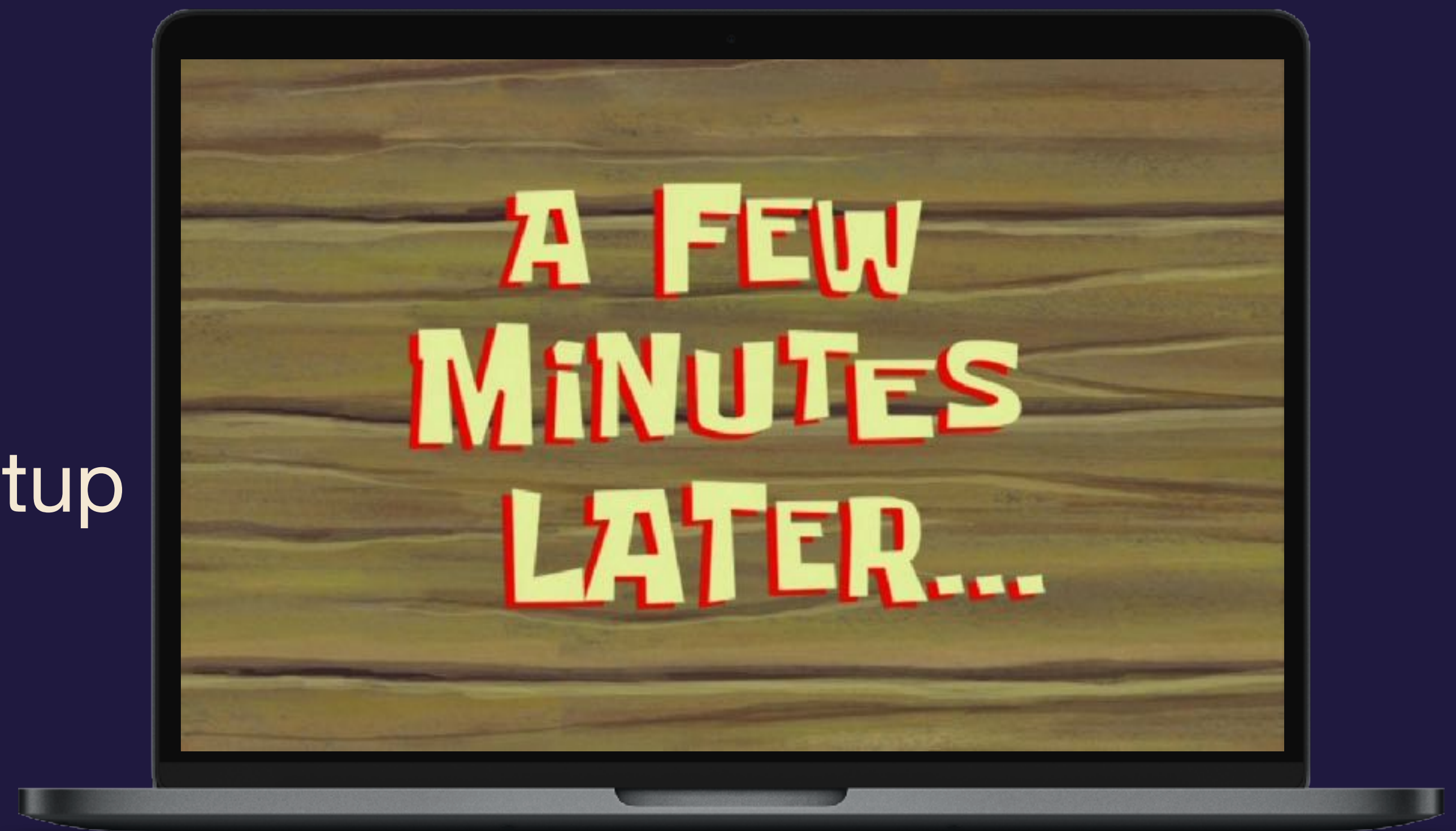
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Beyond Device Management

Big Sur –
Standard User, with delay at Setup
Assistant



Secure Tokens

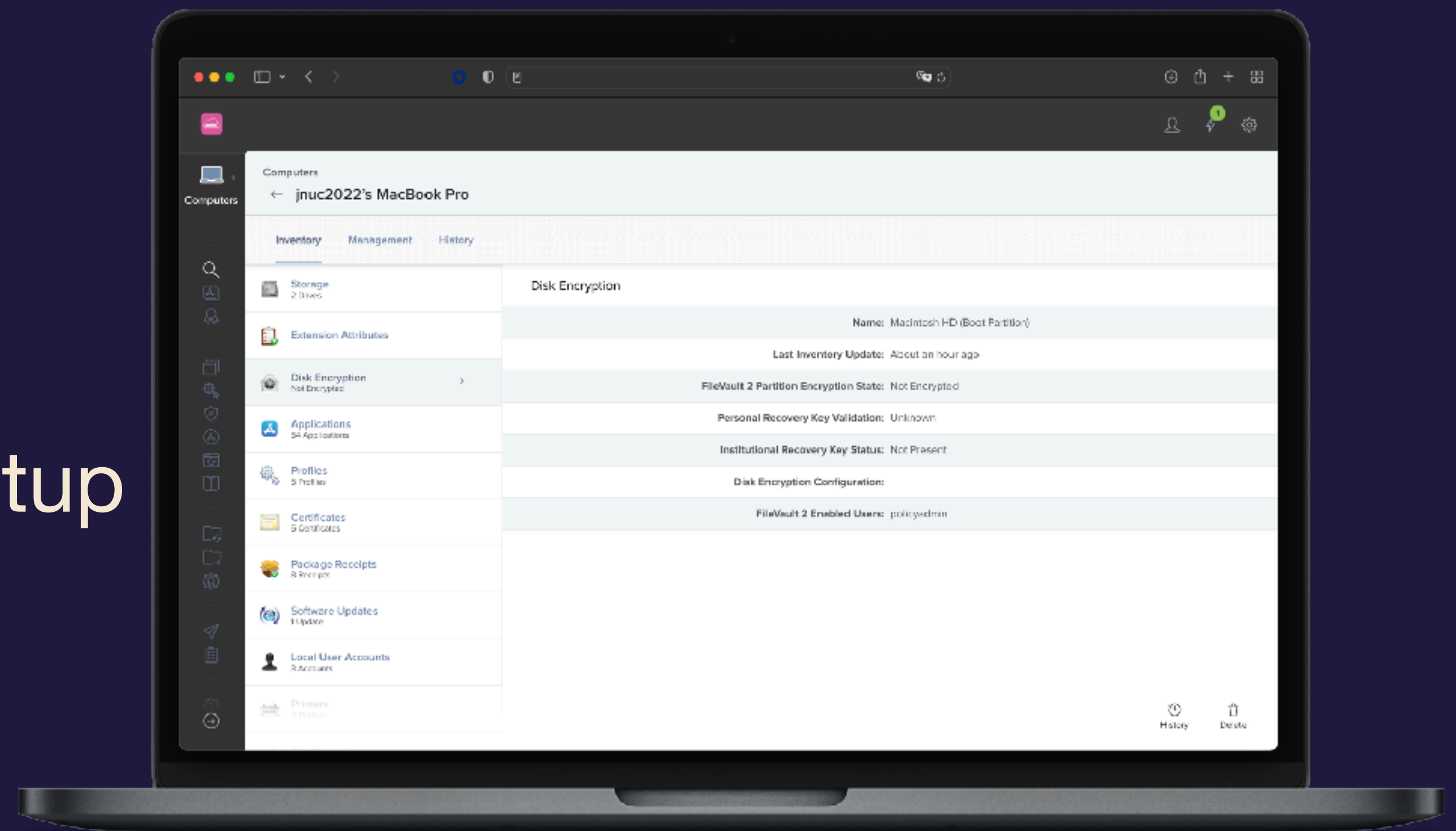
Big Sur -
Standard User, with delay at Setup
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Secure Tokens

dataJAR
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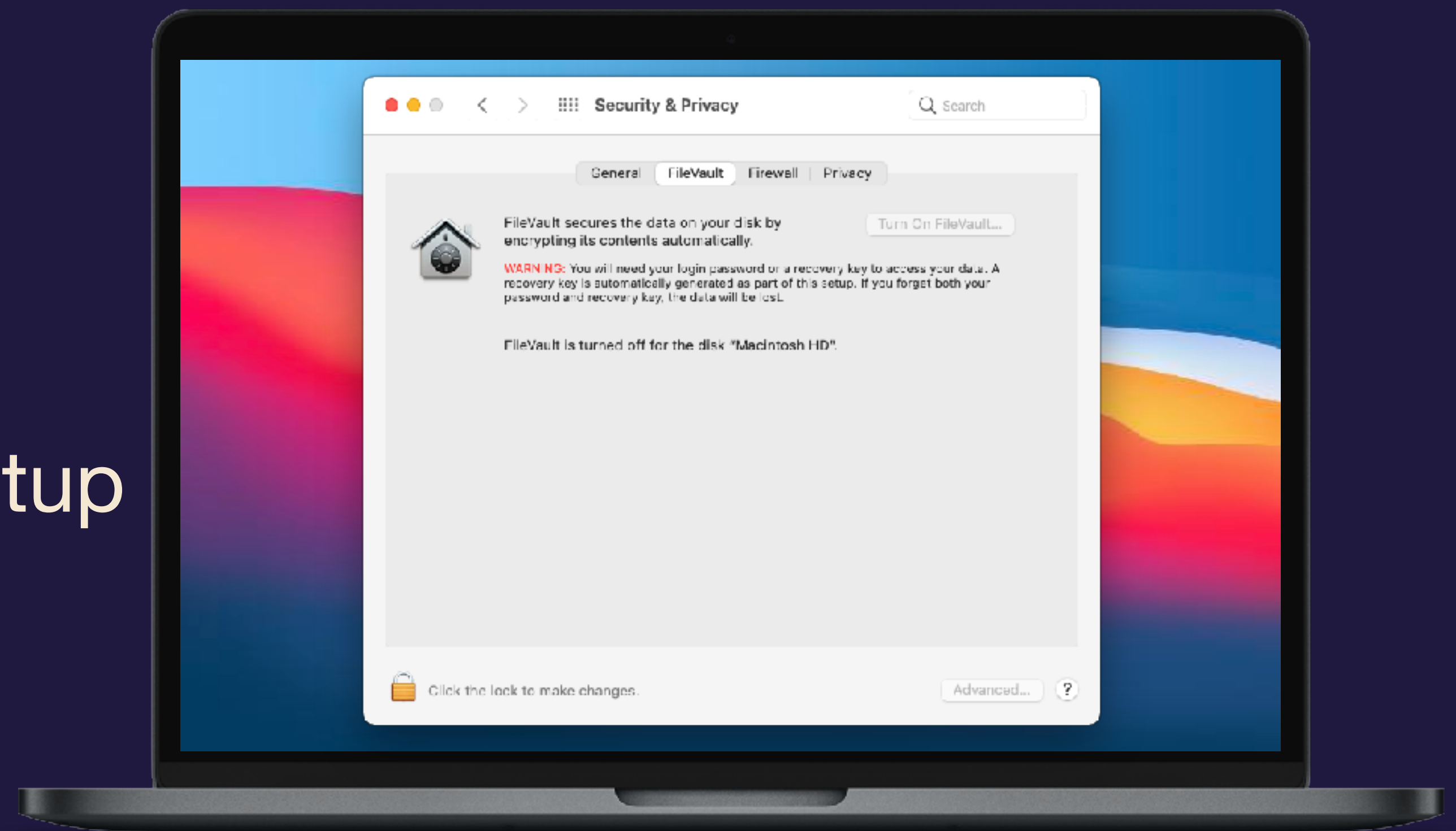
Big Sur –
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Secure Tokens

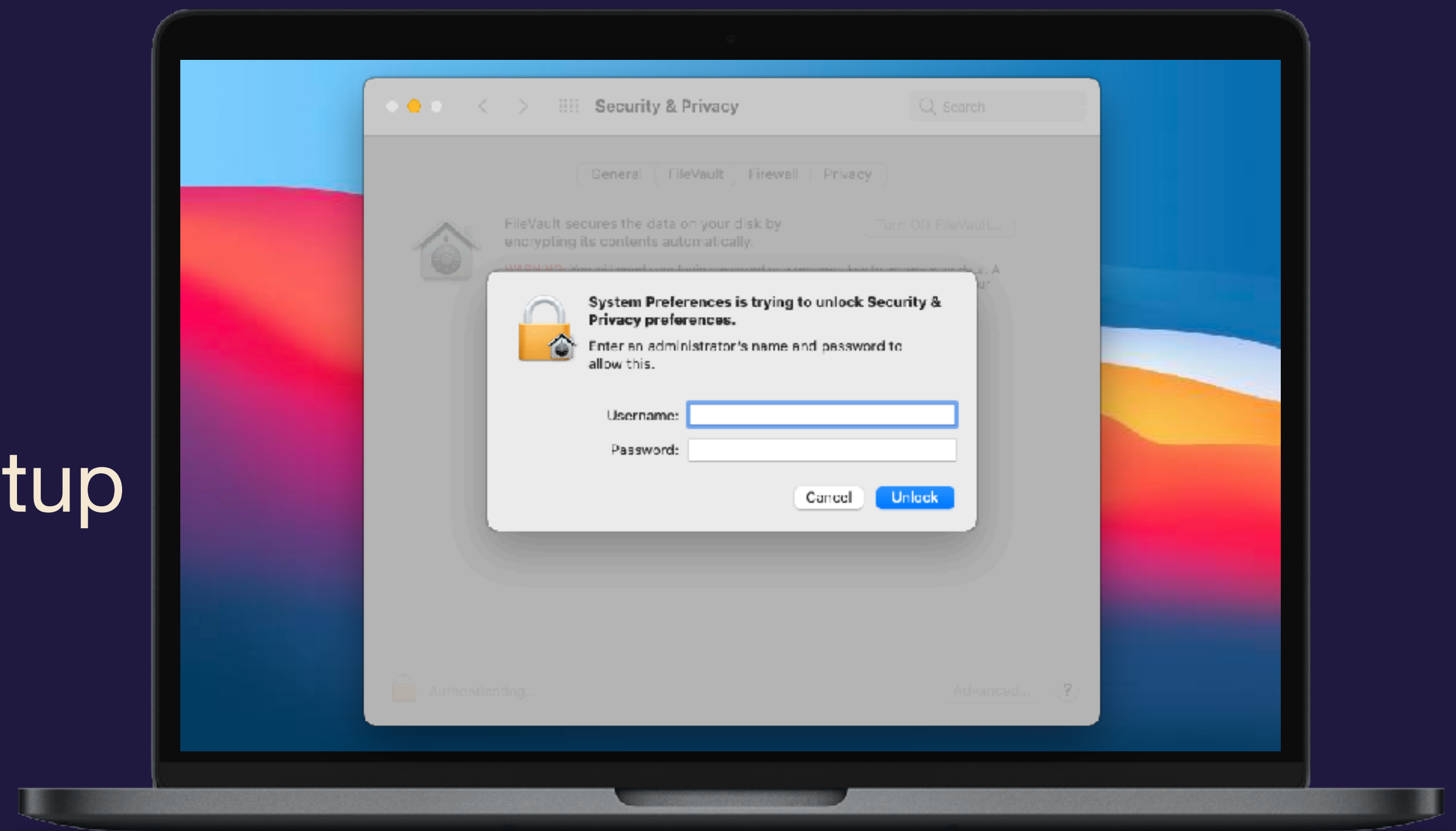
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Secure Tokens

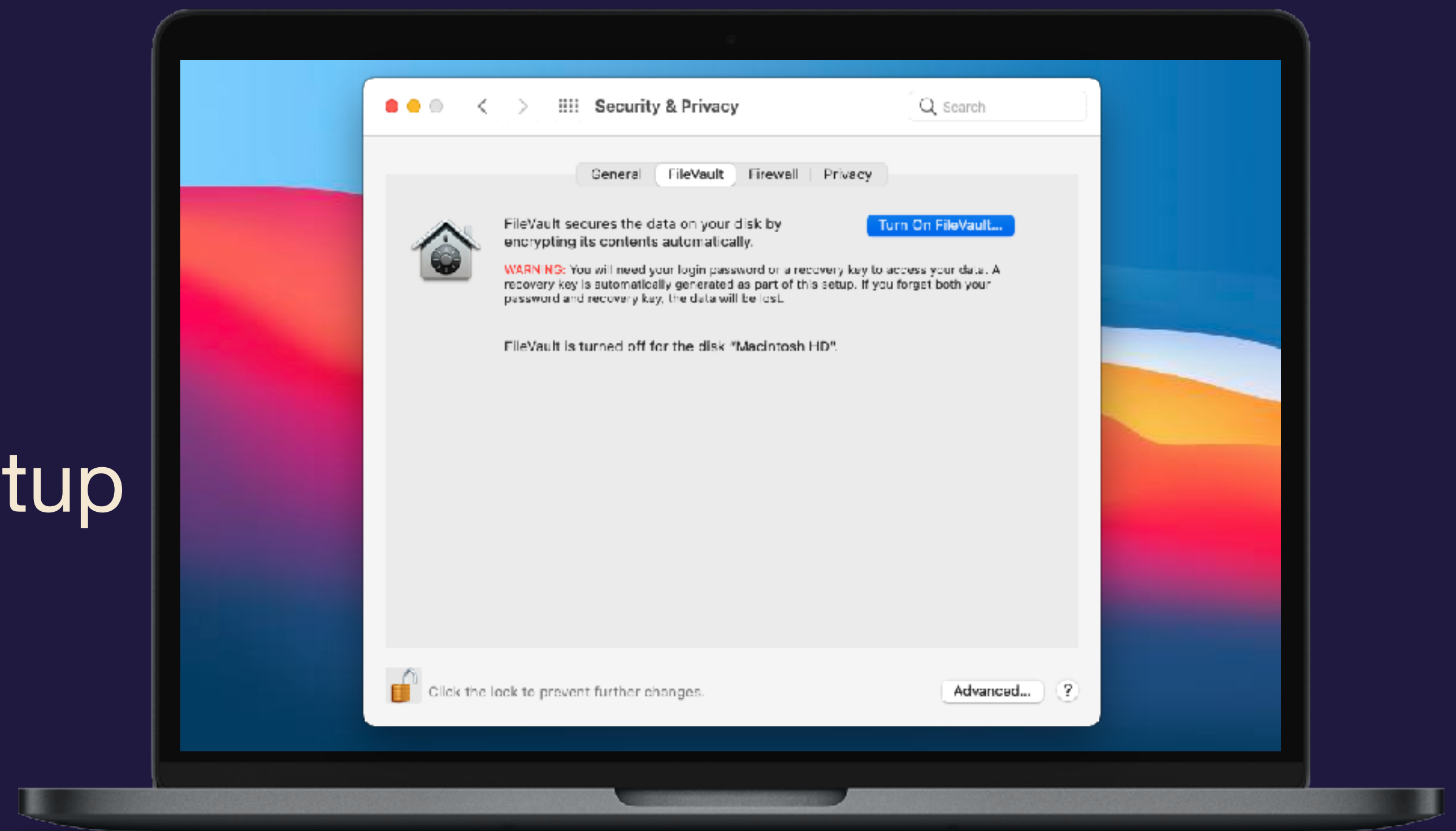
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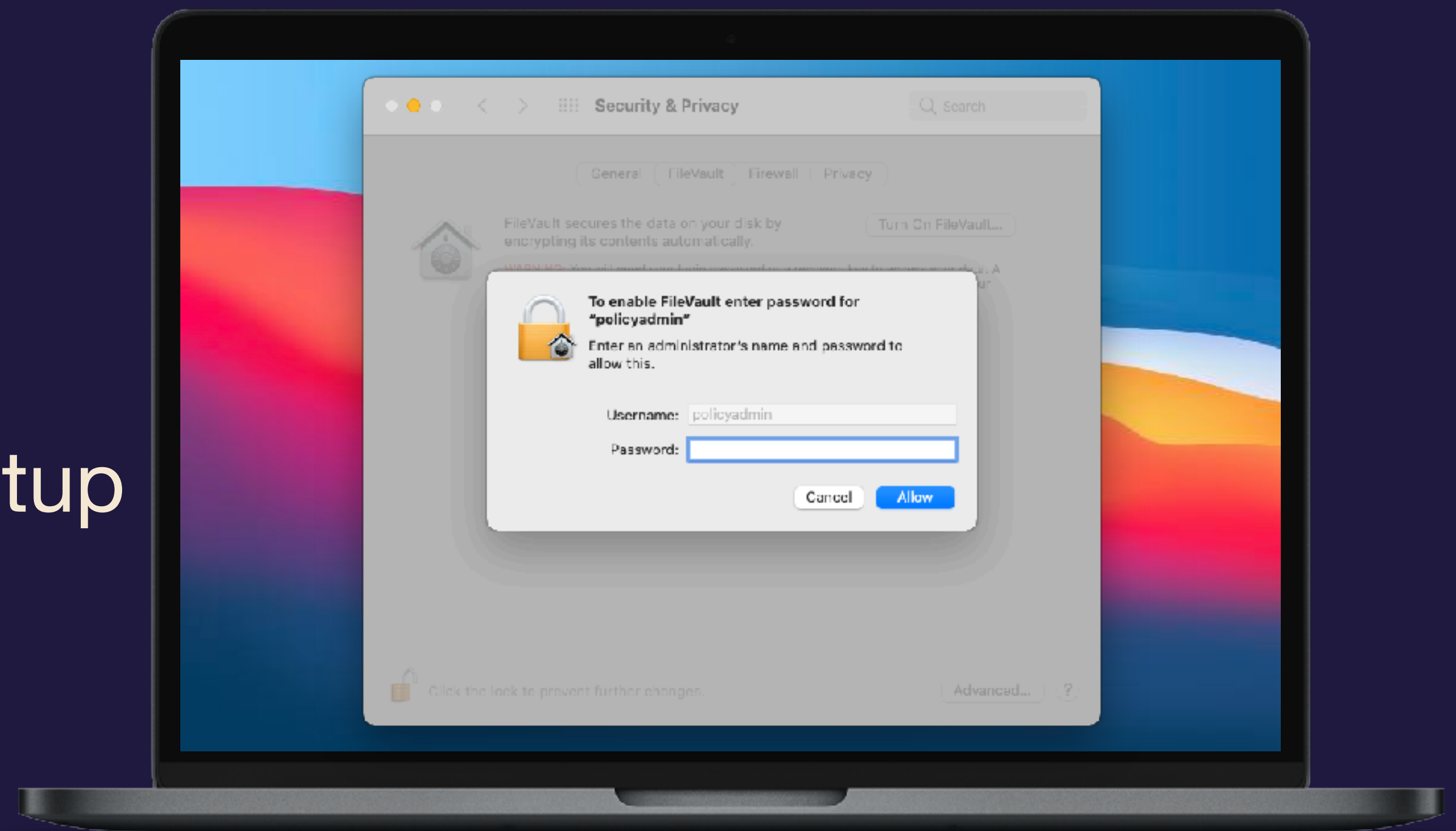
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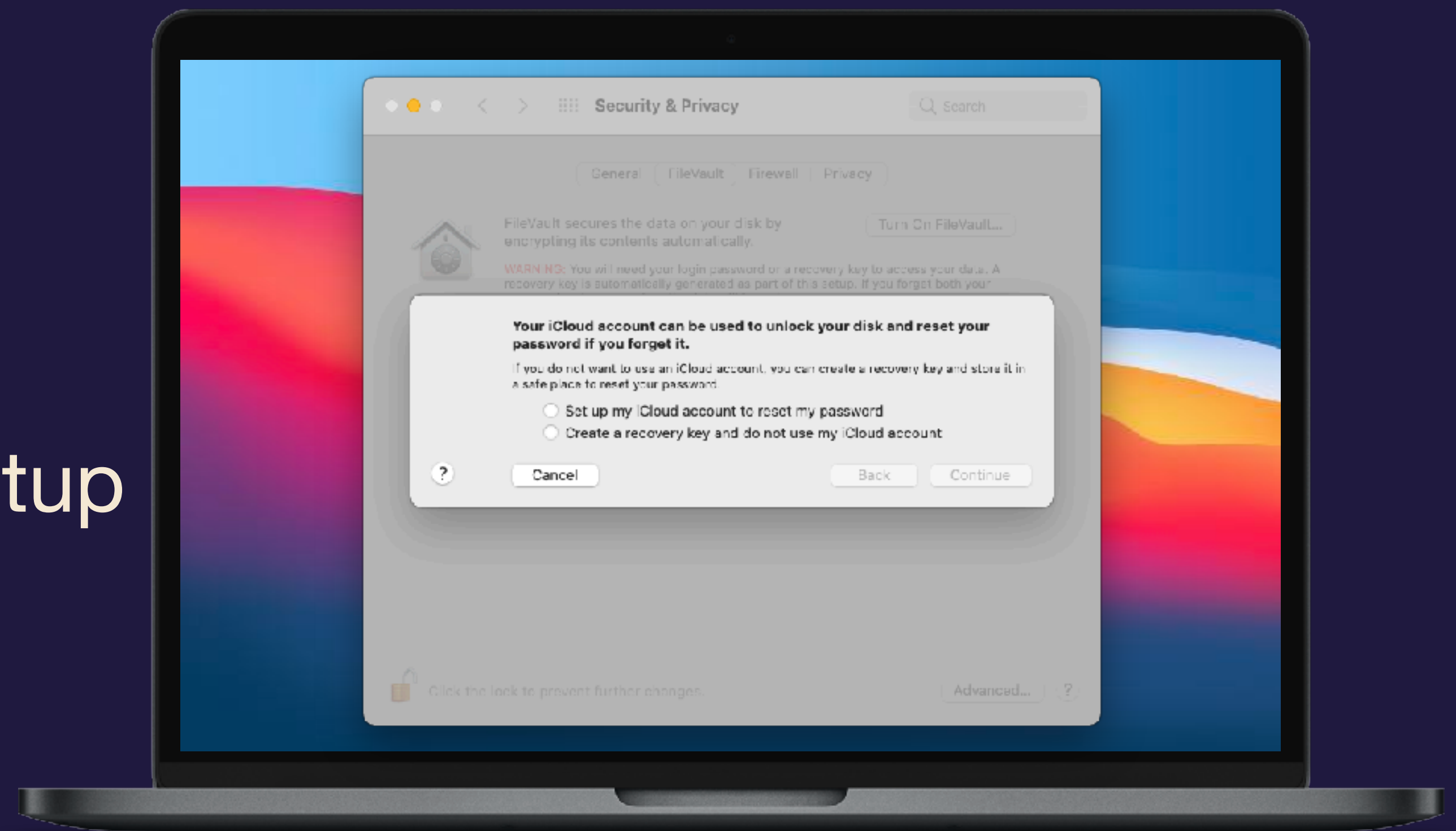
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dataJAR
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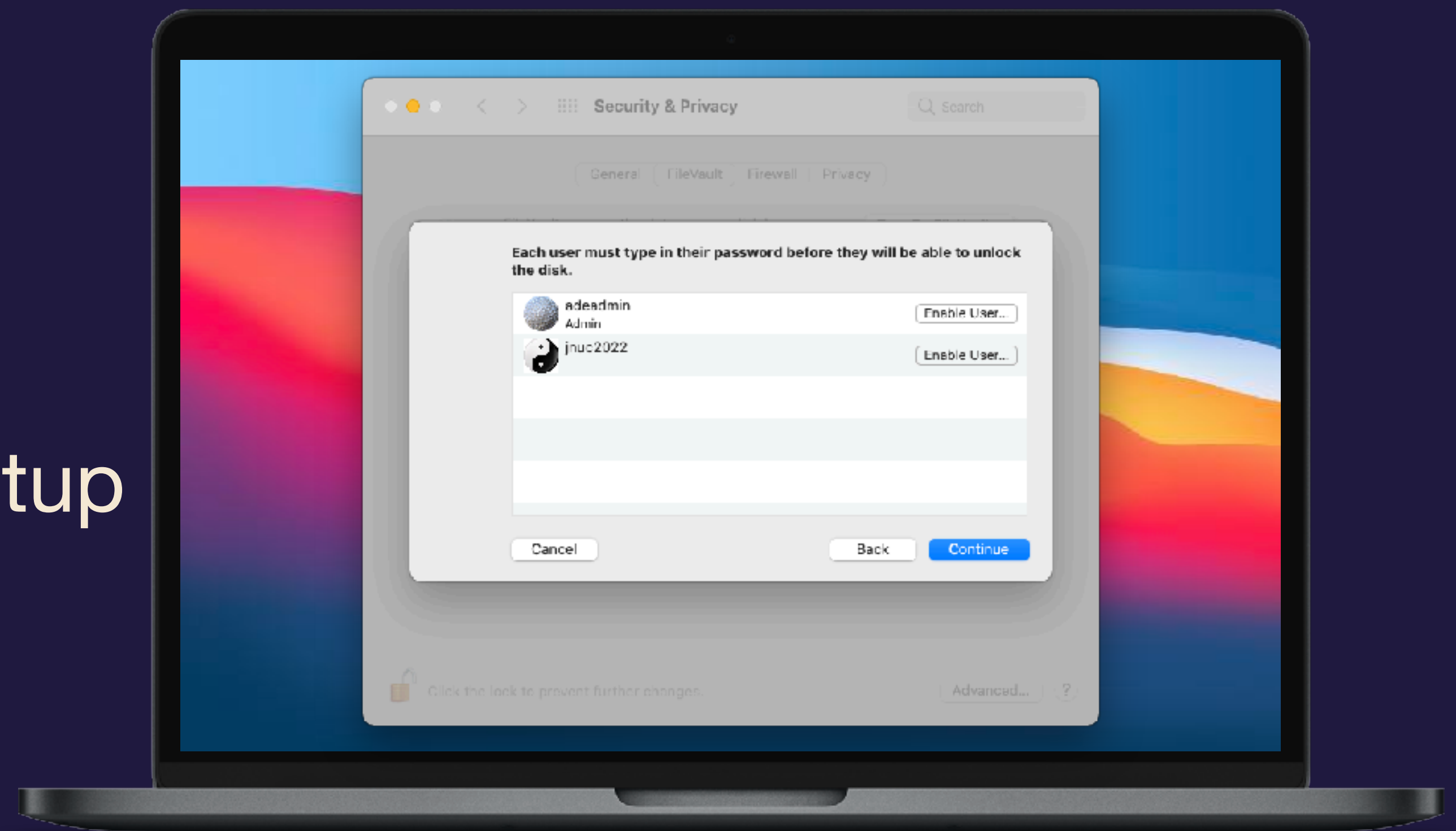
Big Sur –
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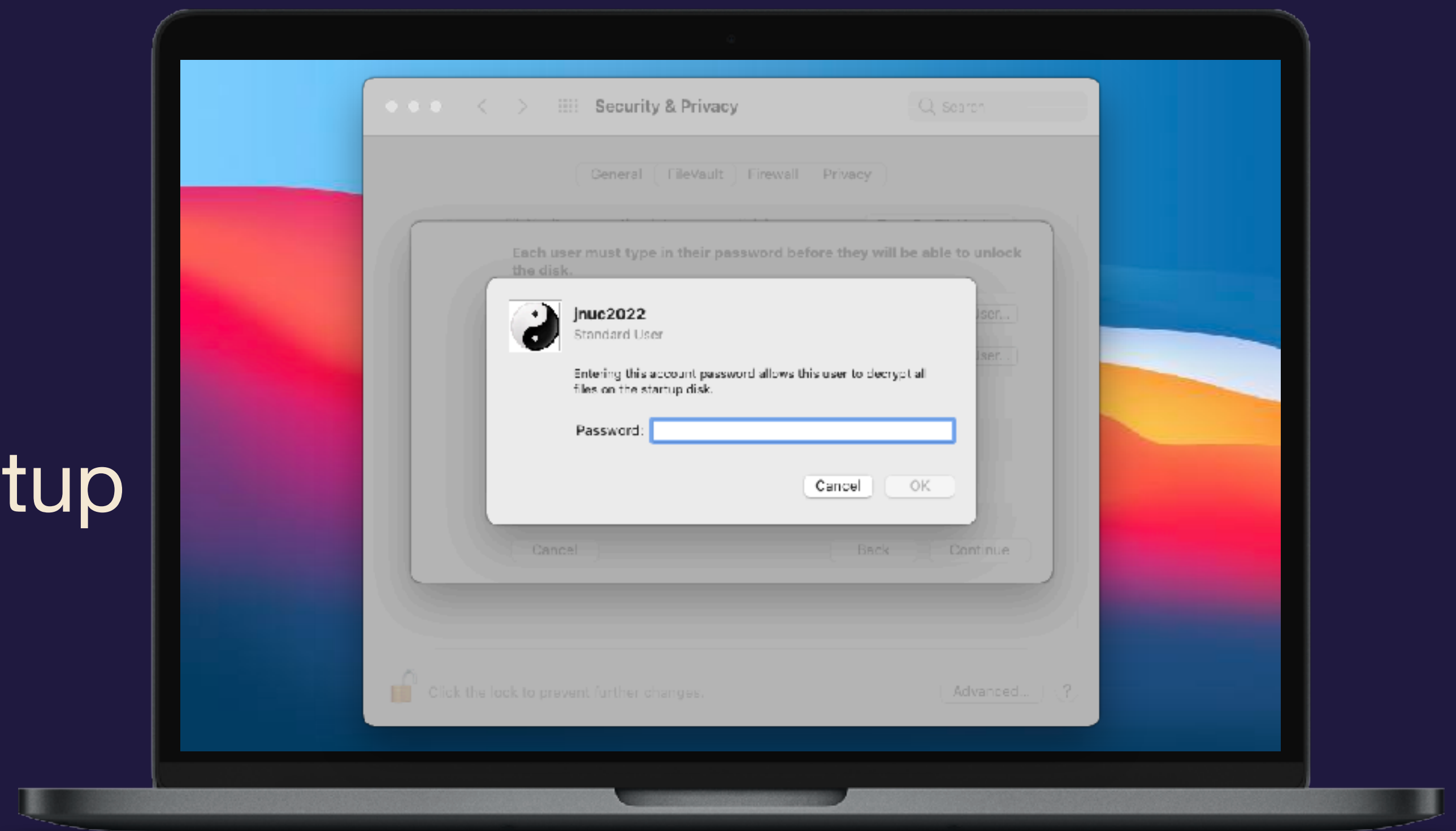
dataJAR
Beyond Device Management

Big Sur -
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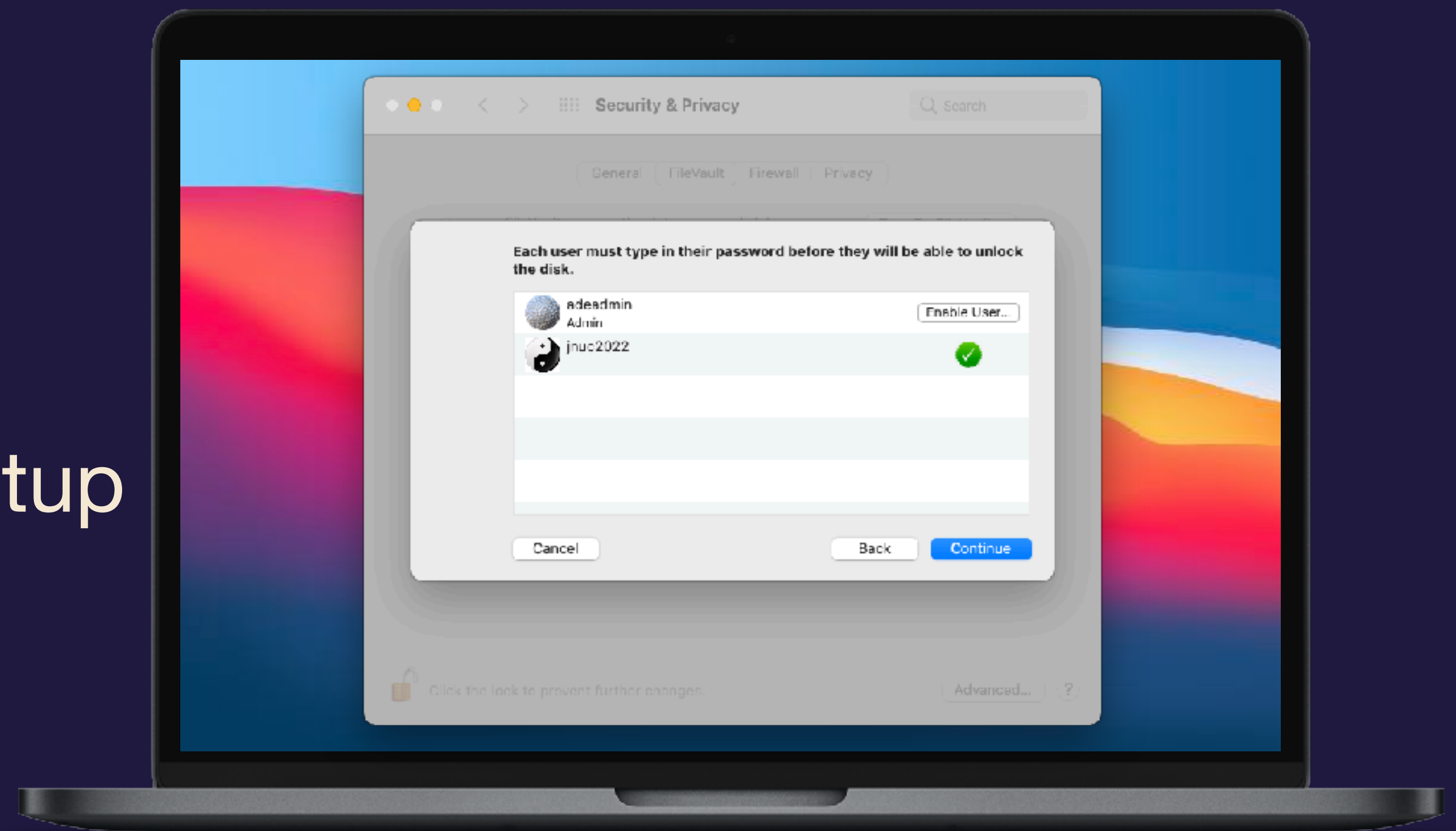
Secure Tokens

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Secure Tokens

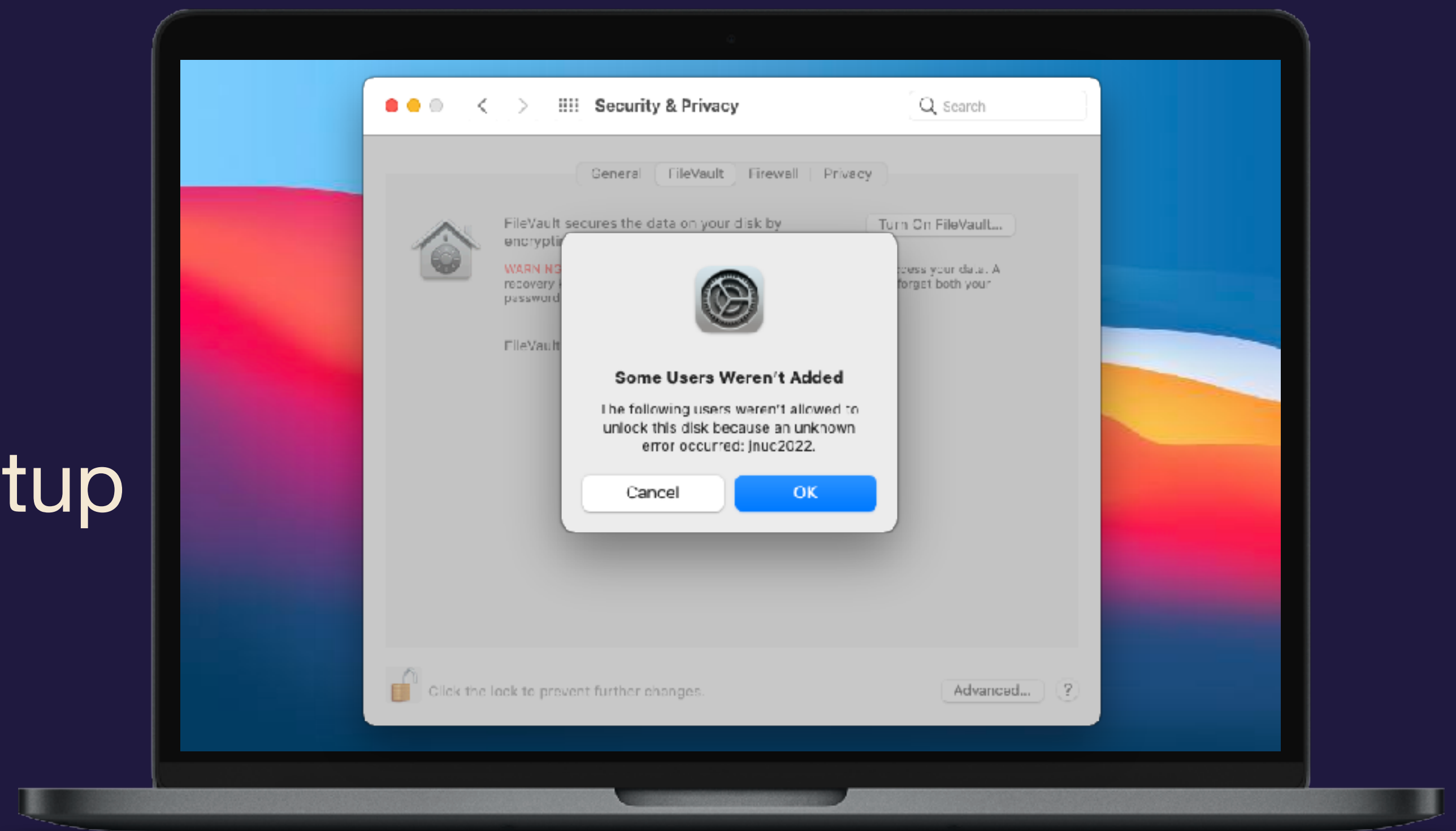
Big Sur –
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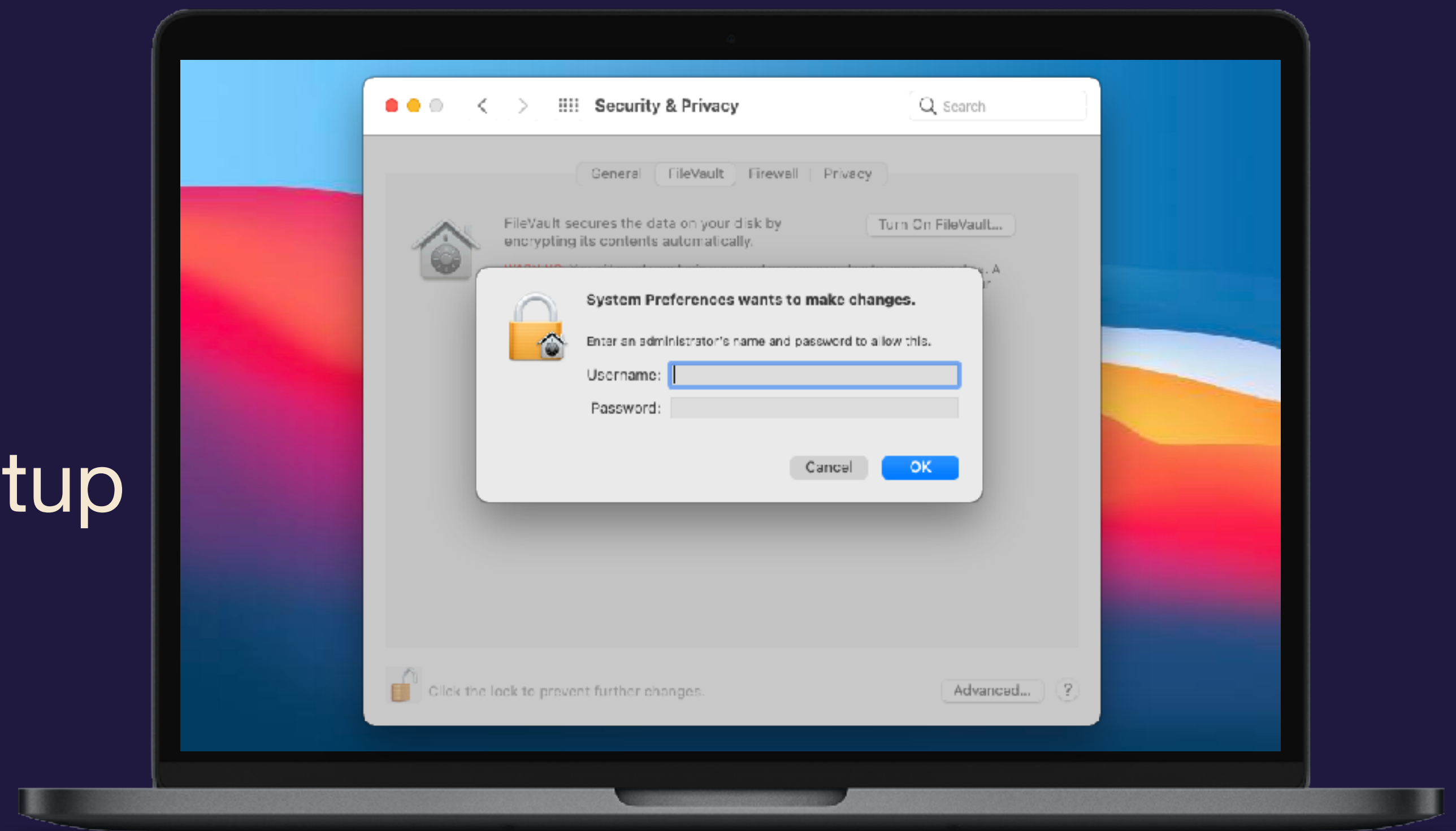
Big Sur –
Standard User, with delay at Setup
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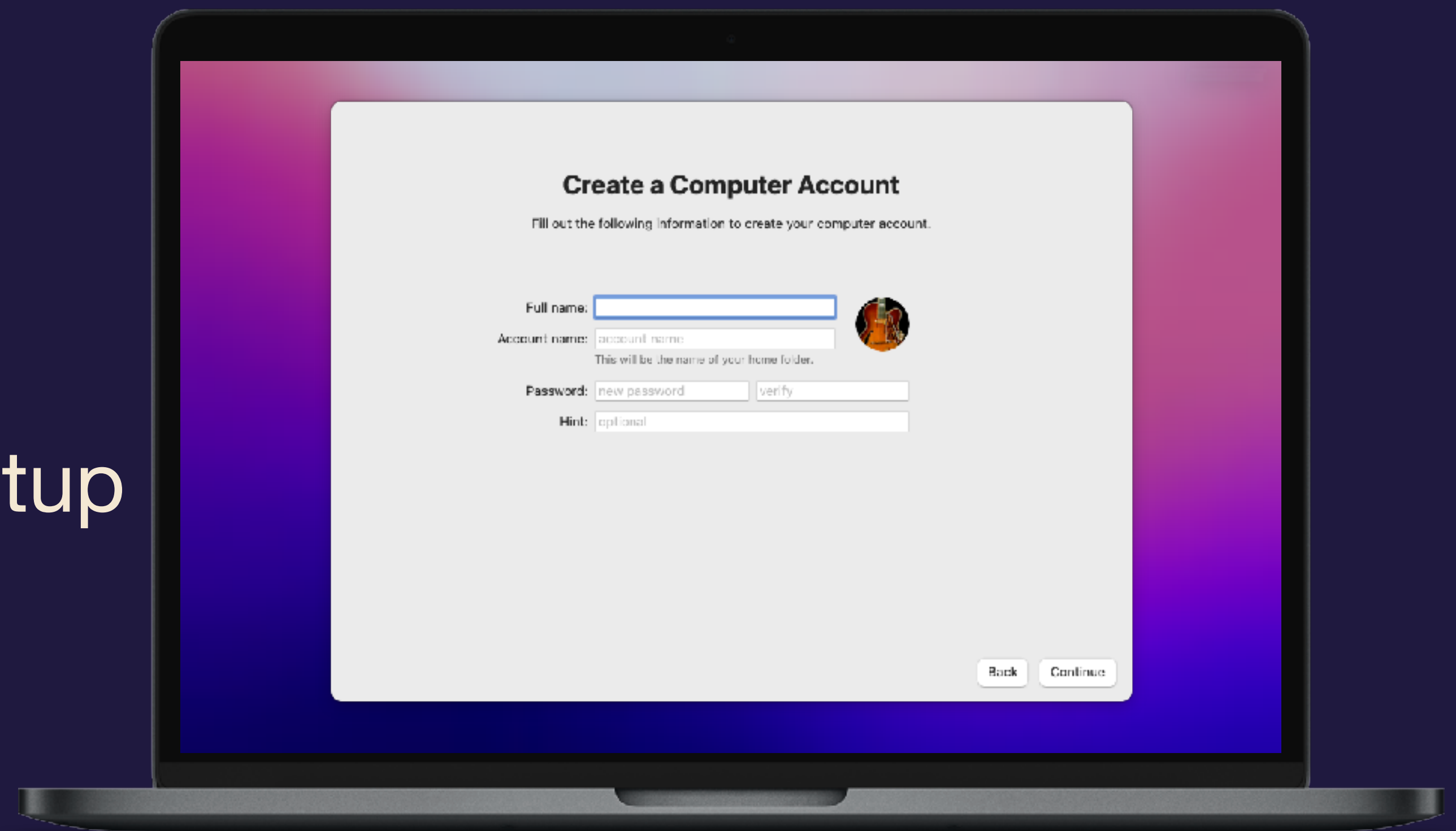
Big Sur –
Standard User, with delay at Setup
Assistant



Secure Tokens

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Beyond Device Management

Monterey -
Standard User, with delay at Setup
Assistant



Secure Tokens

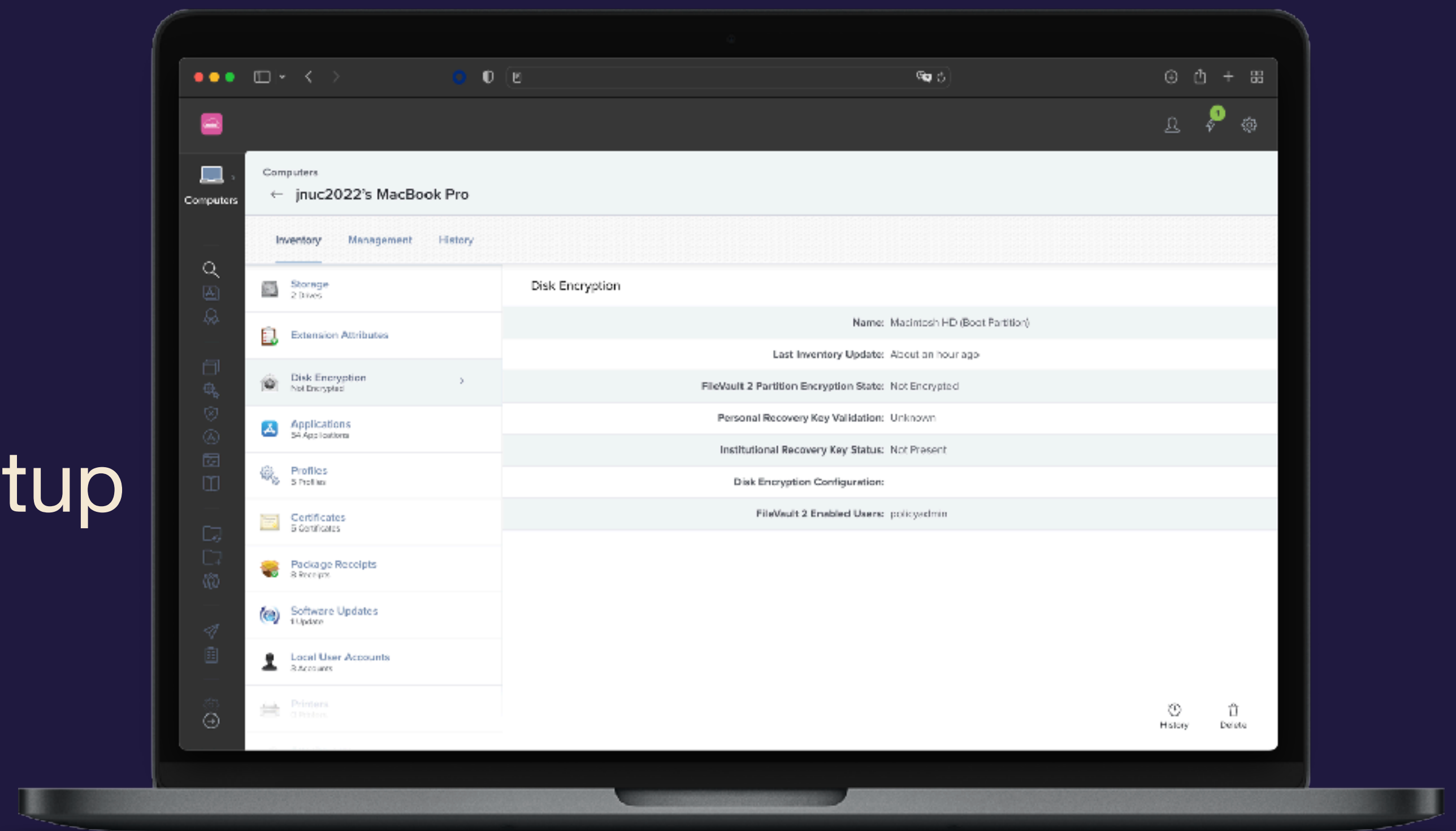
Monterey -
Standard User, with delay at Setup
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dataJAR
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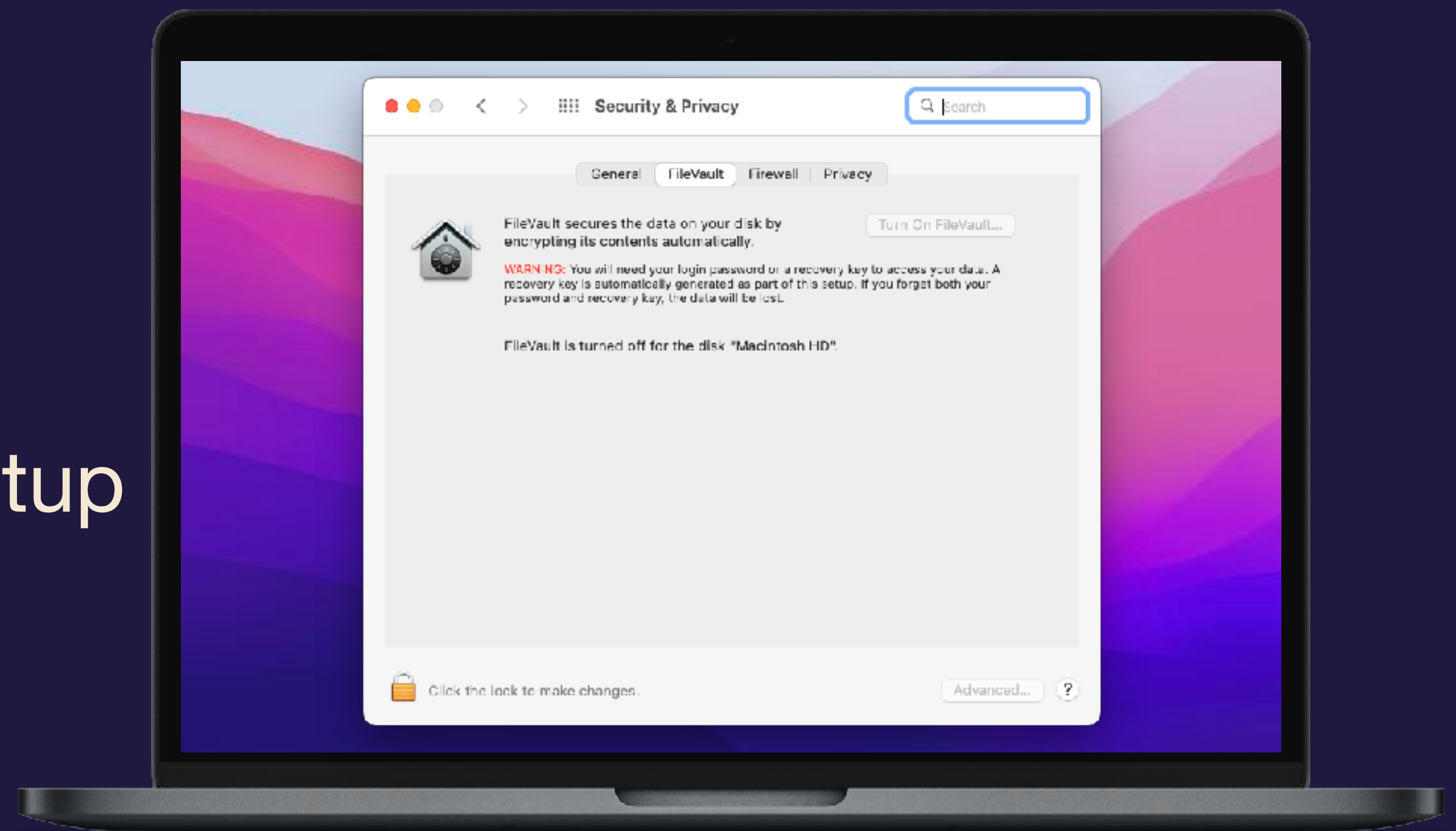
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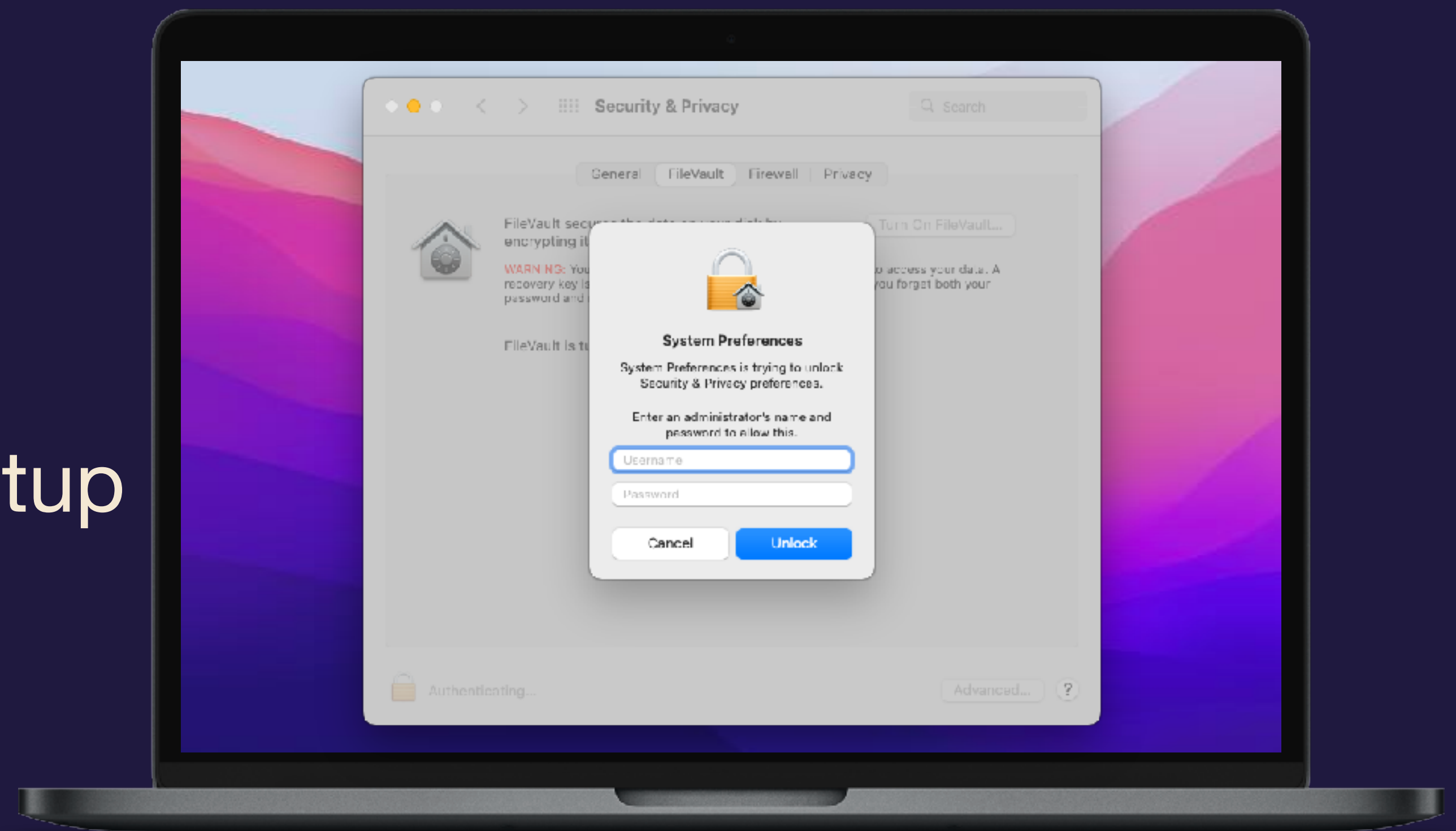
Monterey -
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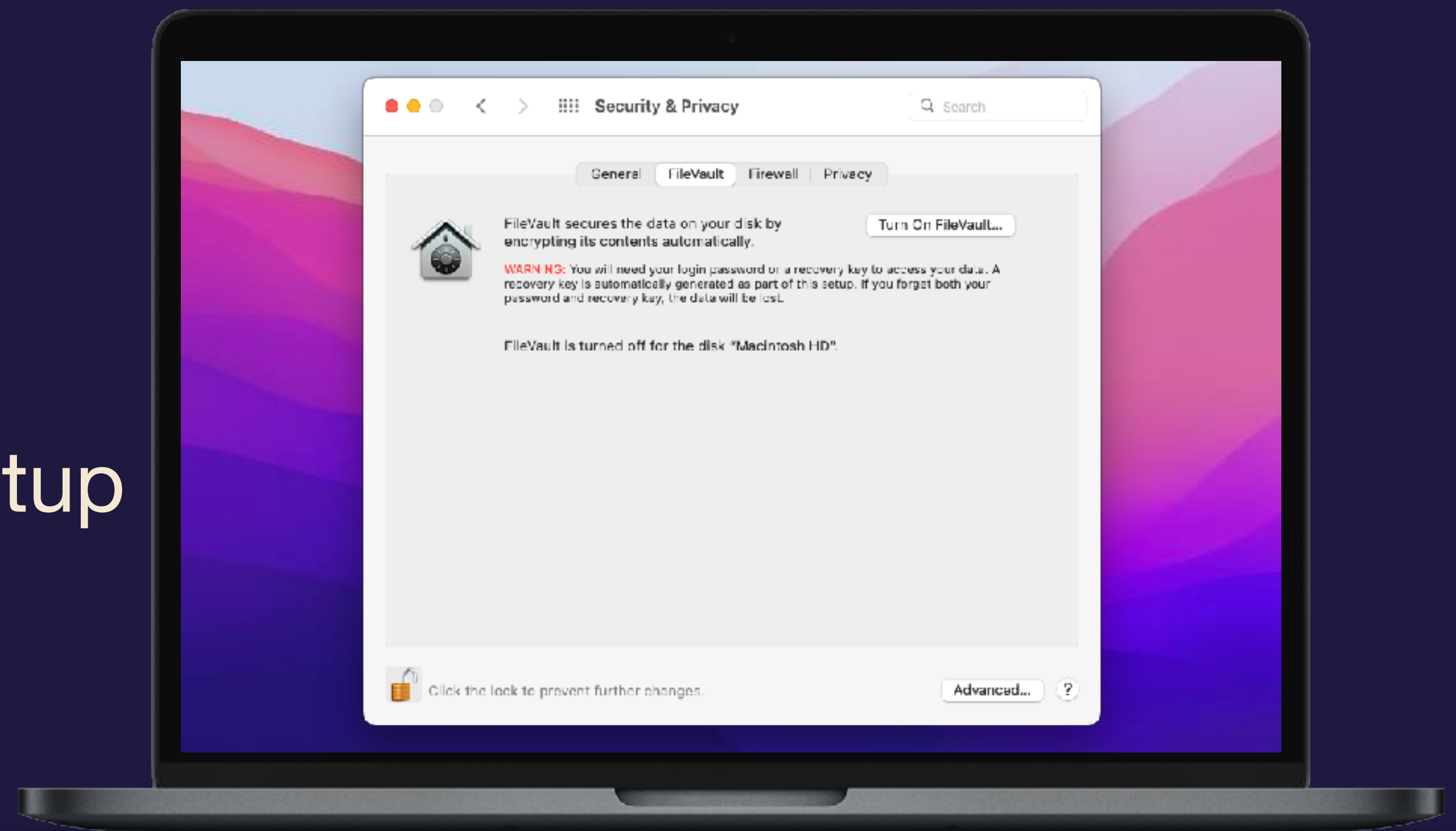
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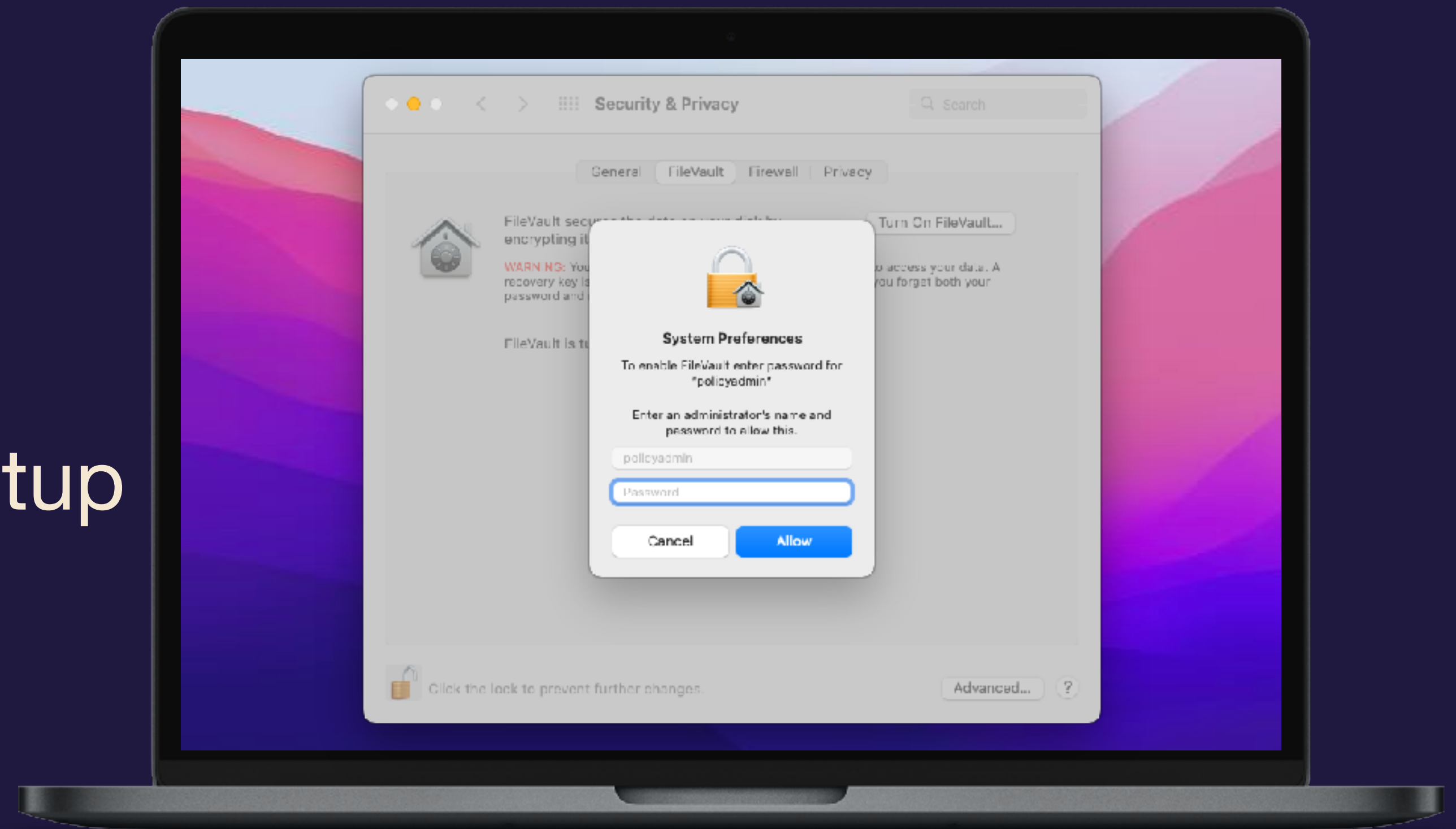
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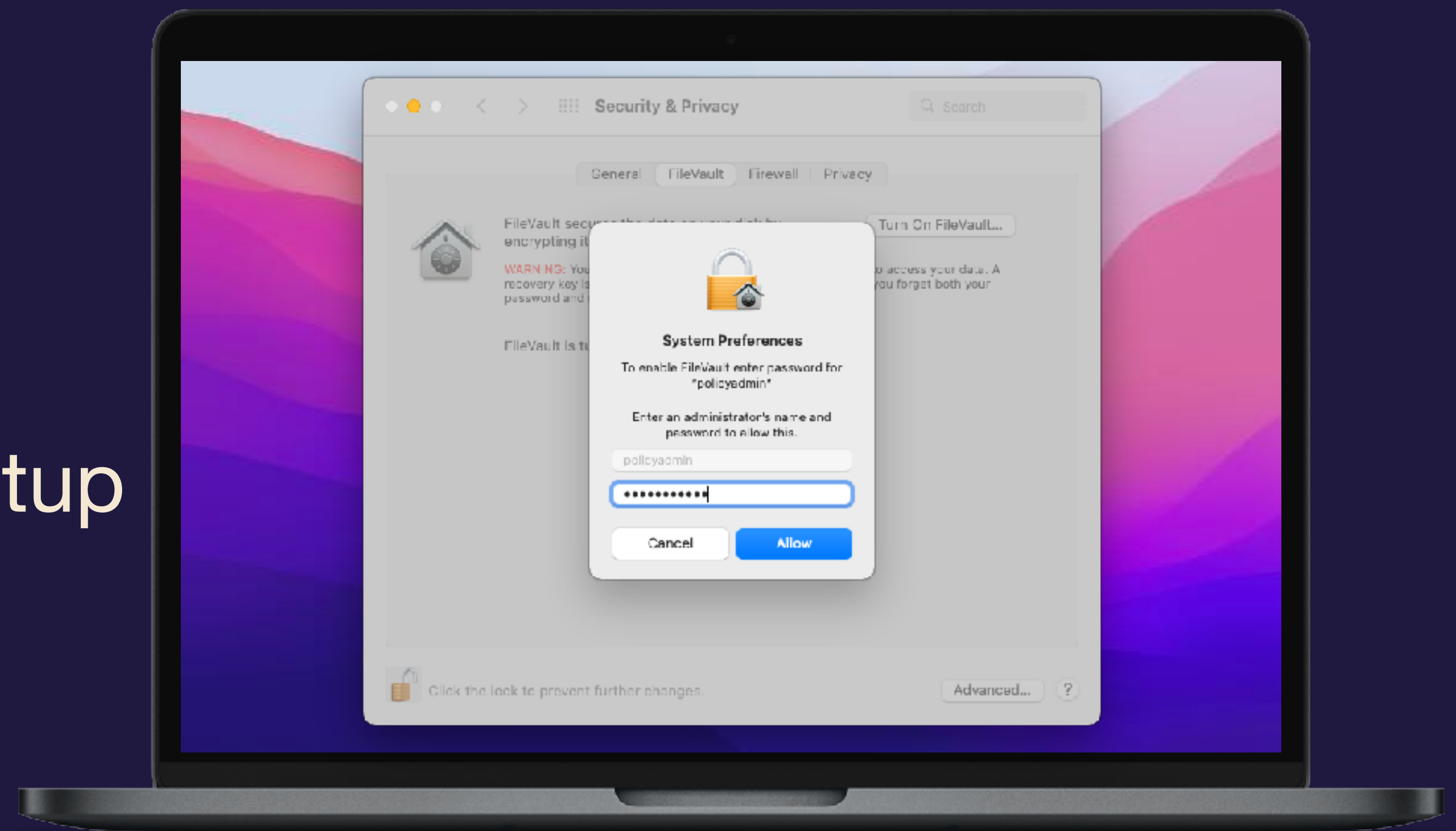
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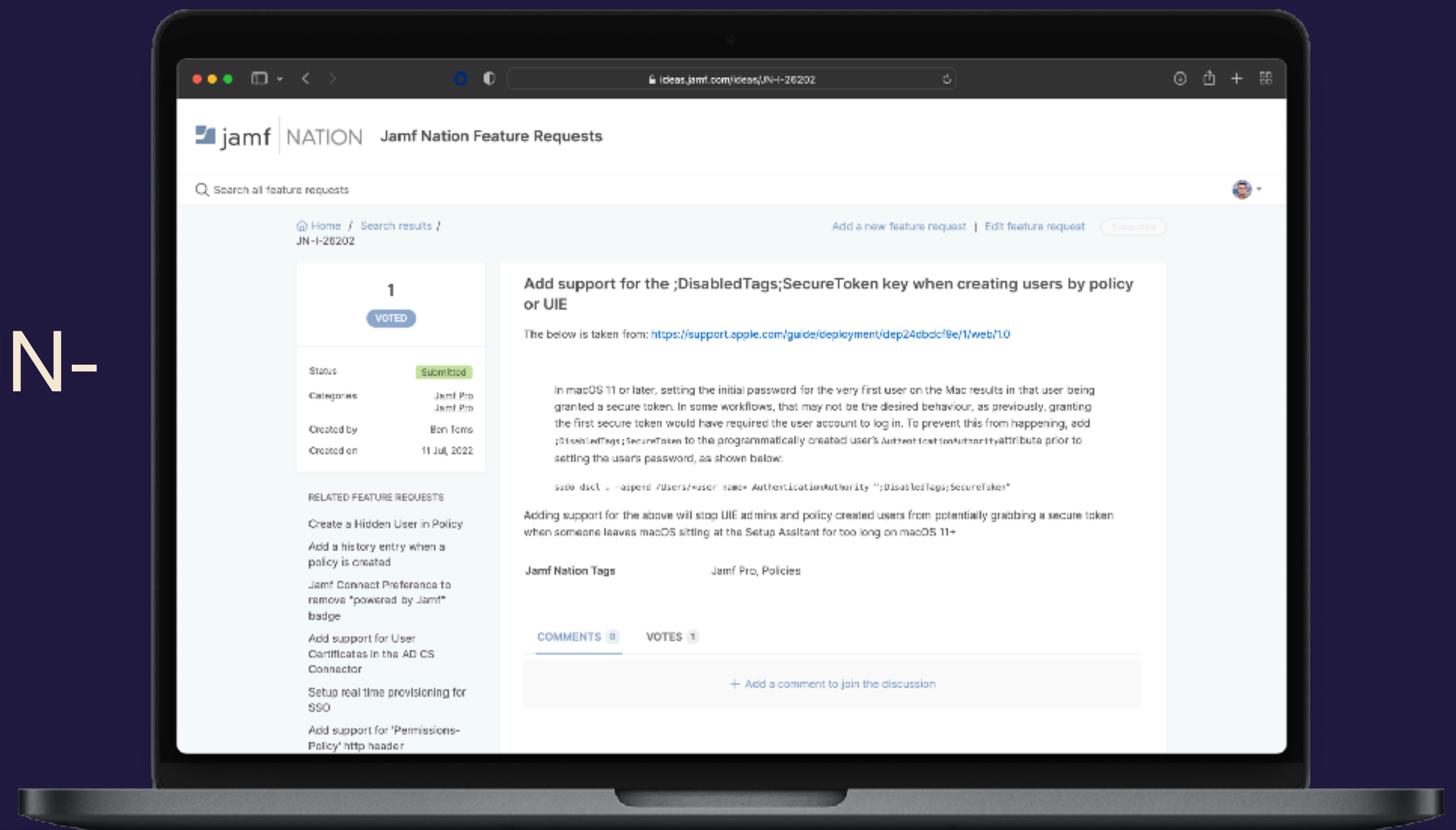
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Secure Tokens

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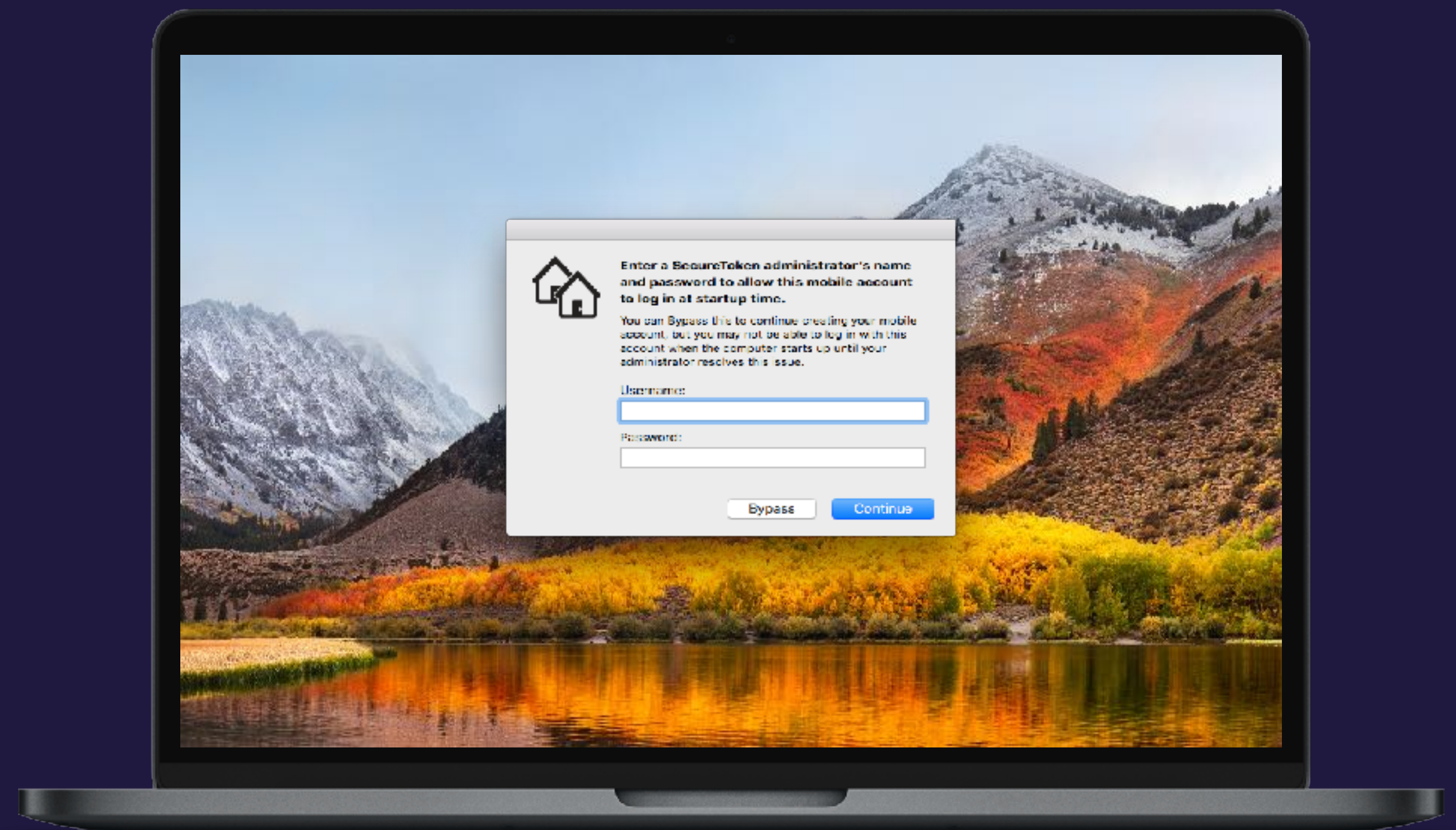
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Secure Tokens

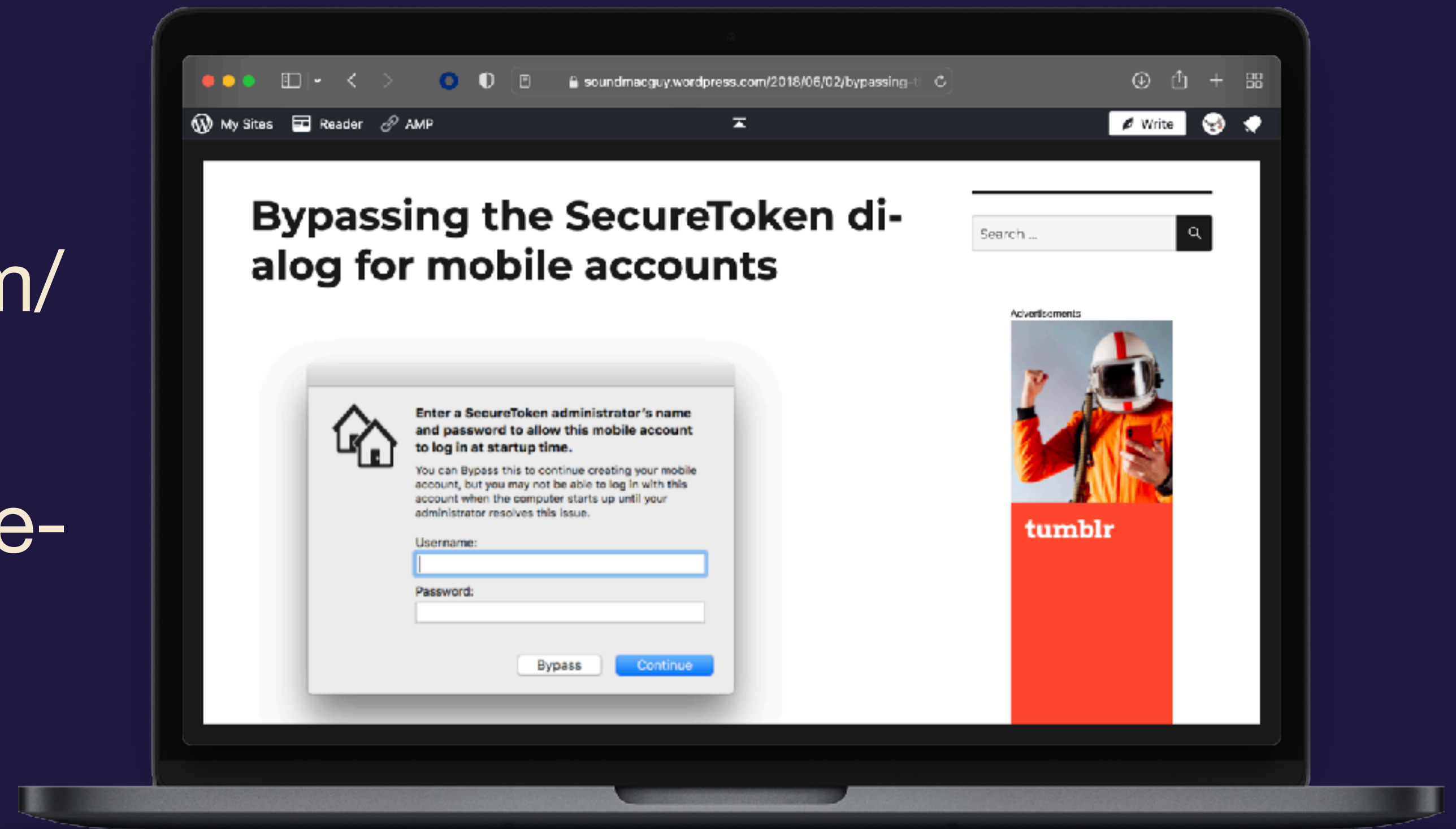
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10.13.4



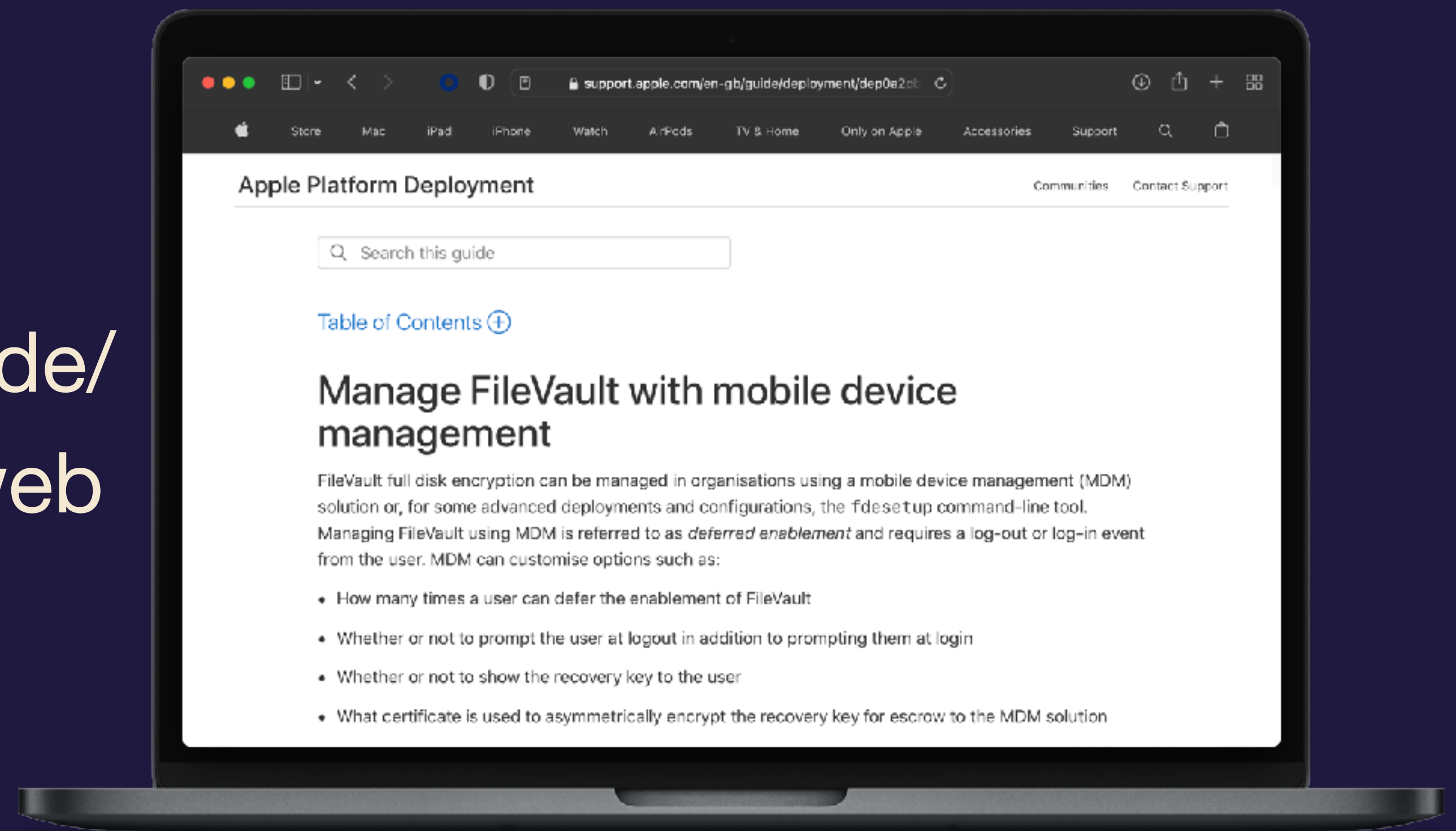
Secure Tokens

[https://
soundmacguy.wordpress.com/
2018/06/02/bypassing-the-
securetoken-dialog-for-mobile-
accounts](https://soundmacguy.wordpress.com/2018/06/02/bypassing-the-securetoken-dialog-for-mobile-accounts)



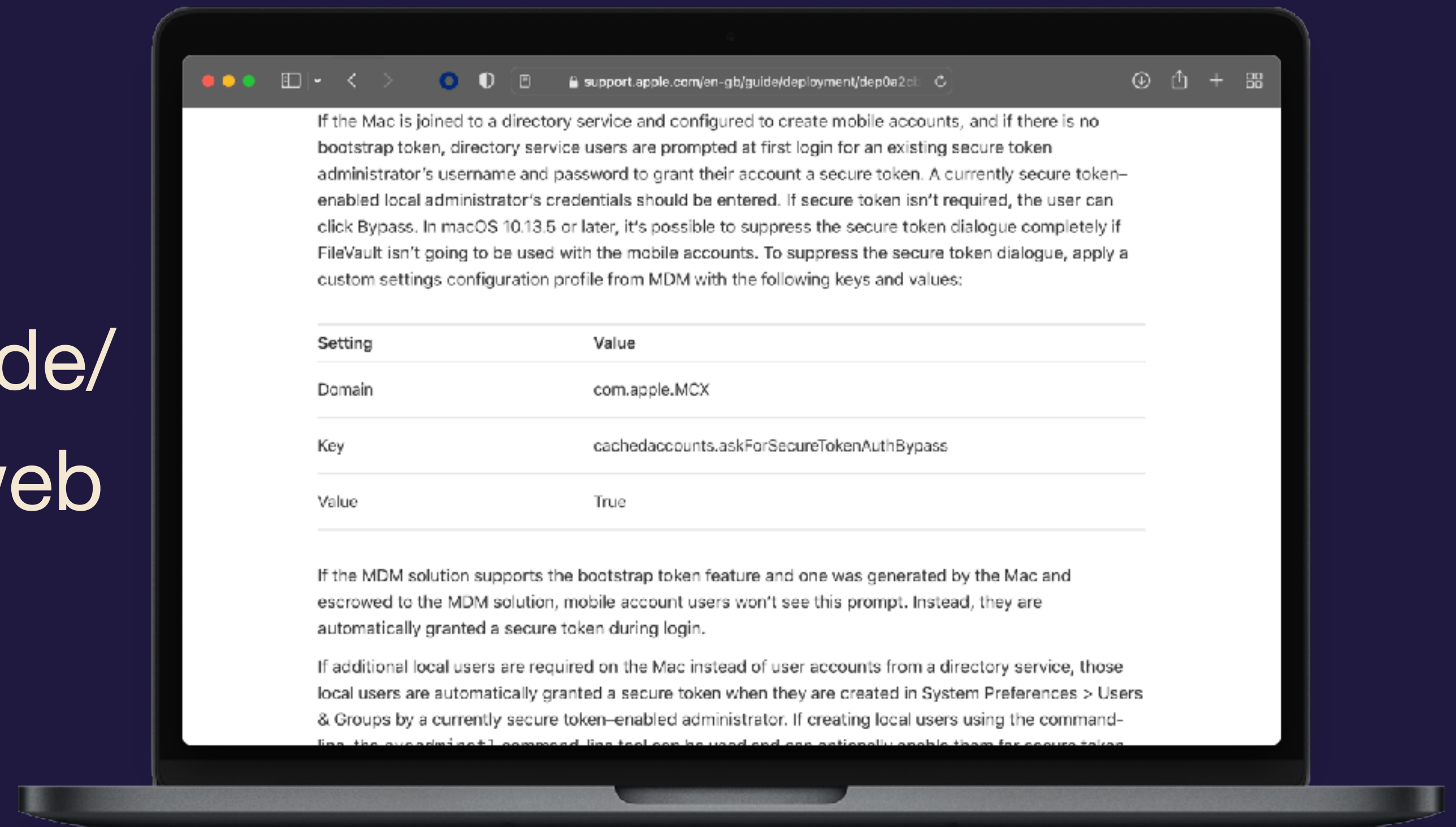
Secure Tokens

<https://support.apple.com/guide/deployment/dep0a2cb7686/web>



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Secure Tokens

“If the Mac is joined to a directory service and configured to create mobile accounts, and if there is no bootstrap token, directory service users are prompted at first login for an existing secure token administrator’s username and password to grant their account a secure token. A currently secure token–enabled local administrator’s credentials should be entered.”



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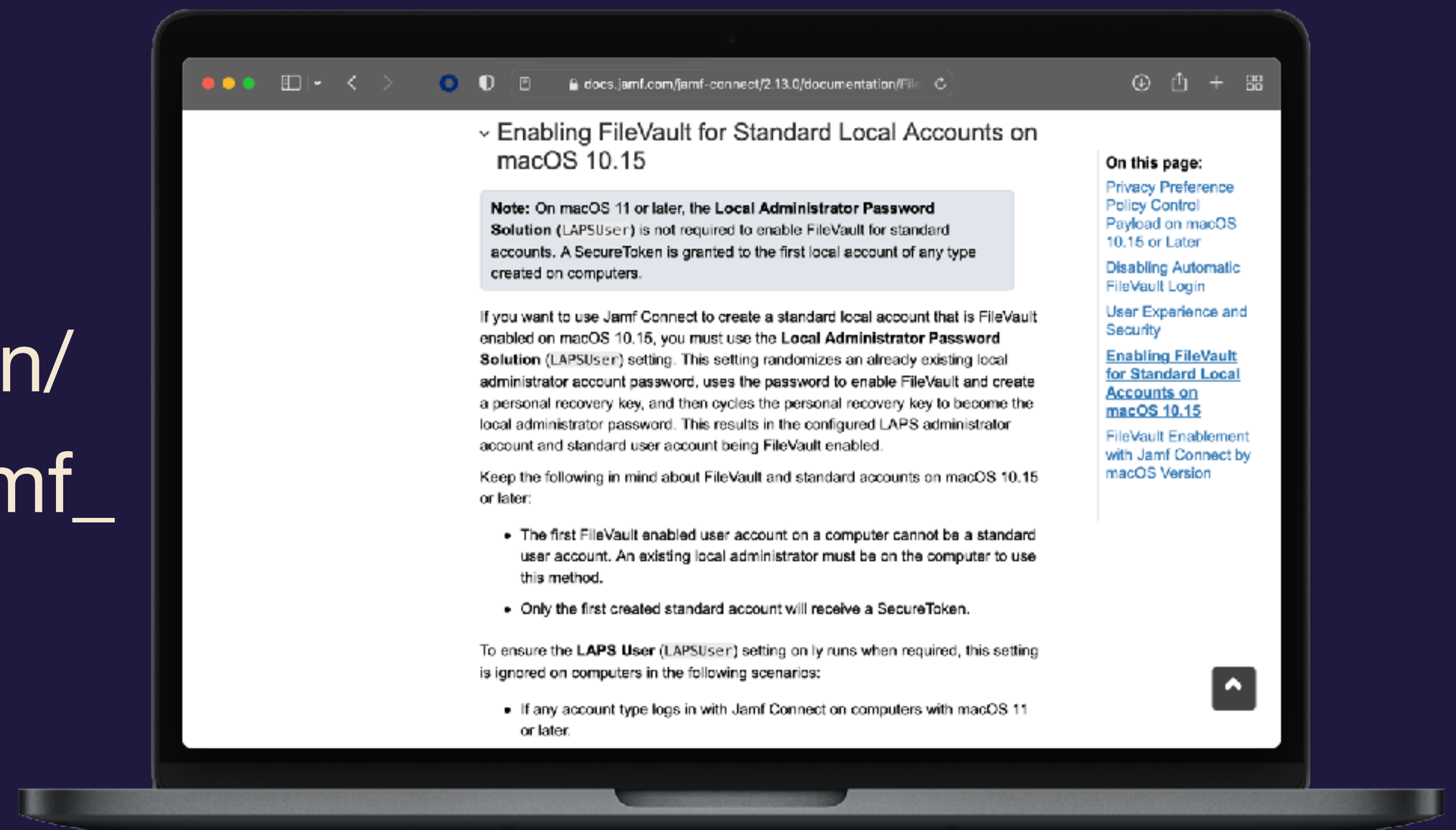
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Secure Tokens

https://docs.jamf.com/jamf-connect/2.13.0/documentation/FileVault_Enablement_with_Jamf_Connect.html



Secure Tokens

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...



Secure Tokens

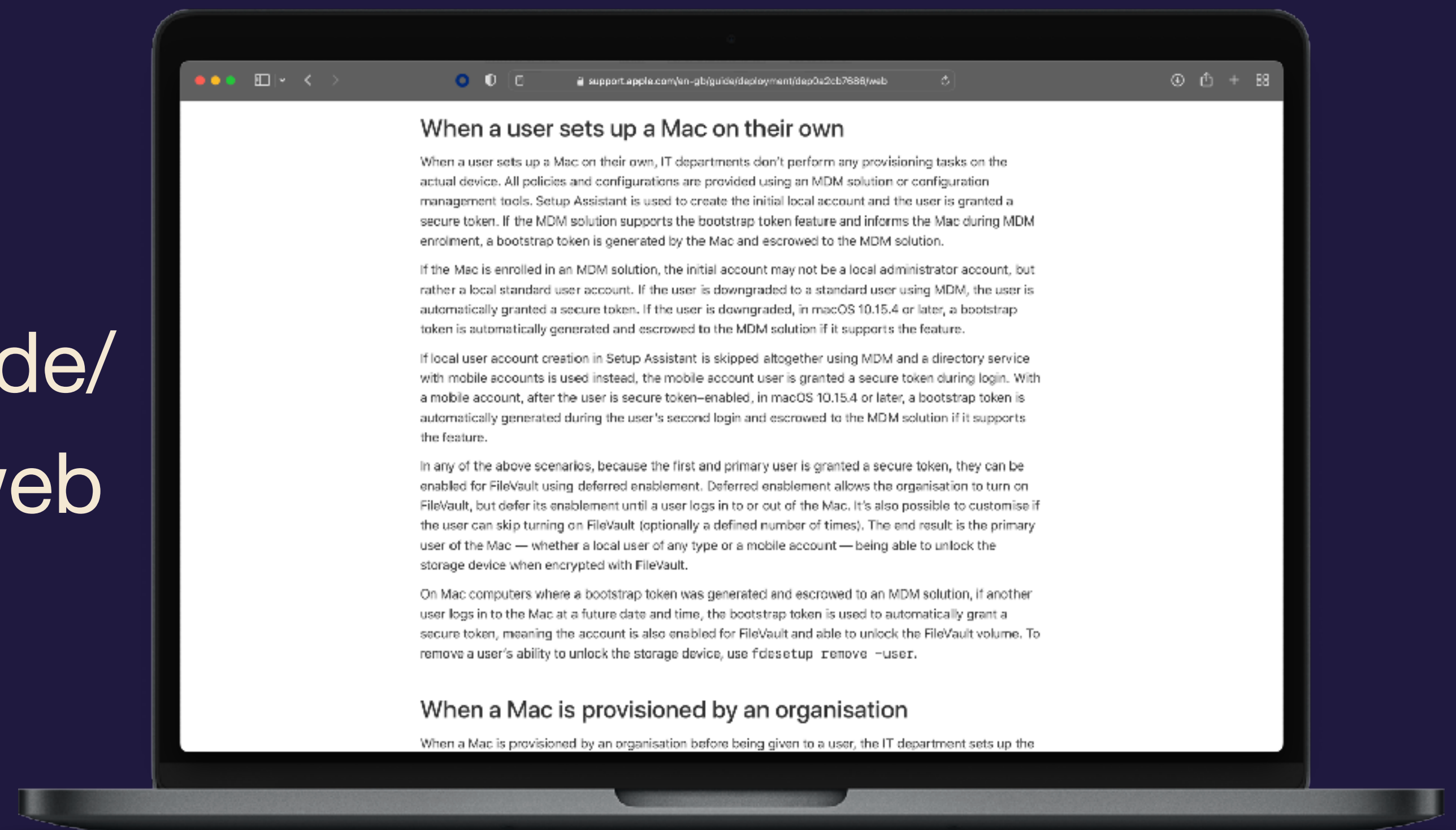
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Secure Tokens

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“Setup Assistant is used to create the initial local account and the user is granted a secure token.”



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Secure Tokens

“When a user sets up a Mac on their own, IT departments don’t perform any provisioning tasks on the actual device.”

“Setup Assistant is used to create the initial local account and the user is granted a secure token.”

“If local user account creation in Setup Assistant is skipped altogether using MDM and a directory service with mobile accounts is used instead, the mobile account user is granted a secure token during login.”



<https://support.apple.com/guide/deployment/dep0a2cb7686/web>

Secure Tokens



<https://support.apple.com/guide/deployment/dep0a2cb7686/web>

Secure Tokens

“In any of the above scenarios, because the first and primary user is granted a secure token, they can be enabled for FileVault using deferred enablement. Deferred enablement allows the organisation to turn on FileVault, but defer its enablement until a user logs in to or out of the Mac.”



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“The end result is the primary user of the Mac — whether a local user of any type or a mobile account — being able to unlock the storage device when encrypted with FileVault.”



<https://support.apple.com/guide/deployment/dep0a2cb7686/web>

Secure Tokens



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Secure Tokens

“MDM configurations or the fdesetup command-line tool can be used to configure FileVault.”



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“In macOS 10.15 or later, using fdesetup to turn on FileVault by providing the user name and password is deprecated and won't be recognised in a future release. The command continues to function but remains deprecated in macOS 11 and macOS 12.0.1.”



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Secure Tokens

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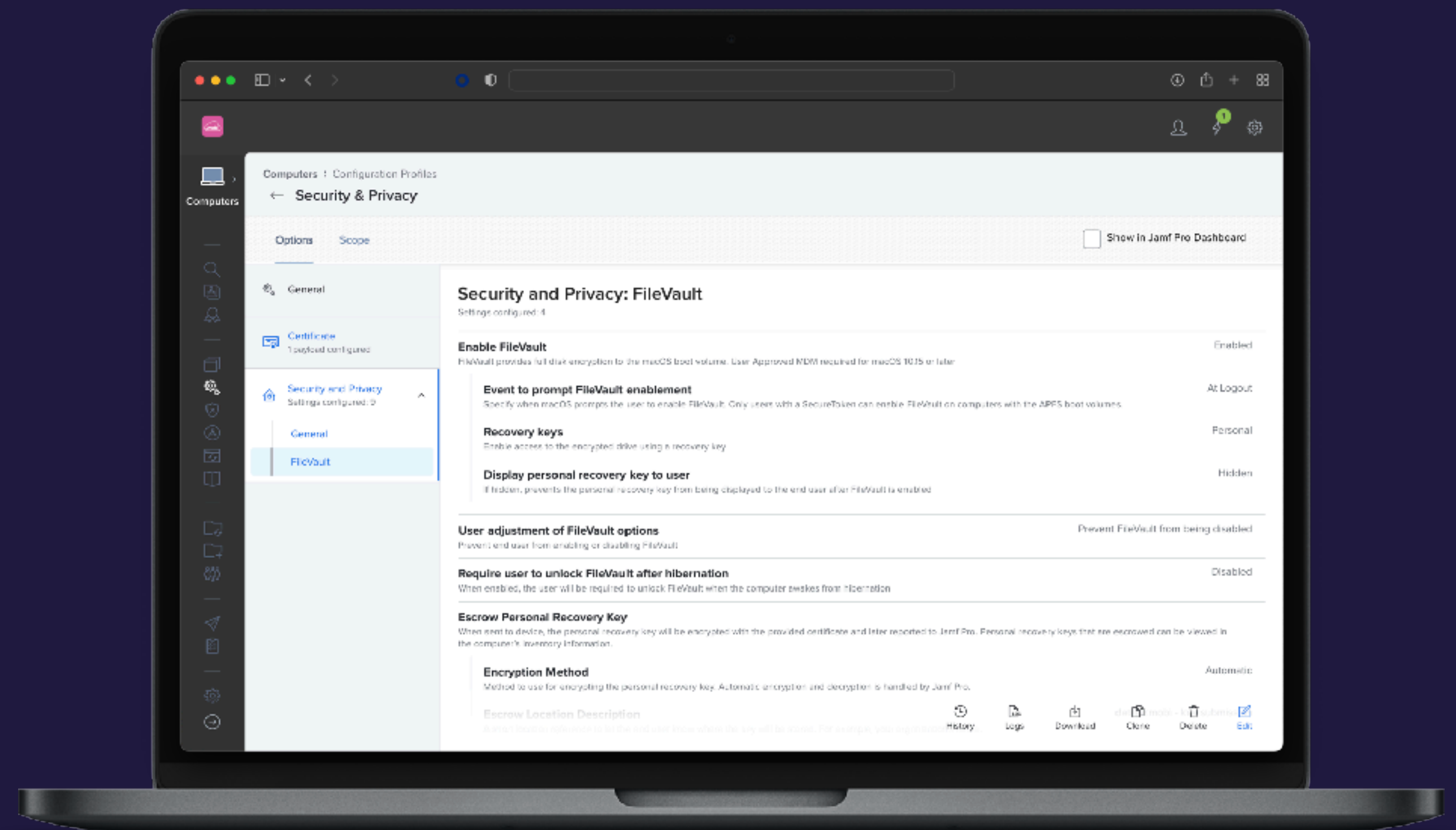
“Consider using deferred enablement using MDM instead.”



<https://support.apple.com/guide/deployment/dep0a2cb7686/web>

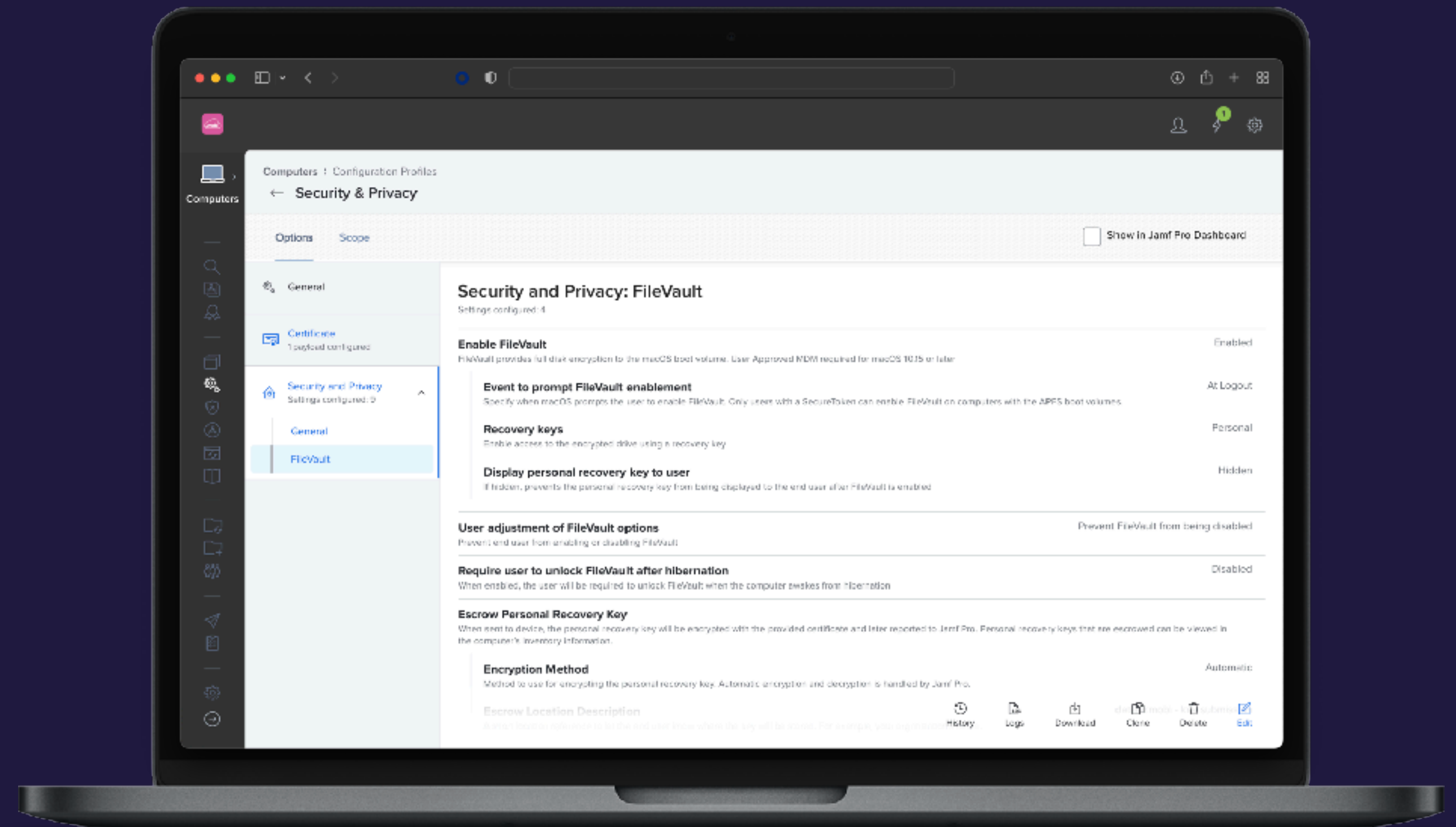
Secure Tokens

dataJAR
Beyond Device Management



Secure Tokens

dataJAR
Beyond Device Management

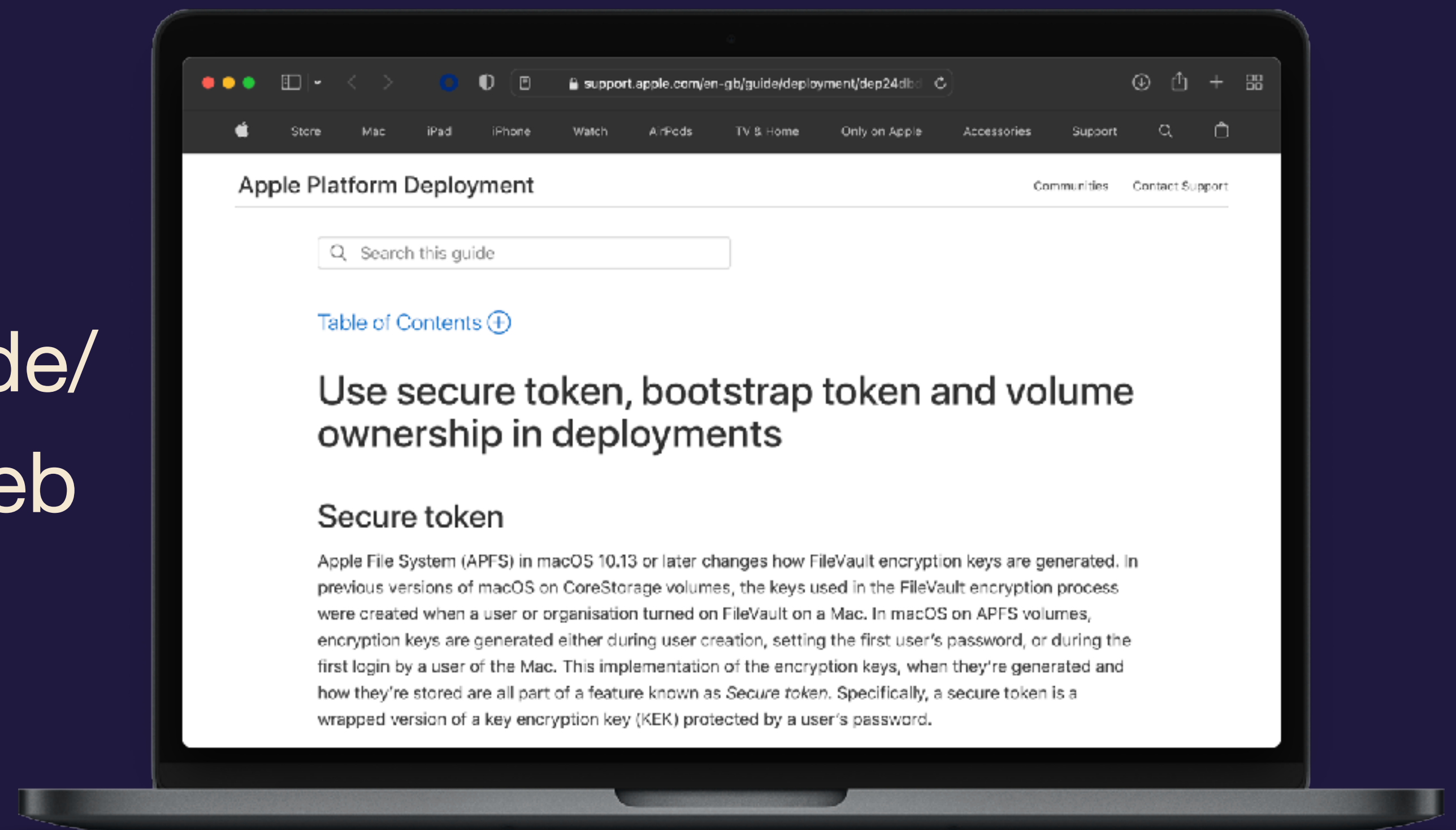


Bootstrap tokens



Bootstrap Tokens

<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>



Bootstrap Tokens



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Bootstrap Tokens

“In macOS 10.15 or later, a *bootstrap token* is used to help with granting a secure token to both mobile accounts and the optional device enrolment-created administrator account (“managed administrator”). In macOS 11 or later, the bootstrap token can grant a secure token to any user logging in to a Mac computer, including local user accounts.”



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“In macOS 10.15.4 or later, when a user who is secure token-enabled logs in for the first time, a bootstrap token is generated and escrowed to MDM. A bootstrap token can also be generated and escrowed to MDM using the profiles command-line tool, if needed.”

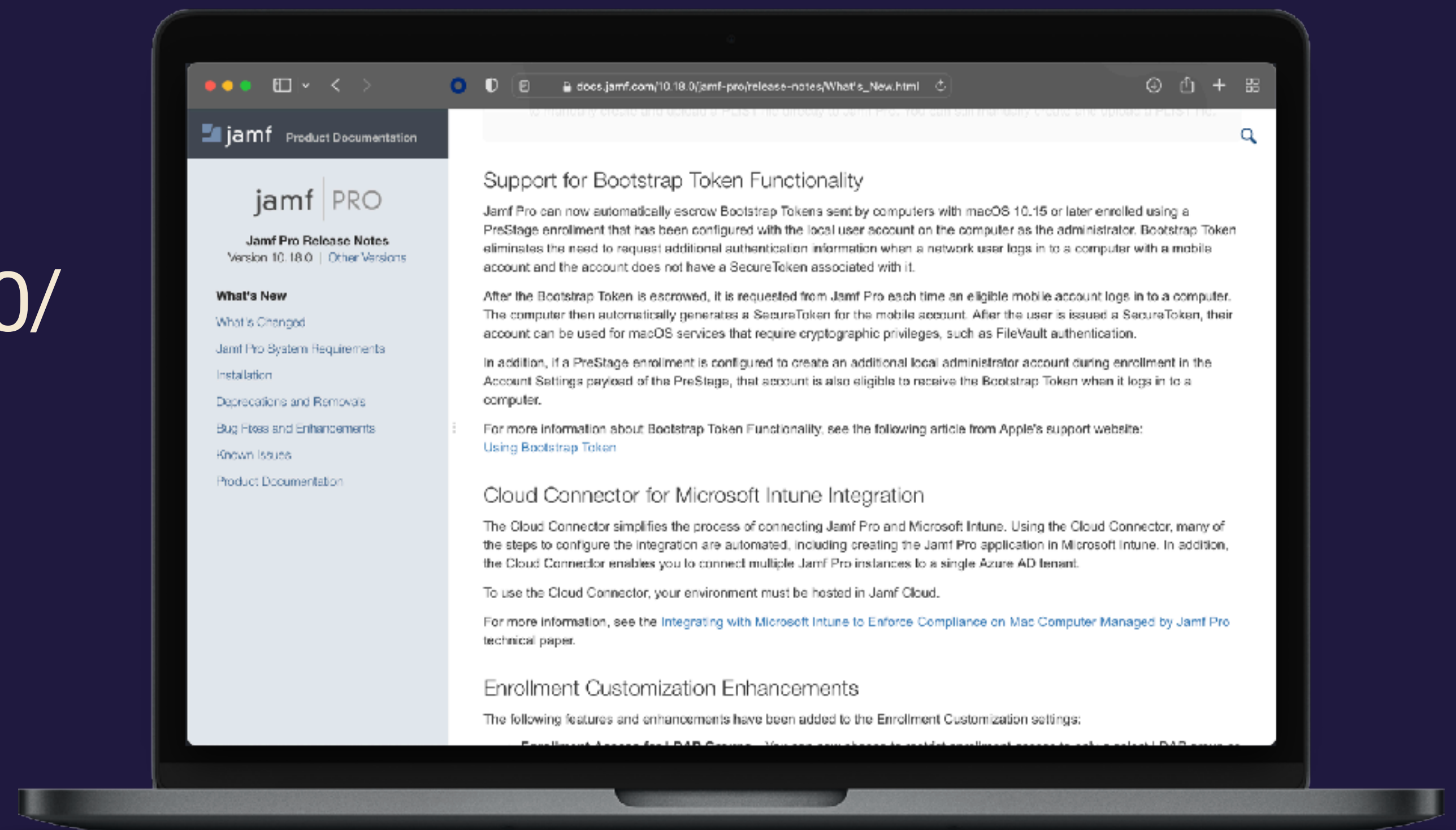
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Bootstrap Tokens

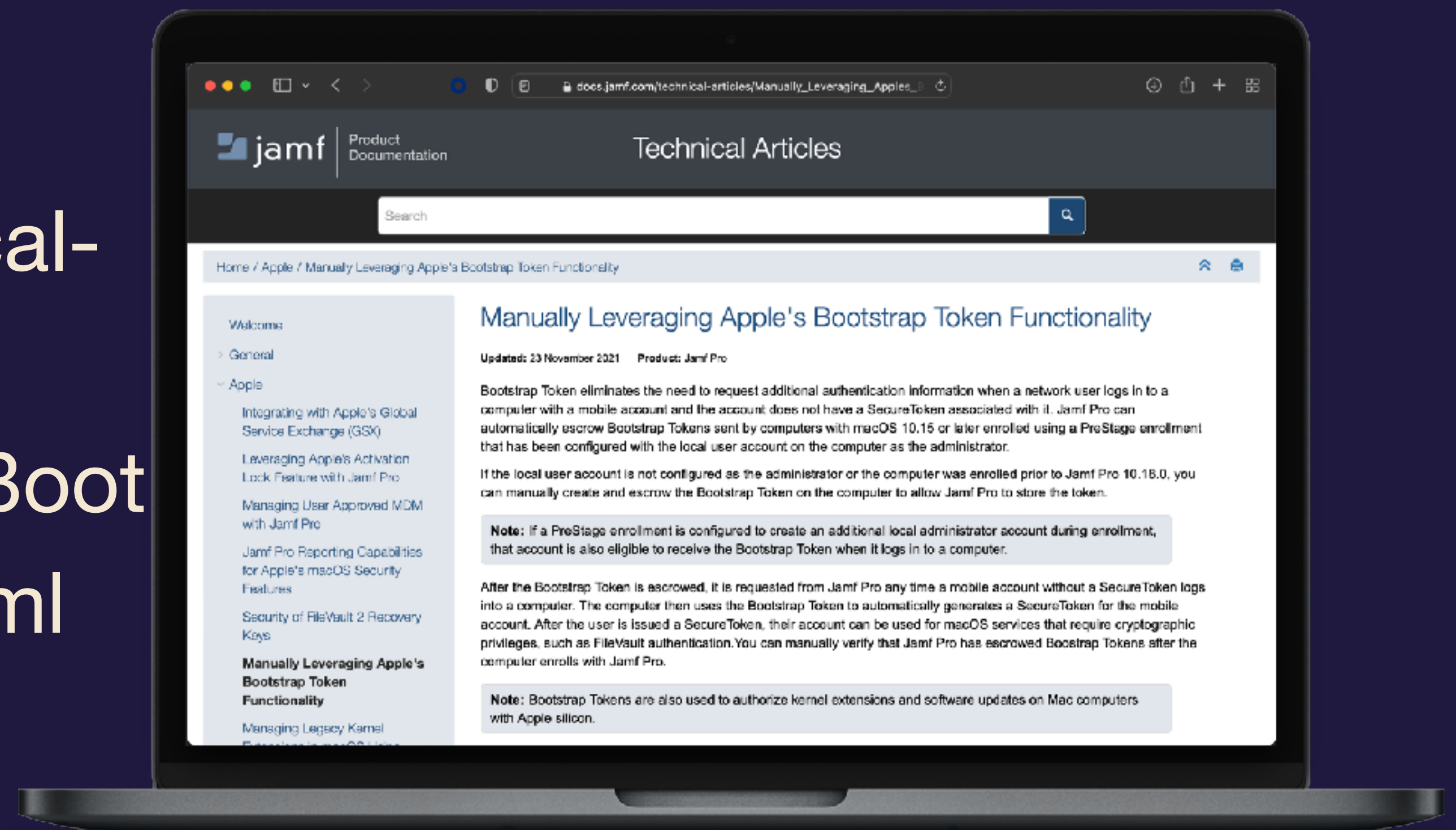
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[https://docs.jamf.com/10.18.0/
jamf-pro/release-notes/
What's_New.html](https://docs.jamf.com/10.18.0/jamf-pro/release-notes/What's_New.html)



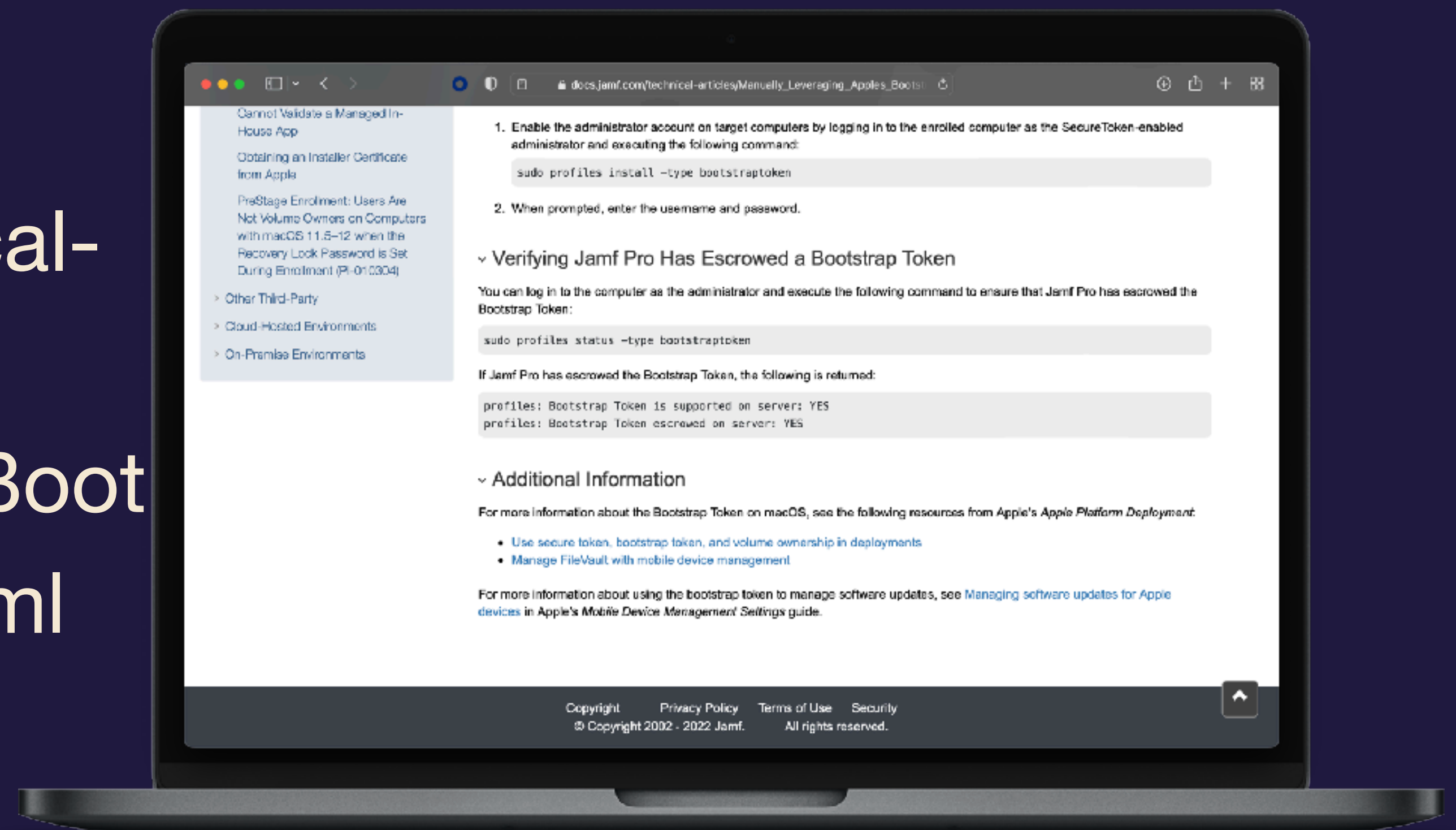
Bootstrap Tokens

https://docs.jamf.com/technical-articles/Manually_Leveraging_Apples_Bootstrap_Token_Functionality.html



Bootstrap Tokens

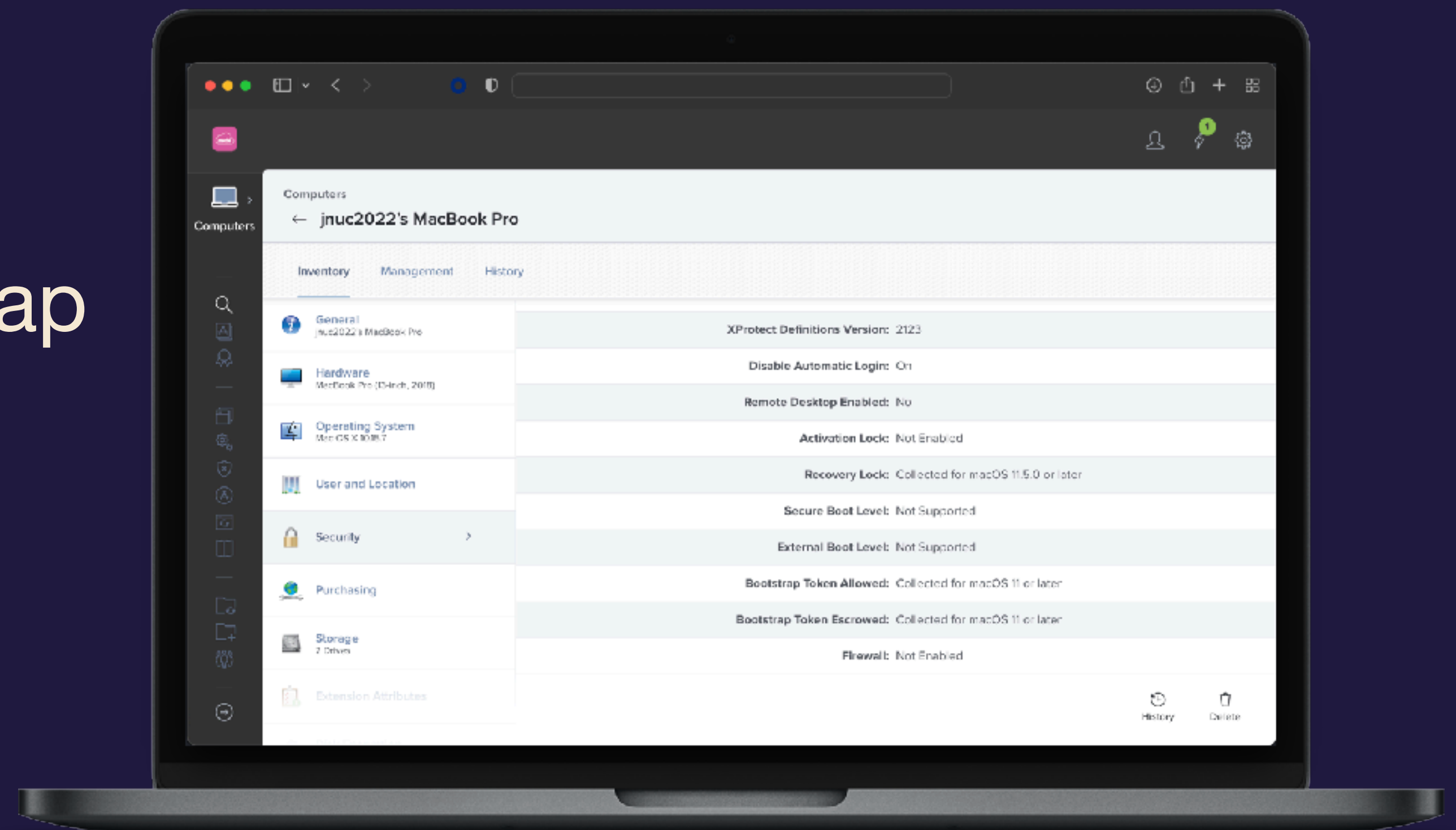
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Bootstrap Tokens

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Beyond Device Management

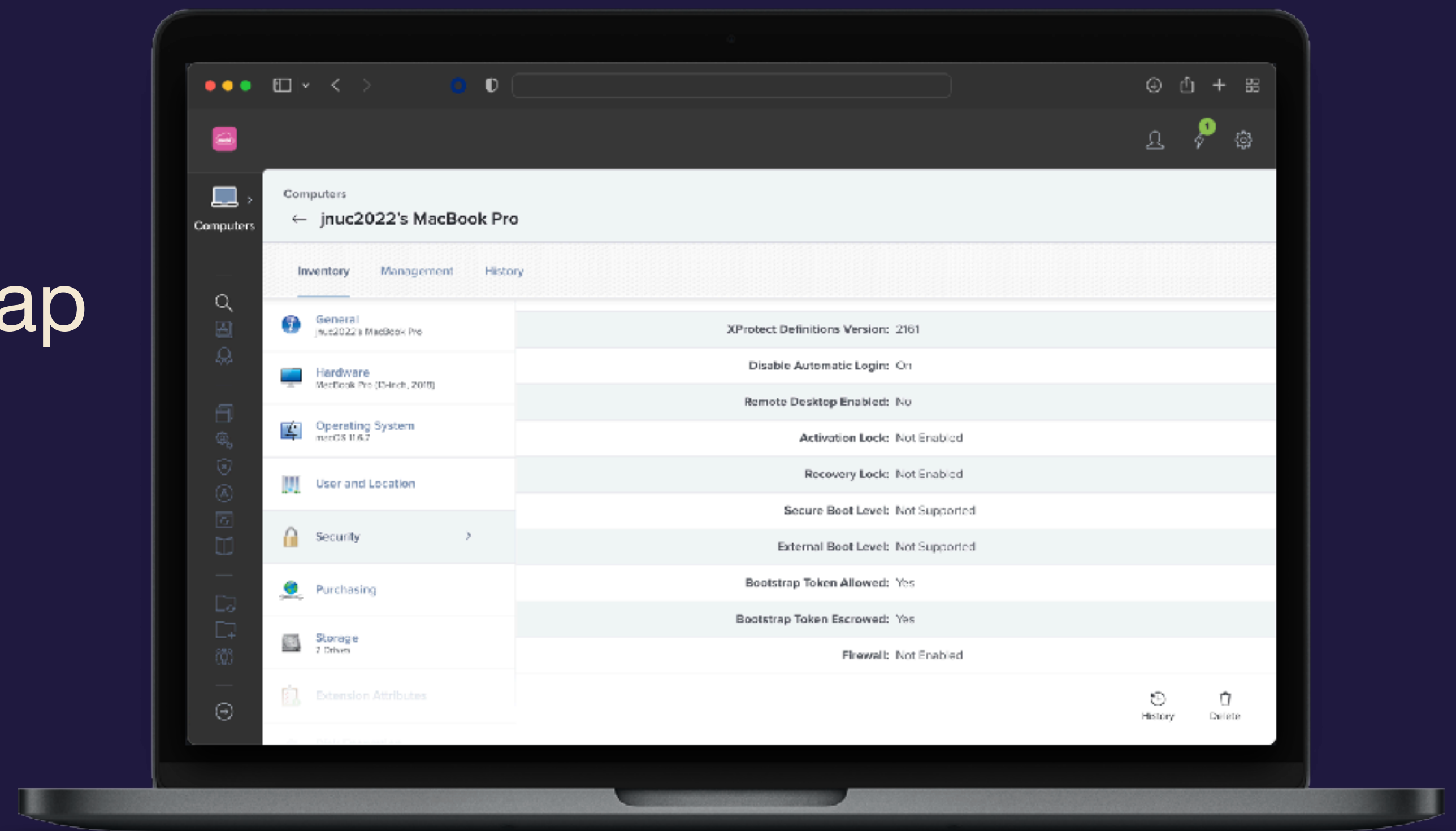
Inventory > Security > Bootstrap
Token Allowed*



Bootstrap Tokens

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Token Allowed*

(* requires macOS 11+)



Bootstrap Tokens



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Bootstrap Tokens

“In macOS 11 or later, the bootstrap token may also be used for more than just granting secure token to user accounts.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Bootstrap Tokens

“In macOS 11 or later, the bootstrap token may also be used for more than just granting secure token to user accounts.”

“On a Mac computer with Apple silicon, the bootstrap token, if available, can be used to authorise the installation of both kernel extensions and software updates when managed using MDM.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Bootstrap Tokens

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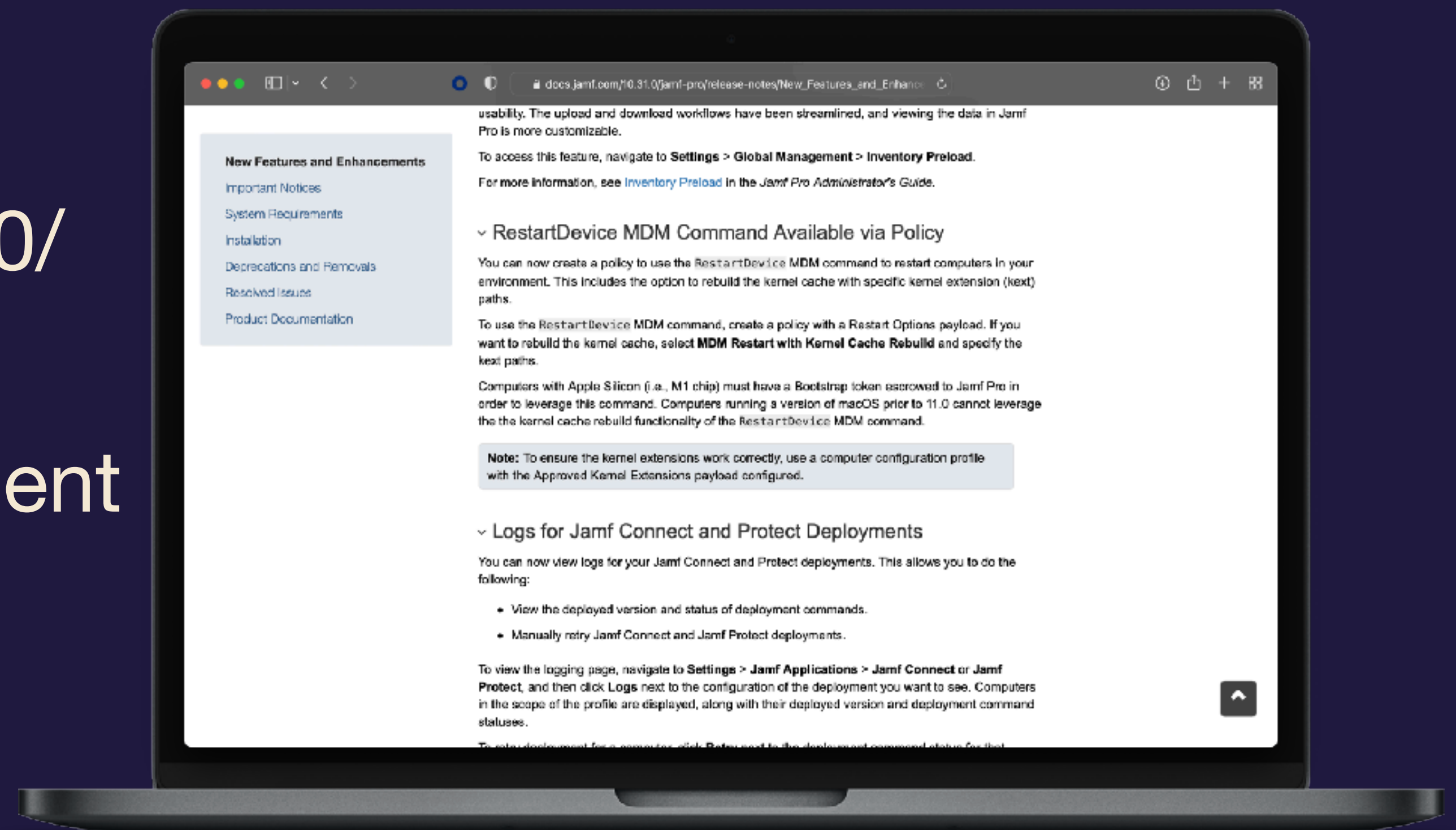
“The bootstrap token is also used to silently authorise an Erase all Content and Settings command when triggered through MDM on macOS 12.0.1 or later.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

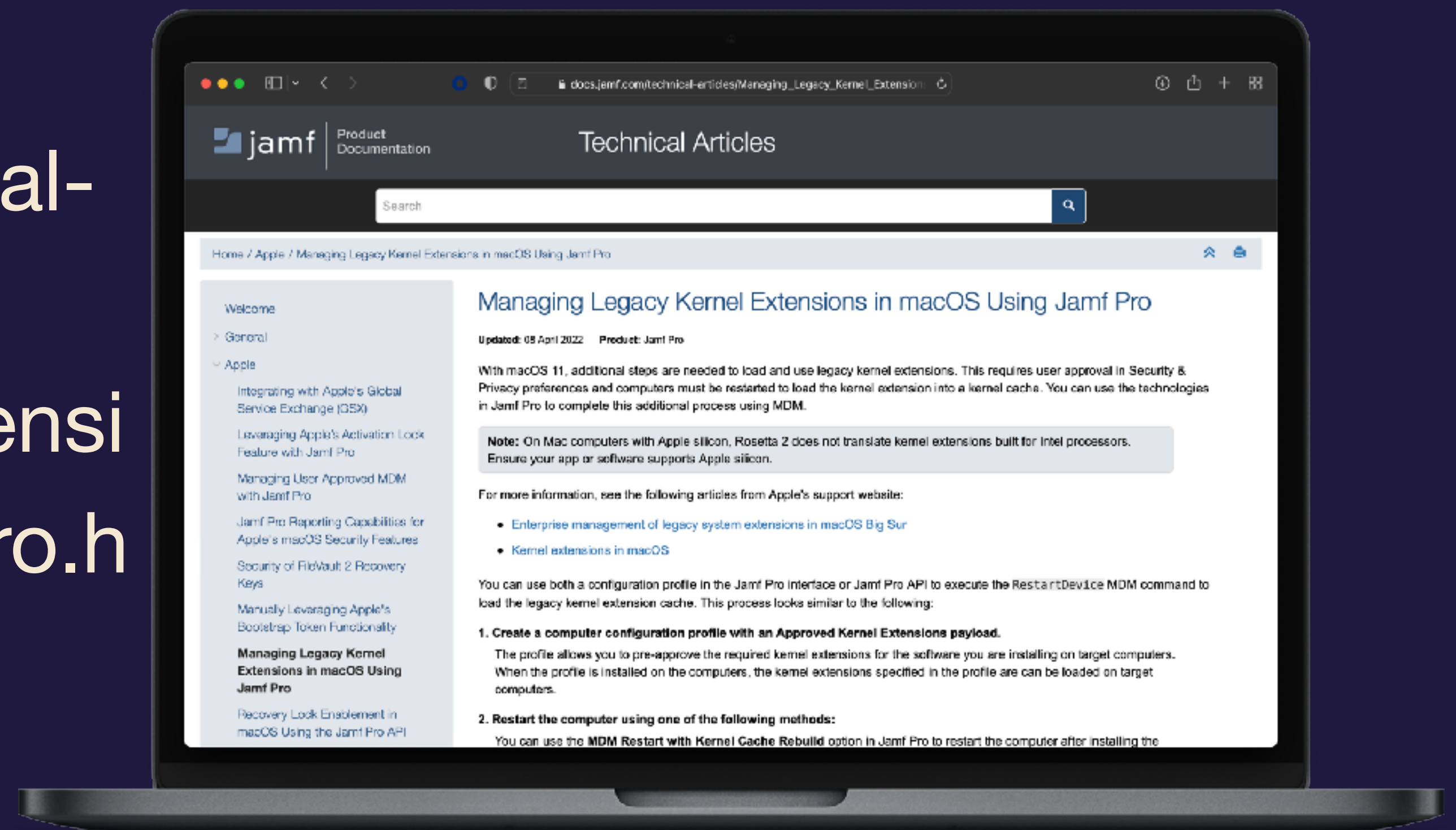
Bootstrap Tokens

[https://docs.jamf.com/10.31.0/
jamf-pro/release-notes/
New_Features_and_Enhancement
s.html](https://docs.jamf.com/10.31.0/jamf-pro/release-notes/New_Features_and_Enhancements.html)



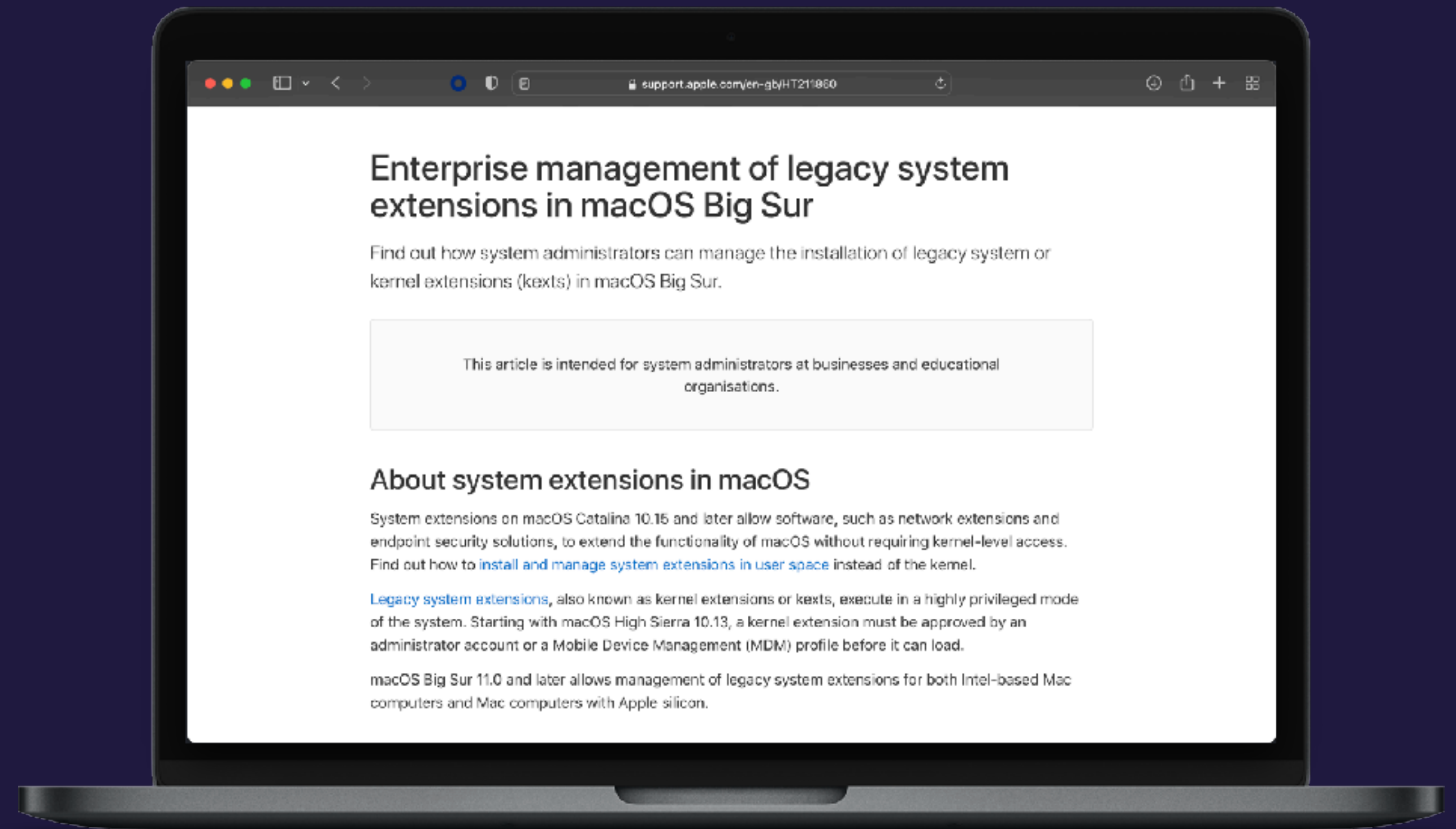
Bootstrap Tokens

https://docs.jamf.com/technical-articles/Managing_Legacy_Kernel_Extensions_in_macOS_Using_Jamf_Pro.html



Bootstrap Tokens

[https://support.apple.com/
HT211860](https://support.apple.com/HT211860)



Bootstrap Tokens



<https://support.apple.com/HT211860>

Bootstrap Tokens

“Before you can install a kernel extension on a Mac computer with Apple silicon, the security policy must be changed in one of the following ways:

- If you have devices enrolled in MDM with Automated Device Enrolment, you can authorise remote management of kernel extensions automatically and change the security policy.*”



<https://support.apple.com/HT211860>

Bootstrap Tokens

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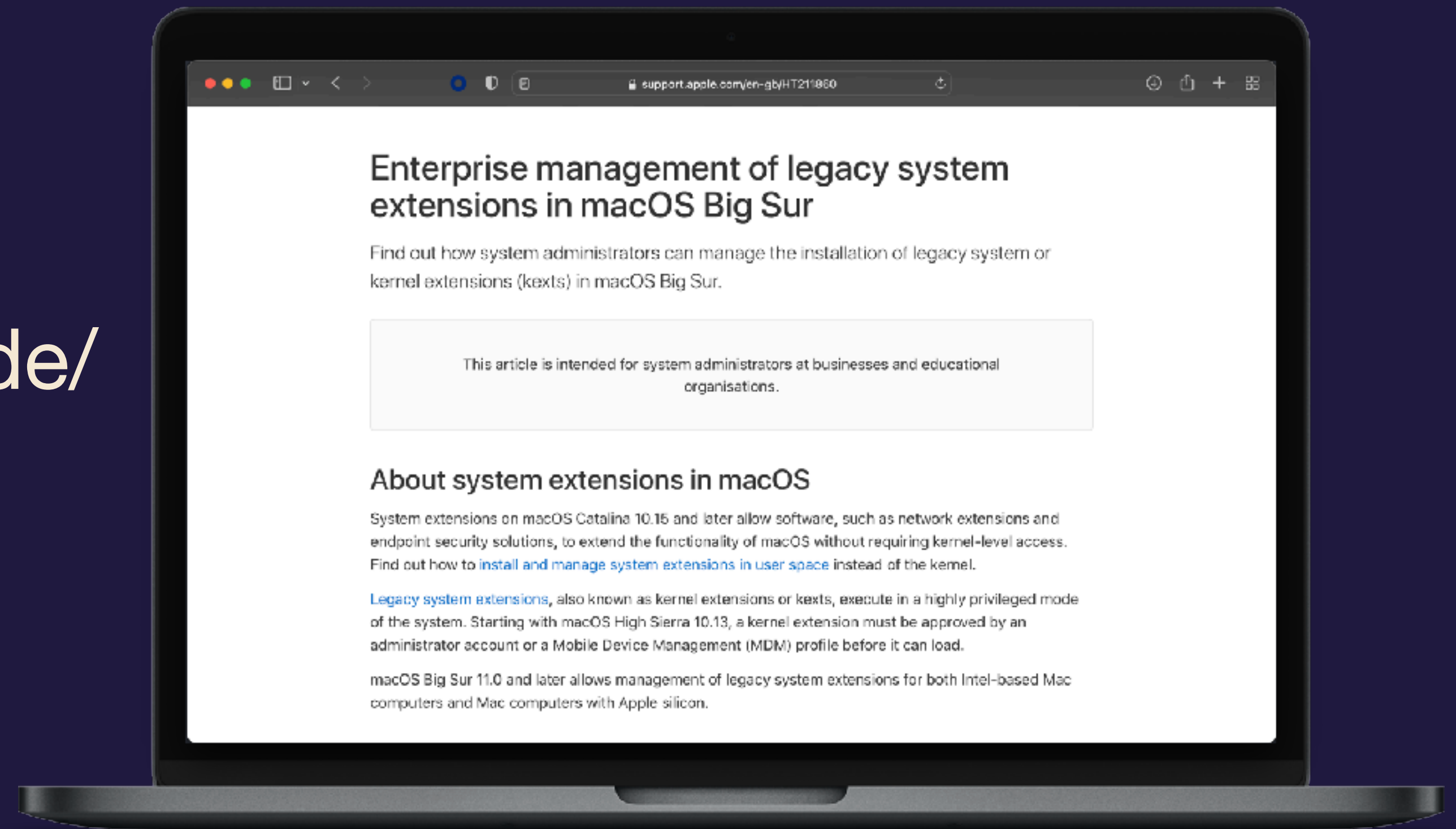
“ The MDM also must be able to support a bootstrap token. Additionally, the client must send the bootstrap token to the MDM server before the MDM has tried to perform an operation that requires the bootstrap token.”*

<https://support.apple.com/HT211860>



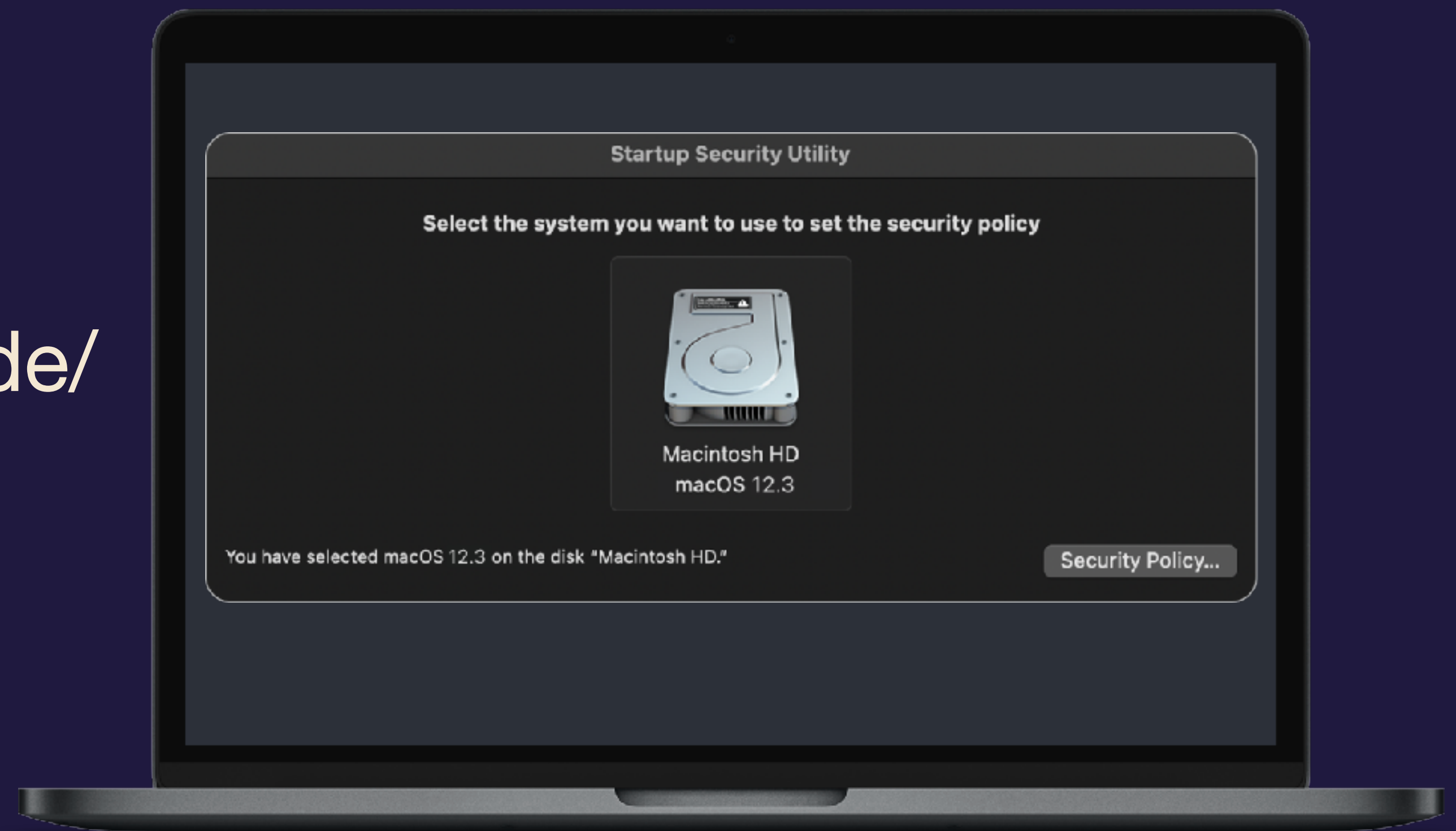
Bootstrap Tokens

<https://support.apple.com/guide/security/sec7d92dc49f/web>



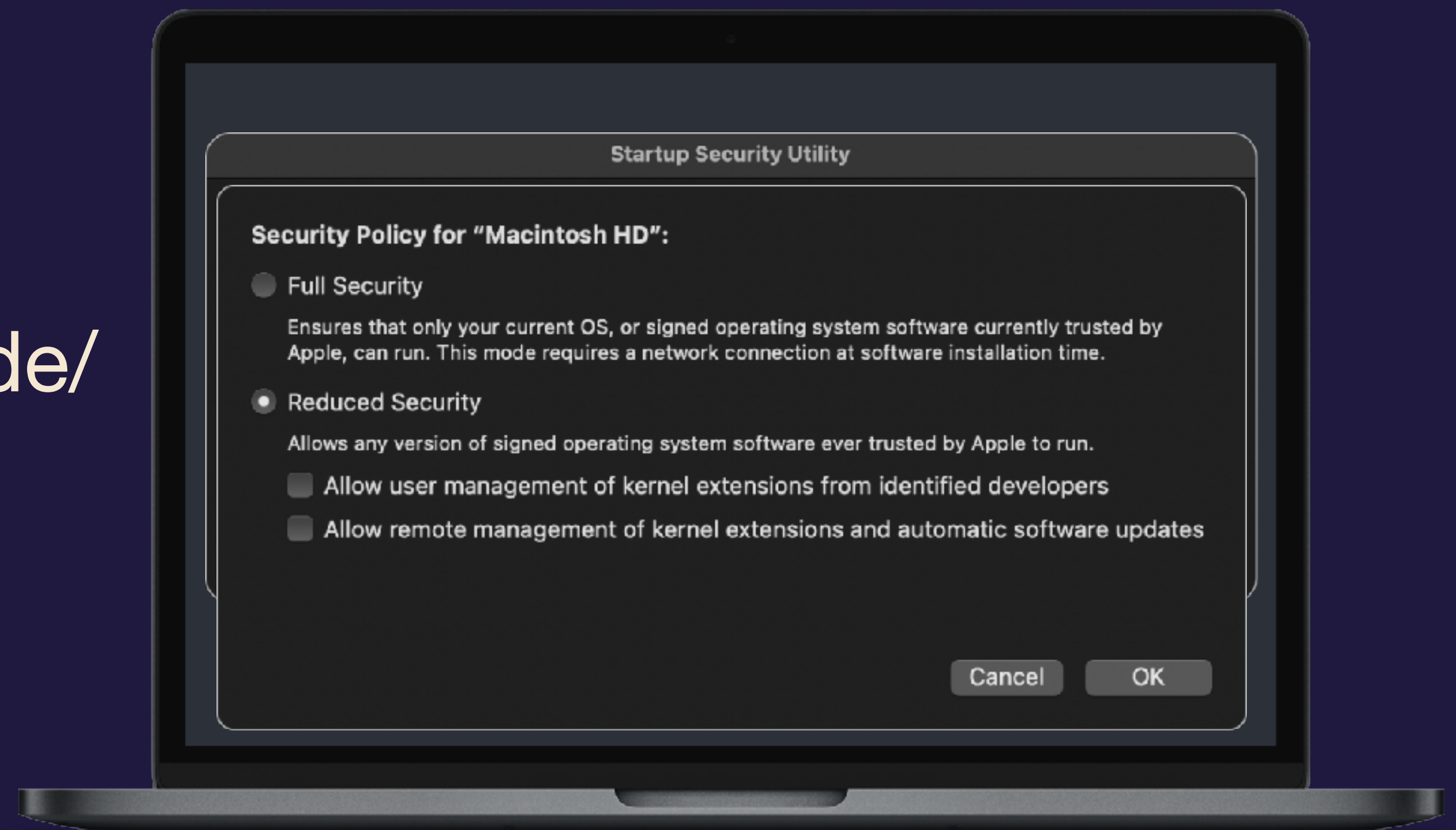
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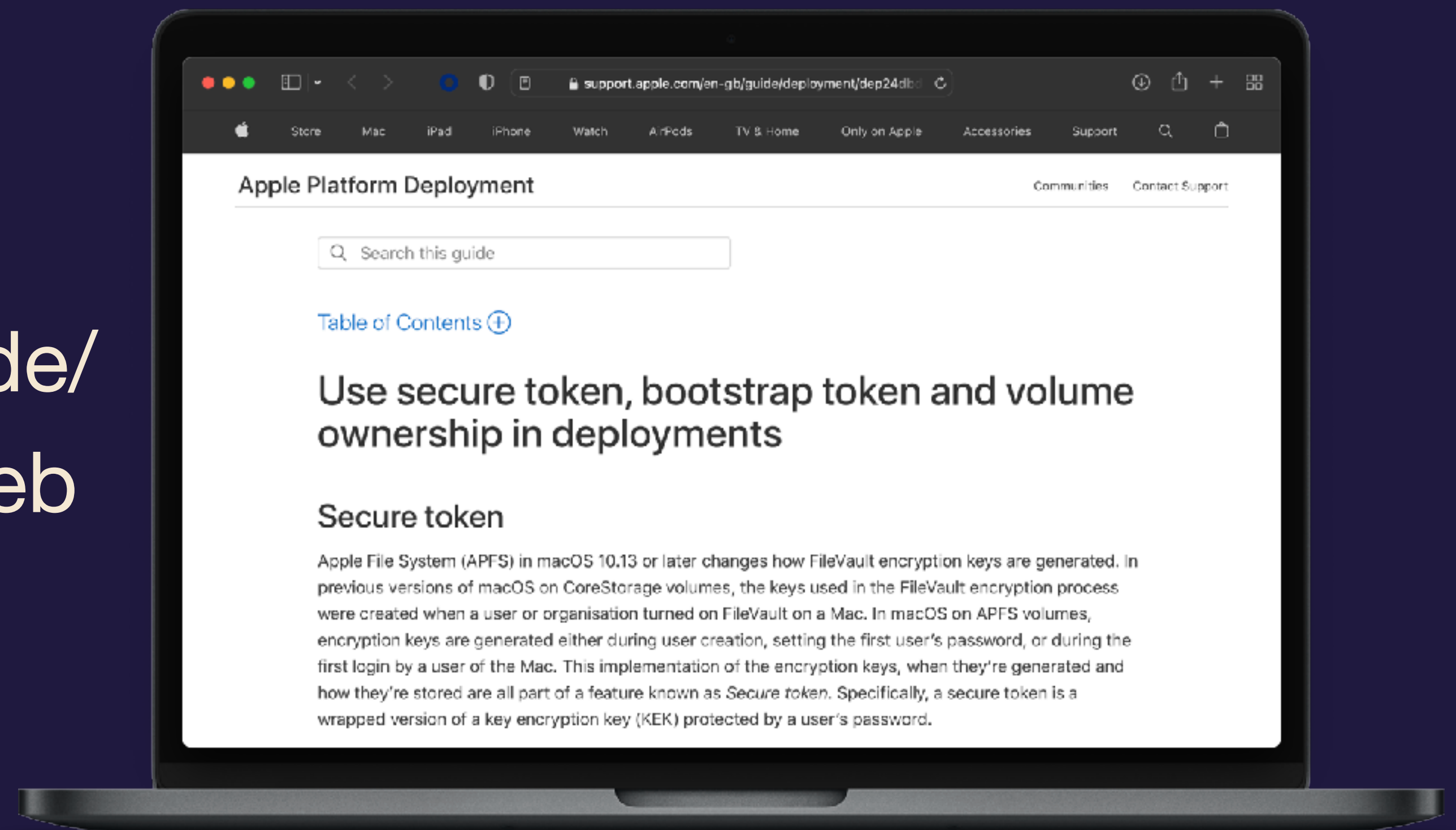


Volume ownership



Volume ownership

<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>



Volume ownership



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership

“The user that first claimed a Mac by configuring it for their use is granted a secure token on a Mac with Apple silicon and becomes the first volume owner.”

“When a bootstrap token is available and in use, it also becomes a volume owner and then grants volume ownership status to additional accounts as it grants them secure tokens.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership

“Because both the first user to be granted a secure token and the bootstrap token become volume owners, as well as the bootstrap token’s ability to grant a secure token to additional users (and thus volume ownership status as well), volume ownership should not be something that needs to be actively managed or manipulated in an organisation.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership

“Previous considerations for managing and granting a secure token should generally align with volume ownership status as well.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership

“It’s possible to be a volume owner and not be an administrator, but certain tasks require checking the ownership of both.”

“For example, modifying startup security settings requires being both an administrator and a volume owner, while authorising software updates is allowed by standard users and only requires ownership.”



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership



<https://support.apple.com/guide/deployment/dep24dbdcf9e/web>

Volume ownership

“Major macOS upgrades require elevated privileges to begin, but standard user ownership credentials can be used with the required user options on the macOS Installer’s startosinstall command-line tool (if it’s already running with elevated privileges).”

“However, if the --eraseinstall option is used, the credentials must be from a user who is both an *owner* and an *administrator*.”



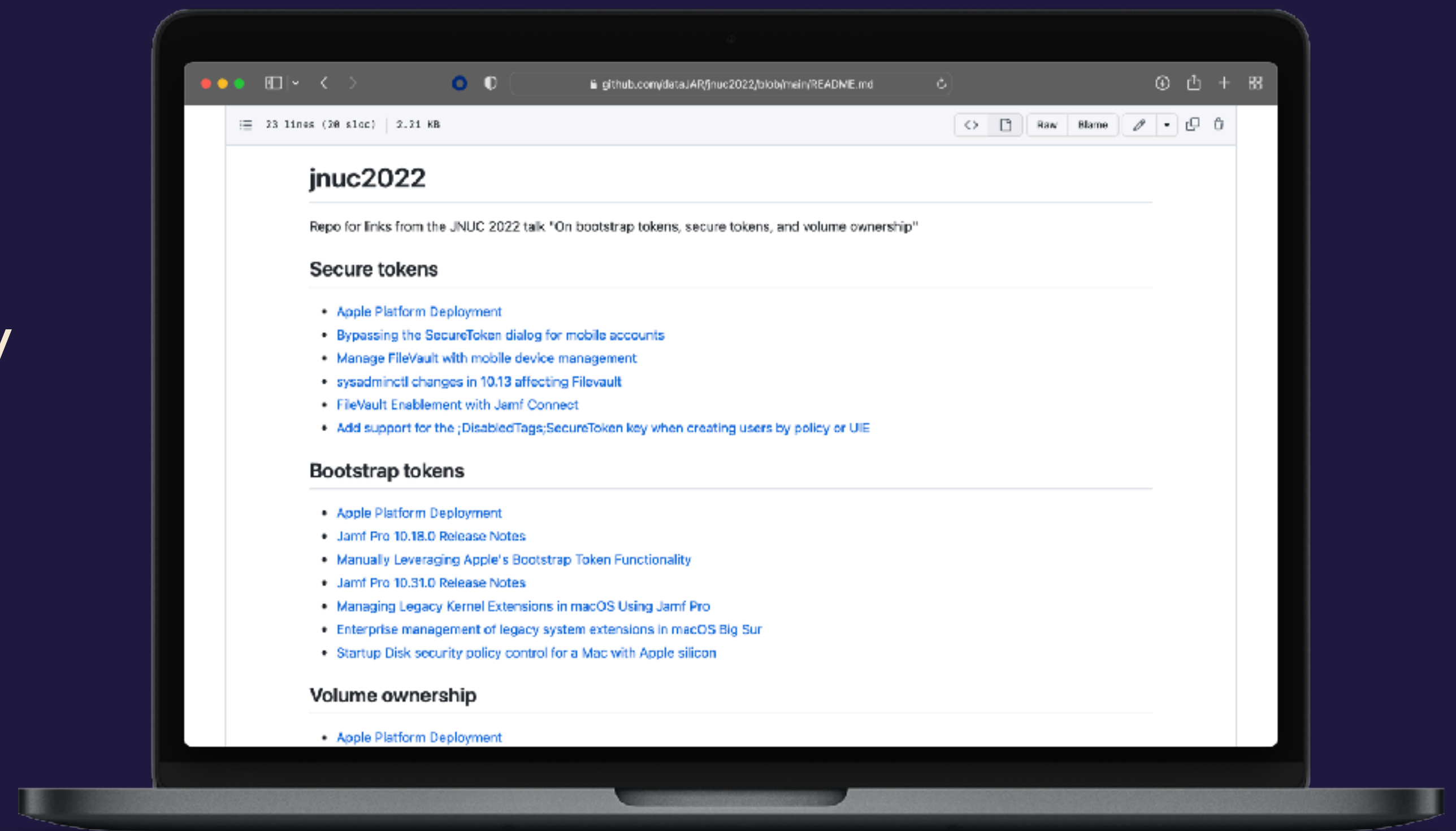
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Recap



Recap

[https://github.com/dataJAR/
jnuc2022](https://github.com/dataJAR/jnuc2022)



**Thank you for
listening!**

